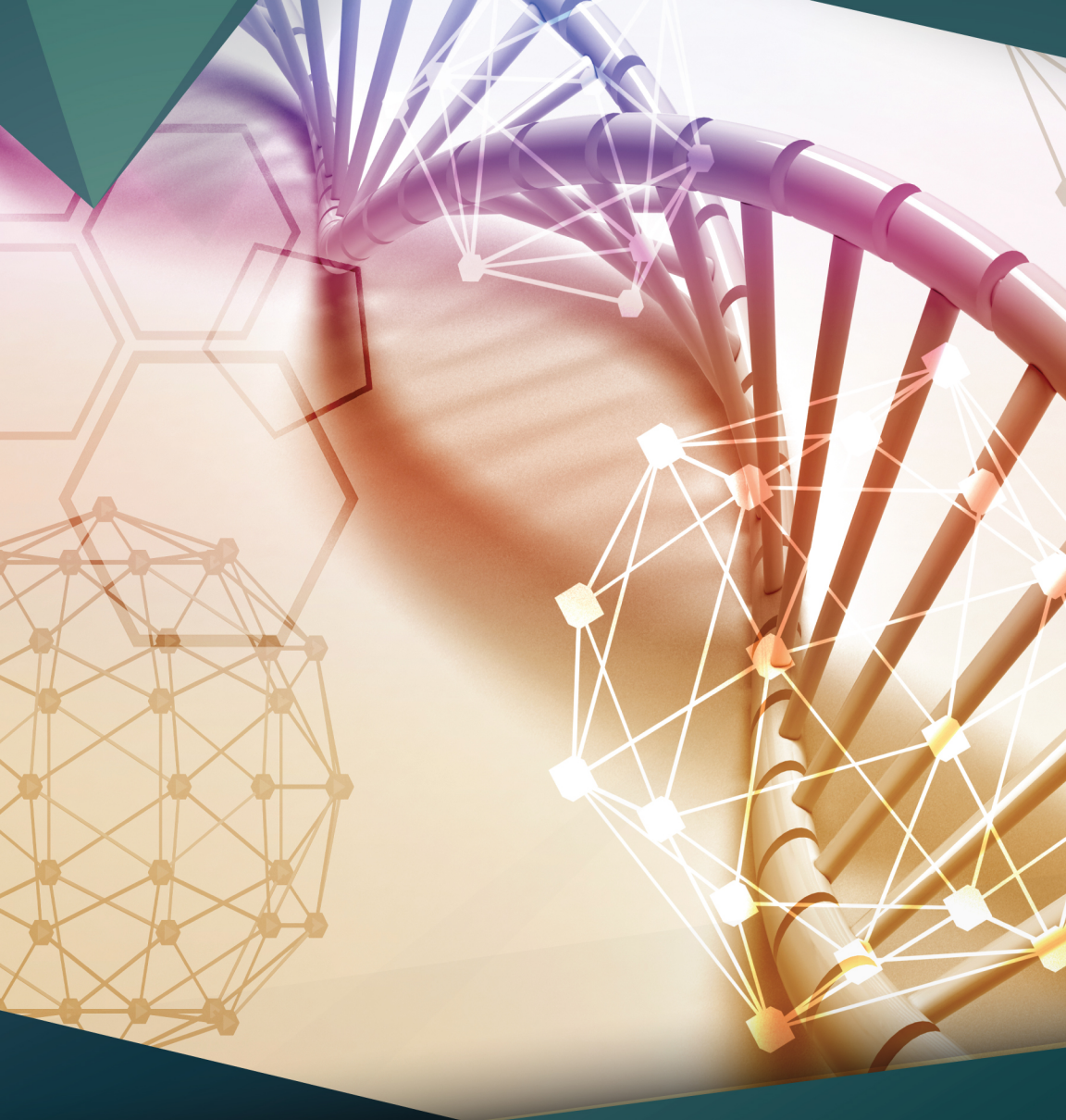




CORE CONCEPTS IN BIOLOGY:

Structure Of DNA

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PREFACE

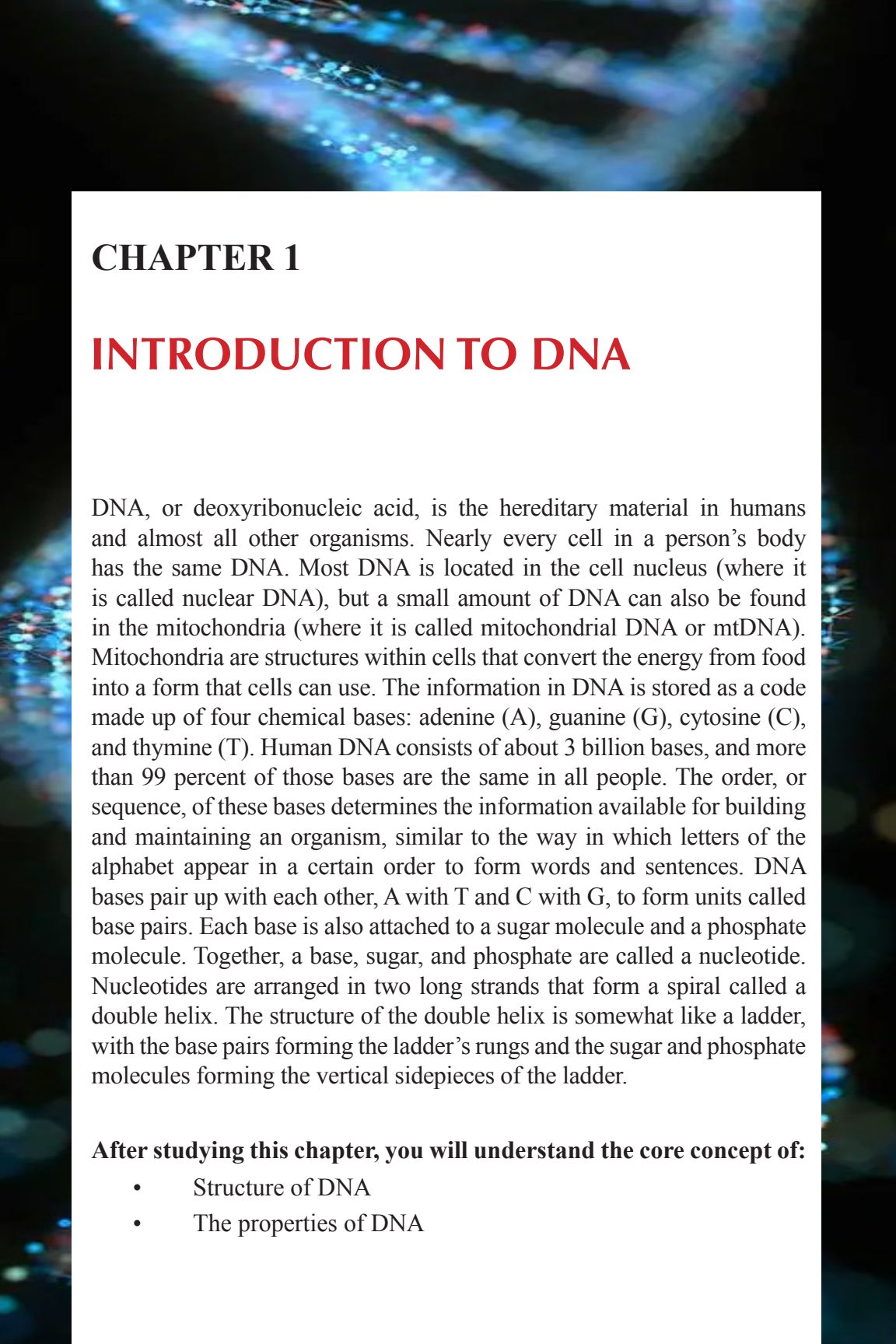
It has been argued that the discovery of DNA, as well as our understanding of its structure and functioning, could well be the most important discovery of the last century. It was hoped that understanding of the structure would show that DNA holds the genetic signals that are repeated as chromosomes split and produce two identical copies of themselves. In the late 1940s and early 1950s, numerous research groups in the United States and Europe have made serious efforts— both collaborative and rival— to explain how the atoms of DNA are bound together by covalent bonds and how the resulting molecules are formed in three-dimensional space.

Not surprisingly, there were initially concerns that DNA might have very complex and strange structures that were radically different from one gene to another. Great relief was therefore demonstrated when the basic DNA structure was found to be a double helix. This taught us that all genes have approximately the same three-dimensional structure and that the variations between the two genes exist in the order and number of their four nucleotide blocks along the complementary strands. The effect of the discovery of DNA on scientific and medical advancement has been immense, whether it is the detection of genes that cause major diseases or the creation and manufacture of drugs to treat these devastating diseases. In addition, the discovery of these genes and their subsequent study in terms of therapeutic treatment has eventually influenced and will continue to influence research in the future. DNA was particularly important in the field of forensic science.

The discovery of DNA ensures that it is possible to determine the guilt or innocence of a person who is being investigated for a crime. It also

means that scant evidence can still provide valuable information as to the perpetrator of the crime. It is also important that victim identification can occur, especially in cases where the condition of the victim is unrecognizable to family or friends. In this sense, DNA has been important in revolutionizing the entire field of forensic science. This impact is felt within the criminal justice system and contributes to the accurate safeguarding of society.

Core Concepts in Biology: Structure of DNA examines the structure of the DNA molecule and explain in general terms how it is able to store hereditary information. Obviously, the DNA and RNA structures are richer and more complex than was initially seen. Yes, there is no single standard DNA and RNA structure. As we will see in this book, there are in fact variations on standard structural themes that emerge from the unique physical, chemical and topological properties of the polynucleotide chain. An important book for living in today's world, this book provides background information that is critical to understanding how genetics is now influencing our everyday lives.



CHAPTER 1

INTRODUCTION TO DNA

DNA, or deoxyribonucleic acid, is the hereditary material in humans and almost all other organisms. Nearly every cell in a person's body has the same DNA. Most DNA is located in the cell nucleus (where it is called nuclear DNA), but a small amount of DNA can also be found in the mitochondria (where it is called mitochondrial DNA or mtDNA). Mitochondria are structures within cells that convert the energy from food into a form that cells can use. The information in DNA is stored as a code made up of four chemical bases: adenine (A), guanine (G), cytosine (C), and thymine (T). Human DNA consists of about 3 billion bases, and more than 99 percent of those bases are the same in all people. The order, or sequence, of these bases determines the information available for building and maintaining an organism, similar to the way in which letters of the alphabet appear in a certain order to form words and sentences. DNA bases pair up with each other, A with T and C with G, to form units called base pairs. Each base is also attached to a sugar molecule and a phosphate molecule. Together, a base, sugar, and phosphate are called a nucleotide. Nucleotides are arranged in two long strands that form a spiral called a double helix. The structure of the double helix is somewhat like a ladder, with the base pairs forming the ladder's rungs and the sugar and phosphate molecules forming the vertical sidepieces of the ladder.

After studying this chapter, you will understand the core concept of:

- Structure of DNA
- The properties of DNA

INTRODUCTION

Deoxyribonucleic acid, or DNA, is a molecule that contains the instructions an organism needs to develop, live and reproduce. These instructions are found inside every cell, and are passed down from parents to their children. DNA is a molecule composed of two chains that coil around each other to form a double helix carrying genetic instructions for the development, functioning, growth and reproduction of all known organisms and many viruses. DNA and ribonucleic acid (RNA) are nucleic acids; alongside proteins, lipids and complex carbohydrates (polysaccharides), nucleic acids are one of the four major types of macromolecules that are essential for all known forms of life. An important property of DNA is that it can replicate, or make copies of itself. Each strand of DNA in the double helix can serve as a pattern for duplicating the sequence of bases. This is critical when cells divide because each new cell needs to have an exact copy of the DNA present in the old cell.





CORE CONCEPT: STRUCTURE OF DNA

DNA (deoxyribonucleic acid) is a type of macromolecule known as a nucleic acid. It is shaped like a twisted double helix and is composed

of long strands of alternating sugars and phosphate groups, along with nitrogenous bases (adenine, thymine, guanine and cytosine). DNA is organized into structures called chromosomes and housed within the nucleus of our cells. DNA is also found in cell mitochondria.

DNA contains the genetic information necessary for the production of cell components, organelles, and for the reproduction of life. Protein production is a vital cell process that is dependent upon DNA. Information contained within the genetic code is passed from DNA to RNA to the resulting proteins during protein synthesis.

DNA is made up of molecules called nucleotides. Each nucleotide contains a phosphate group, a sugar group and a nitrogen base. The four types of nitrogen bases are adenine (A), thymine (T), guanine (G) and cytosine (C). The order of these bases is what determines DNA's instructions, or genetic code. Human DNA has around 3 billion bases, and more than 99 percent of those bases are the same in all people, according to the U.S. National Library of Medicine (NLM).

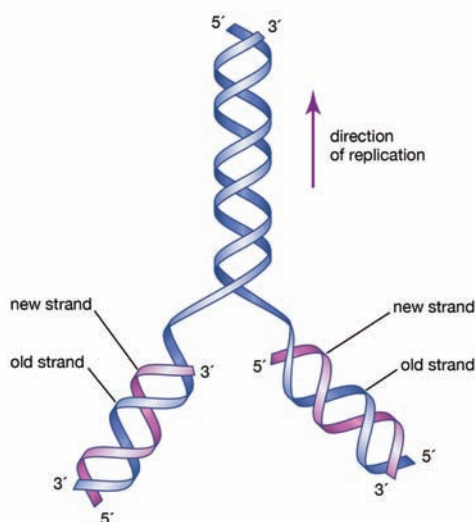
DNA Shape and Structure

DNA is composed of a sugar-phosphate backbone and nitrogenous bases. In double-stranded DNA, the nitrogenous bases pair up. Adenine pairs with thymine (A-T) and guanine pairs with cytosine (G-C). The shape of DNA resembles that of a spiral staircase. In this double helical shape, the sides of the staircase are formed by strands of deoxyribose sugar and phosphate molecules. The stair steps are formed by the nitrogenous bases.

Chromatin is a mass of genetic material composed of DNA and proteins that condense to form chromosomes during eukaryotic cell division.

The twisted double helix shape of DNA helps to make this biological molecule more compact. DNA is further compressed into structures called chromatin so that it can fit within the nucleus.

Chromatin is composed of DNA that is wrapped around small proteins known as histones. Histones help to organize DNA into structures called nucleosomes, which form chromatin fibers. Chromatin fibers are further coiled and condensed into chromosomes.



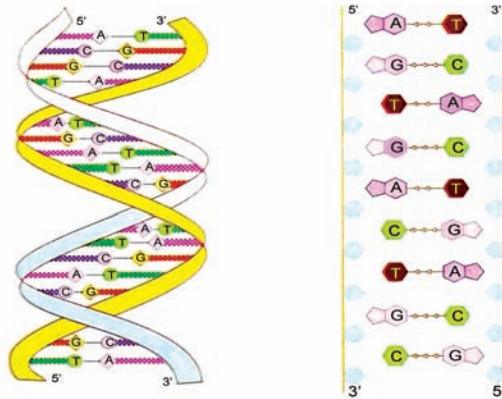
DNA or deoxyribonucleic acid is a type of molecule known as a nucleic acid. It consists of a 5-carbon deoxyribose sugar, a phosphate, and a nitrogenous base. Double-stranded DNA consists of two spiral nucleic acid chains that are twisted into a double helix shape. This twisting allows DNA to be more compact. In order to fit within the nucleus, DNA is packed into tightly coiled structures called chromatin. Chromatin condenses to form chromosomes during cell division. Prior to DNA replication, the chromatin loosens giving cell replication machinery access to the DNA strands. Deoxyribonucleic acid, also abbreviated as DNA, is the principal informational macromolecule

of the cell, which stores, translates and transfers the genetic information. In the prokaryotes, the DNA is found mostly in the nuclear zone. In eukaryotes it is found in the nucleus, mitochondria and chloroplast. The present understanding of the storage and utilization of the cell's genetic information is based upon the discovery of the structure of DNA by Watson and Crick in 1953.

Double Helical Structure of DNA (Watson and Crick Model)

The three dimensional structure of DNA as proposed by Watson and Crick and the recent advances in it are summarized here:

1. DNA is made of two helical chains coiled around the same axis, to form a right-handed double helix.
2. The two chains in the helix are anti-parallel to each other, i.e., the 5'-end of one polynucleotide chain and the 3'-end of the other polynucleotide chain is on the same side and close together.

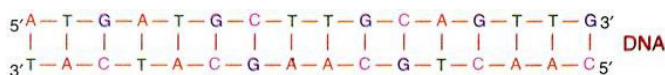


Double helical structure of DNA

3. The distance between each turn is 3.6 nm (formerly 3.4 nm).
4. There are 10.5 nucleotides per turn (formerly 10 nucleotides).

Cytosine is one of the four main bases found in DNA and RNA, along with adenine, guanine, and thymine (uracil in RNA).

5. The spatial relationship between the two strands creates major and minor grooves between the two strands. In these grooves some proteins interact.
6. The hydrophilic backbones of alternating deoxyribose and negatively charged phosphate groups are on the outside of the double helix.
7. The hydrophobic pyrimidine and purine bases are inside the double helix, which stabilizes the double helix of the DNA.
8. The double helix is also stabilized by inter-chain hydrogen bond formed between a purine and pyrimidine base.
9. A particular purine base, pairs by hydrogen bonds, only with a particular pyrimidine base, i.e., Adenine (A) pairs with Thymine (T) and Guanine (G) pairs with **Cytosine** (C) only.
10. Two hydrogen bonds pairs Adenine and Thymine ($A = T$), whereas three hydrogen bonds pairs Guanine and Cytosine ($G \equiv C$).
11. The base pairs $A = T$ and $G \equiv C$ are known as complementary base pairs.
12. Due to the presence of complementary base pairing, the two chains of the DNA double helix are complementary to each other.



Hence the number of A' bases are equal to the number of T' bases (or 'G' is equal to 'C') in a given double stranded DNA.

13. One of the strands in the double helix is known as sense strand, i.e., which codes for RNA/proteins and the other strand is known as antisense strand.

Different Structural Forms of DNA

The DNA molecules exist in four different structural forms or organizations under different physiological conditions or in different cells or at different points in the same DNA

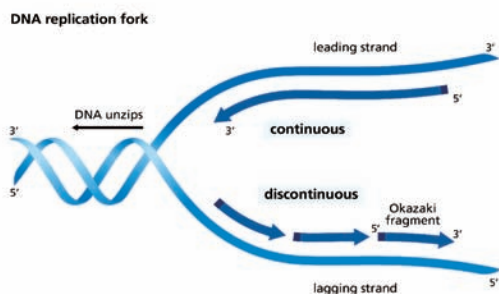
Comparison between difference structure of DNA

A		B	Z	H
I.	Shorter and wider	Normal reference strand– Watson and Crick model	Longer and thinner	It is a long stretch (part) of DNA with alternating T and C or polypurine/ polypyrimidine
	Right-handed double helix	Right-handed double helix	Left-handed double helix	Triple helix
	Distance between each turn is 2.3 nm	Distance between each turn is 3.6 nm	Distance between each turn is 3.8 nm	—
	11 base pairs per turn	10.5 base pairs per turn	12 base pairs per turn	—
	Stable in solutions devoid of water	Most stable under physiological conditions	Doubtful existence in physiological state	Helps in gene regulation

DNA Replication

The double helix shape of DNA makes DNA replication possible. In replication, DNA makes a copy of itself in order to pass genetic information on to newly formed daughter cells. In order for replication to take place, the DNA must unwind to allow cell replication machinery to copy each strand. Each replicated molecule is composed of a strand from the original DNA molecule and a newly formed strand. Replication produces genetically identical DNA molecules. DNA replication occurs in interphase, a stage prior to the start of the division processes of mitosis and meiosis.

DNA replication is the process by which DNA makes a copy of itself during cell division.



DNA is the genetic material that defines every cell. Before a cell duplicates and is divided into new daughter cells through either mitosis or meiosis, biomolecules and organelles must be copied to be distributed among the cells. DNA, found within the nucleus, must be replicated in order to ensure that each new cell receives the correct number of chromosomes. The process of DNA duplication is called DNA replication. Replication follows several steps that involve multiple proteins called replication enzymes and RNA. In eukaryotic cells, such as animal cells and plant cells, **DNA replication** occurs in the S phase of interphase during the cell cycle. The process of DNA replication is vital for cell growth, repair, and reproduction in organisms.

Preparation for Replication

- **Step 1: Replication Fork Formation:** Before DNA can be replicated, the double stranded molecule must be “unzipped” into two single strands. DNA has four bases called adenine (A), thymine (T), cytosine (C) and guanine (G) that form pairs between the two strands. Adenine only pairs with thymine and cytosine only binds with guanine. In order to unwind DNA, these interactions between base pairs must be broken. This is performed by an

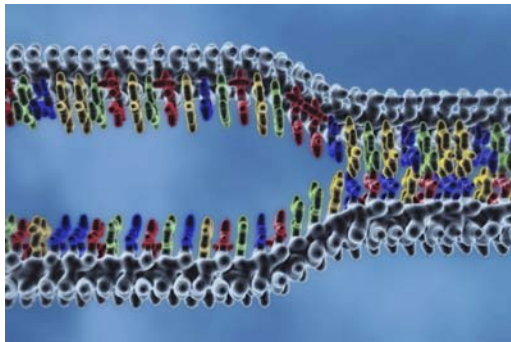
enzyme known as DNA helicase. DNA helicase disrupts the hydrogen bonding between base pairs to separate the strands into a Y shape known as the replication fork. This area will be the template for replication to begin.

DNA is directional in both strands, signified by a 5' and 3' end. This notation signifies which side group is attached the DNA backbone. The 5' end has a phosphate (P) group attached, while the 3' end has a hydroxyl (OH) group attached. This directionality is important for replication as it only progresses in the 5' to 3' direction. However, the replication fork is bi-directional; one strand is oriented in the 3' to 5' direction (leading strand) while the other is oriented 5' to 3' (lagging strand). The two sides are therefore replicated with two different processes to accommodate the directional difference.

- **Step 2: Primer Binding:** The leading strand is the simplest to replicate. Once the DNA strands have been separated, a short piece of RNA called a primer binds to the 3' end of the strand. The primer always binds as the starting point for replication. Primers are generated by the enzyme DNA primase.
- **Step 3: Elongation:** Enzymes known as DNA polymerases are responsible creating the new strand by a process called elongation. There are five different known types of DNA polymerases in bacteria and human cells. In bacteria such as *E. coli*, polymerase III is the main replication enzyme, while polymerase I, II, IV and V are responsible for error checking and repair. DNA polymerase III binds to the strand at the site of the primer and begins adding new base pairs complementary to the strand during replication. In eukaryotic cells, polymerases alpha, delta, and epsilon are the primary polymerases involved in DNA replication. Because replication proceeds in the 5' to 3' direction on the leading strand, the newly formed strand is continuous.

The lagging strand begins replication by binding with multiple primers. Each primer is only several bases apart. DNA polymerase then adds pieces of DNA, called Okazaki fragments, to the strand between primers. This process of replication is discontinuous as the newly created fragments are disjointed.

- **Step 4: Termination:** Once both the continuous and discontinuous strands are formed, an enzyme called exonuclease removes all RNA primers from the original strands. These primers are then replaced with appropriate bases. Another exonuclease “proofreads” the newly formed DNA to check, remove and replace any errors. Another enzyme called DNA ligase joins Okazaki fragments together forming a single unified strand. The ends of the linear DNA present a problem as DNA polymerase can only add nucleotides in the 5' to 3' direction. The ends of the parent strands consist of repeated DNA sequences called telomeres. Telomeres act as protective caps at the end of chromosomes to prevent nearby chromosomes from fusing. A special type of DNA polymerase enzyme called telomerase catalyzes the synthesis of telomere sequences at the ends of the DNA. Once completed, the parent strand and its complementary DNA strand coils into the familiar double helix shape. In the end, replication produces two DNA molecules, each with one strand from the parent molecule and one new strand.



DNA replication would not occur without enzymes that catalyze various steps in the process. Enzymes that participate in the eukaryotic DNA replication process include:

- **DNA helicase** - unwinds and separates double stranded DNA as it moves along the DNA. It forms the replication fork by breaking hydrogen bonds between nucleotide pairs in DNA.
- **DNA primase** - a type of RNA polymerase that generates RNA primers. Primers are short RNA molecules that act as templates

for the starting point of DNA replication.

- ***DNA polymerases*** - synthesize new DNA molecules by adding nucleotides to leading and lagging DNA strands.
- ***Topoisomerase or DNA Gyrase*** - unwinds and rewinds DNA strands to prevent the DNA from becoming tangled or supercoiled.
- ***Exonucleases*** - group of enzymes that remove nucleotide bases from the end of a DNA chain.
- ***DNA ligase*** - joins DNA fragments together by forming phosphodiester bonds between nucleotides.

DNA replication is the production of identical DNA helices from a single double-stranded DNA molecule. Each molecule consists of a strand from the original molecule and a newly formed strand. Prior to replication, the DNA uncoils and strands separate. A replication fork is formed which serves as a template for replication. Primers bind to the DNA and DNA polymerases add new nucleotide sequences in the 5' to 3' direction. This addition is continuous in the leading strand and fragmented in the lagging strand. Once elongation of the DNA strands is complete, the strands are checked for errors, repairs are made, and telomere sequences are added to the ends of the DNA.

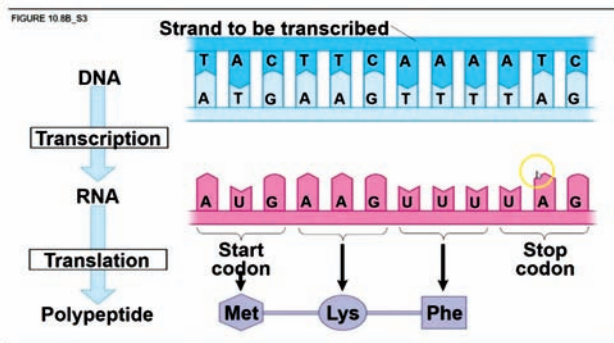
DNA Translation

DNA translation is the process for the synthesis of proteins. Segments of DNA called genes contain genetic sequences or codes for the production of specific proteins. In order for translation to occur, the DNA must first unwind and allow DNA transcription to take place. In transcription, the DNA is copied and an RNA version of the DNA code (RNA transcript) is produced. With the help of cell ribosomes and transfer

Protein synthesis, a messenger RNA (mRNA) molecule copied from DNA provides the instruction for the synthesis of a specific protein.

RNA, the RNA transcript undergoes translation and **proteins synthesis**.

DNA transcription is a process that involves transcribing genetic information from DNA to RNA. The transcribed DNA message, or RNA transcript, is used to produce proteins. DNA is housed within the nucleus of our cells. It controls cellular activity by coding for the production of proteins. The information in DNA is not directly converted into proteins, but must first be copied into RNA. This ensures that the information contained within the DNA does not become tainted.



DNA consists of four nucleotide bases that are paired together to give DNA its double helical shape. These bases are: adenine (A), guanine (G), cytosine (C), and thymine (T). Adenine pairs with thymine (A-T) and cytosine pairs with guanine (C-G). Nucleotide base sequences are the genetic code or instructions for protein synthesis.

There are three main steps to the process of DNA transcription:

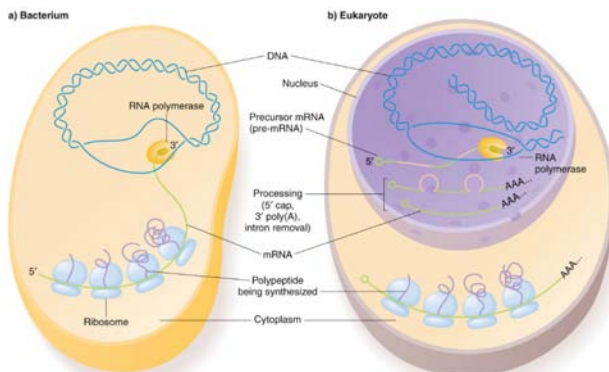
- **RNA Polymerase Binds to DNA:** DNA is transcribed by an enzyme called RNA polymerase. Specific nucleotide sequences tell RNA polymerase where to begin and where to end. RNA polymerase attaches to the DNA at a specific area called the promoter region. The DNA in the promoter region contains specific sequences that allow RNA polymerase to bind to the DNA.
- **Elongation:** Certain enzymes called transcription factors unwind the DNA strand and allow RNA polymerase to transcribe only a single strand of DNA into a single stranded RNA polymer called messenger RNA (mRNA). The strand that serves as the template is called the antisense strand. The strand that is not transcribed is called the sense strand.

Like DNA, RNA is composed of nucleotide bases. RNA however, contains the nucleotides adenine, guanine, cytosine, and uracil (U). When RNA polymerase transcribes the DNA, guanine pairs with cytosine (G-C) and adenine pairs with uracil (A-U).

- **Termination:** RNA polymerase moves along the DNA until it reaches a terminator sequence. At that point, RNA polymerase releases the mRNA polymer and detaches from the DNA.

Transcription in Prokaryotic and Eukaryotic Cells

While transcription occurs in both prokaryotic and eukaryotic cells, the process is more complex in eukaryotes. In prokaryotes, such as bacteria, the DNA is transcribed by one RNA polymerase molecule without the assistance of transcription factors. In eukaryotic cells, transcription factors are needed for transcription to occur and there are different types of RNA polymerase molecules that transcribe the DNA depending on the type of genes. Genes that code for proteins are transcribed by RNA polymerase II, genes coding for ribosomal RNAs are transcribed by RNA polymerase I, and genes that code for transfer RNAs are transcribed by RNA polymerase III. In addition, organelles such as mitochondria and chloroplasts have their own RNA polymerases which transcribe the DNA within these cell structures.



From Transcription to Translation

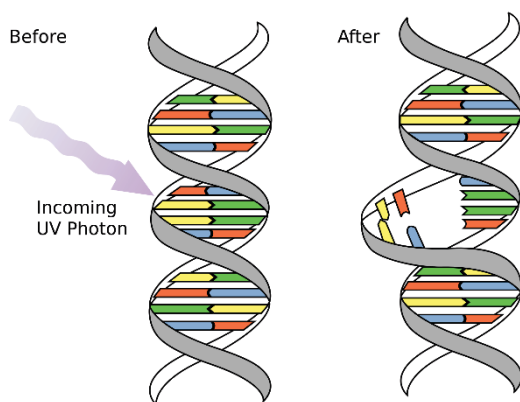
Since proteins are constructed in the cytoplasm of the cell, mRNA must cross the nuclear membrane to reach the cytoplasm in eukaryotic cells. Once in the cytoplasm, ribosomes and another RNA molecule called

RNA polymerase is an enzyme that is responsible for copying a DNA sequence into an RNA sequence, during the process of transcription.

transfer RNA work together to translate mRNA into a protein. This process is called translation. Proteins can be manufactured in large quantities because a single DNA sequence can be transcribed by many **RNA polymerase** molecules at once.

DNA Mutation

Any change in the sequence of nucleotides in DNA is known as a gene mutation. These changes can affect a single nucleotide pair or larger gene segments of a chromosome. Gene mutations are caused by mutagens such as chemicals or radiation, and can also result from errors made during cell division.



DNA Modeling

DNA stands for deoxyribonucleic acid. It is housed within the nucleus of our cells and contains the genetic information for the reproduction of life. The structure of DNA was discovered by James Watson and Francis Crick in the 1950's.

DNA is a type of macromolecule known as a nucleic acid. It is shaped like a twisted double helix and is composed of long strands of alternating sugars and phosphate groups, as well as nitrogenous bases

(adenine, thymine, guanine and cytosine). DNA controls cellular activity by coding for the production of enzymes and proteins. The information in DNA is not directly converted into proteins, but must first be copied into RNA in a process called transcription.

DNA as the Genetic Material

Deoxyribonucleic acid carries the genetic information encoded in the sequence of the four bases—adenine, guanine, cytosine, and thymine. The information in DNA is transferred to its daughter molecules through replication (the duplication of DNA molecules) and subsequent cell division. DNA directs the synthesis of proteins through the intermediary molecule RNA. The DNA code is transferred to RNA by a process known as transcription. The RNA code is then translated into a sequence of amino acids during protein synthesis. This is the central dogma of molecular biology: DNA makes RNA makes protein.

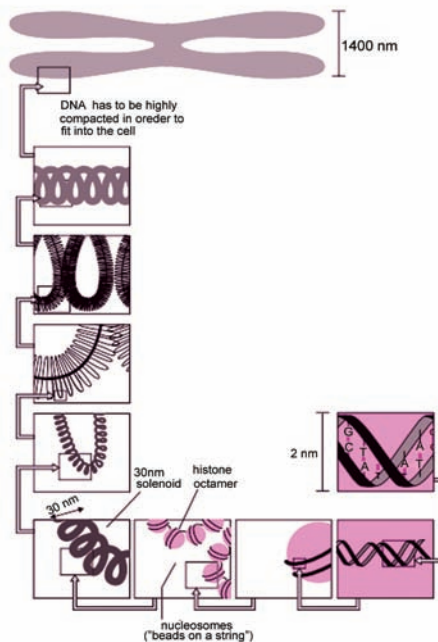
Retroviruses such as human immunodeficiency virus, the cause of AIDS, are an exception to this rule. As their name suggests, they reverse the normal order of data transfer. Inside the virus coat is a molecule of RNA plus an enzyme that can make DNA from an RNA template by the process known as reverse transcription.

DNA models can be constructed from almost anything including candy, paper, and even jewelry. An important thing to remember when constructing your model is to identify the components you will use to represent the nucleotide bases, sugar molecule, and phosphate molecule. When connecting the nucleotide base pairs be sure to connect the ones that pair naturally in DNA.

Packaging of DNA Molecules into Chromosomes

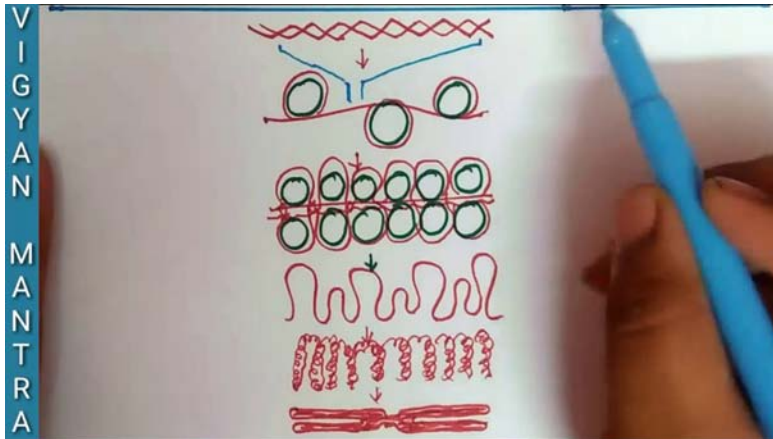
A human cell contains 46 chromosomes (23 pairs), each of which is a single DNA molecule bundled up with various proteins. On average, each human chromosome contains about 1.3×10^8 base pairs (BP) of DNA. If the DNA in a human chromosome were stretched as far as it would go without breaking it would be about 5 cm long, so the 46 chromosomes in all represent about 2 m of DNA. The nucleus in which this DNA must be

contained has a diameter of only about $10\mu\text{m}$, so large amounts of DNA must be packaged into a small space. This represents a formidable problem that is dealt with by binding the DNA to proteins to form chromatin. The DNA double helix is packaged at both small and larger scales. In the first stage, the DNA double helix with a diameter of 2 nm is bound to proteins known as histones. Histones are positively charged because they contain high amounts of the amino acids arginine and lysine and bind tightly to the negatively charged phosphates on DNA. A 146 bp length of DNA is wound around a protein complex composed of two molecules each of four different histones—H2A, H2B, H3, and H4—to form a nucleosome. Because each nucleosome is separated from its neighbor by about 50 bp of linker DNA, this unfolded chromatin state looks like beads on a string when viewed in an electron microscope. Nucleosomes undergo further packaging. A fifth type of histone, H1, binds to the linker DNA and pulls the nucleosomes together helping to further coil the DNA into chromatin fibers 30 nm in diameter, which are referred to as 30-nm solenoids. The fibers then form loops with the help of a class of proteins known as non-histones, and this further condenses the DNA into a higher order set of coils in a process called supercoiling.



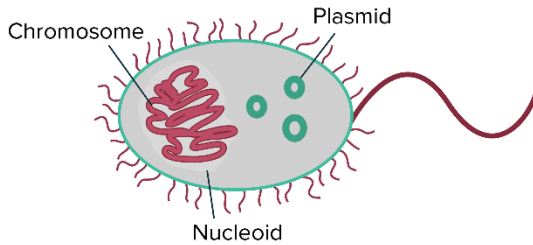
In a normal interphase cell about 10% of the chromatin is highly compacted and visible under the light microscope. This form of chromatin is called heterochromatin and is the portion of the genome where no RNA synthesis is occurring. The remaining interphase chromatin is less compacted and is known as chromatin.

Chromatin is in its most compacted form when the cell is preparing for mitosis. The chromatin folds and condenses further to form the 1400-nm-wide chromosomes we see under the light microscope. Because the cell is to divide, the DNA has been replicated, so that each **chromosome** is now formed by two chromatids, each one a DNA double helix. This means the progeny cell, produced by division of the progenitor cell, will receive a full set of 46 chromosomes.



Prokaryotic Chromosomes

The chromosome of the bacterium *E. coli* is a single circular DNA molecule of about 4.5×10^6 base pairs. It has a circumference of 1 mm, yet must fit into the 1- μ m cell, so like eukaryotic chromosomes it is coiled, supercoiled, and packaged with basic proteins that are similar to eukaryotic histones. However, an ordered nucleosome structure similar to the “beads on a string” seen in eukaryotic cells is not observed in prokaryotes. Prokaryotes do not have nuclear envelopes so the condensed chromosome together with its associated proteins lies free in the cytoplasm, forming a mass that is called the nucleoid to emphasize its functional equivalence to the eukaryotic nucleus.



Plasmids

Plasmids are small circular minichromosomes found in bacteria and some eukaryotes. They are several thousand base pairs long and are probably tightly coiled and supercoiled inside the cell. Plasmids often code for proteins that confer resistance to a particular antibiotic.

Viruses

Viruses rely on the host cell to make more virus. Once viruses have entered cells, the cells' machinery is used to copy the viral genome. Depending on the virus type, the genome may be single- or double-stranded DNA, or even RNA. A viral genome is packaged within a protective protein coat. Viruses that infect bacteria are called bacteriophages. One of these, called lambda, has a fixed-size DNA molecule of 4.5×10^4 base pairs. In contrast, the bacteriophage M13 can change its chromosome size, its protein coat expanding in parallel to accommodate the chromosome. This makes M13 useful in genetic engineering.

Different Forms of DNA

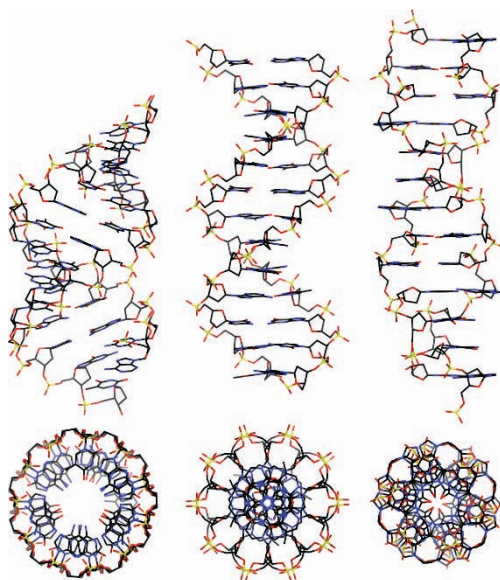
The most stable and natural form of right-handed DNA, whose structure was proposed by Watson and Crick, is also called B-form DNA. This DNA is present in almost all the living organisms. But the B-form DNA can take on other forms too under certain conditions. In all alternative forms of DNA, major changes involve helix geometry.

To recall the helix geometry of B-form DNA, it is described once again:

- The diameter of the helix is 20 Å.

- The pitch, i.e., the length of helix needed to complete one turn, is $34 \text{ \AA}$.
- The distance between two base pairs is $3.4 \text{ \AA}$.
- In the physiologic solution, there are about 10.5 base pairs in one pitch rather than 10.0 found in fiber.
- The axial rise of helix per base pair is $3.37 \text{ \AA}$.
- The tilt of a base pair is 6.3° .

Conversion of B-form DNA to other different forms of DNA takes place due to lowering of water activity, for example by addition of ethanol, by a low humidity, or by a high salt concentration. Otherwise, it retains its normal B-form without changing the physiologic conditions. The alternative forms have also been crystallized and subjected to X-ray structure determinations.



These forms are described below:

A-DNA or Right-Handed DNA

The A-structure is induced in DNA in 70% to 75% ethanol and is found in fibers of DNA in a dehydrated state. The primary difference between A and B helices lies in the sugar ring conformation (pucker). The sugars are C'_3 -endo in the A-form but C'_2 endo in the B-form.

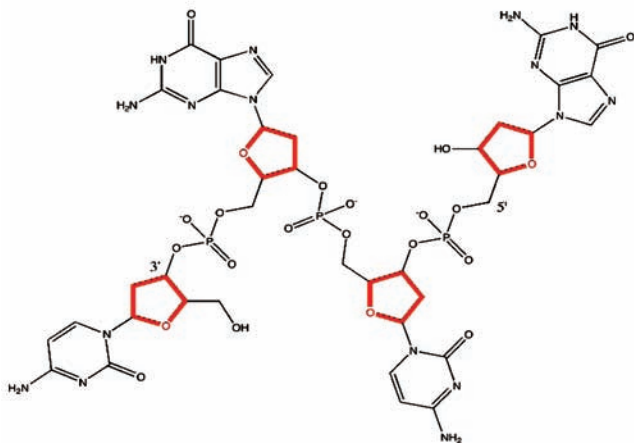
This altered sugar pucker shortens the distance between adjacent phosphates on one strand by about 1 Å. Thus A-DNA has between 11 and 12 base pairs per helix pitch rather than the 10.5 typical of B-form.

The second major difference in A-DNA is that the base pairs are displaced from the central helix axis, towards the major grooves. The base pair in B-DNA are essentially centered over the helix axis. In the A-structure, they are shifted by side nearly 5 Å from the center, resulting in a ribbon-shaped helix with a cylindrical open core and a very deep but narrow major groove.

The shape of the functional groups of the bases within these grooves undoubtedly render A and B forms clearly distinguishable by protein interacting with the DNA. Base stacking—both intra and inter strand—is the key element in stabilizing the A helix.

Now the helix geometry of A-DNA at a glance is:

- The length of one pitch is 28.15 Å.
- There are 11-12 base pairs per pitch.
- The axial rise per base pair is 2.56 Å.
- There is a tilting of base pairs from the axis of the helix of 20.2°.



Z – DNA or Left-Handed DNA

Z-DNA is a new family of DNA structure which is left-handed, i.e., the strands of DNA are twisted towards left-side instead of the usual right-

handed twisting. B-form is twisted clockwise whereas Z-form is twisted anti-clockwise. Therefore two possible helical forms of DNA are mirror images of each other.

Alexander Rich and his colleagues of USA have discovered the Z-form of DNA. Soon the left-handed DNA was observed in other laboratories, but only under certain conditions. Factors that promote the Z-structure include methylation, bromination, specific DNA binding protein and sufficient torsional stress as in negatively supercoiled DNA. The Z-DNA was so-named because of the zigzag course of the backbones.

Z-DNA differs from the A and B forms not only in its left-handedness, but also in the orientation of the glycosyl bonds. The syn and anti-orientations alternate, in contrast with the all-anti conformation in A- and B- DNA.

The pyrimidine nucleotides are in the standard anti conformation with a C'₂'-endo sugar pucker, while the purine residues are syn and contain a C'₃'-endo conformation. The Z-form is more likely to occur in alternating purine-pyrimidine sequences because steric repulsion makes it less-favorable for a pyrimidine to adopt the syn conformation.

The crystal structures have a strong dinucleotide repeat unit due to the large alternation in helix twist between the C and G in a CpG step which is about 15° while that between the G and C in the subsequent GpC is close to 45°.

Both B-DNA and A-DNA have two types of grooves—one major and one minor. But Z- DNA has the single groove which is quite deep, extending to the axis of the helix. The diameter of Z-DNA is about 18 Å⁰ and the length of pith is 45 Å⁰.

The basic helix geometry of Z-DNA is:

- The length of one pitch is 45 Å⁰.
- Number of base pairs/pitch is 12.
- The axial rise per base is 3.7 Å⁰.
- Tilting of base pairs from the axis of helix is 7°.
- Diameter of the helix is 18 Å⁰.

Z-DNA is a matter of intense current interest. It also occurs in nature. Its presence has also been established in the chromosome of the fruit-fly *Drosophila melanogaster* and also in the chromosomes of some

other organisms. There is growing evidence that certain DNA regions rich in guanine and cytosine within long molecules do adopt a Z-like conformation. This peculiar form of DNA is presumed to have a role in the regulation of gene activity. It could be a recognition signal for some important function of DNA. To understand the function of Z-DNA in *Drosophila melanogaster* chromosomes, research on synthetic DNA was conducted.

It has been demonstrated that purine-pyrimidine sequences can undergo transitions between right-handed B-conformation depending on the salt concentration or on chemical modification of the bases. The reversibility suggests that Z- DNA has some regulatory role.

For example, some nucleotide sequences can be established in the Z-form by being methylated. Methylation and de-methylation are very important in controlling the activity of genes.

There could be many possibilities by which the Z- DNA controls the gene regulations such as when certain control sites are established in the Z-form by methylation—regulatory protein binds to the site and keeps gene turned off; de-methylation might switch the site to the B-conformation, causing the regulatory molecule to let go.

Identification of Z-DNA

Rich et al could raise antibodies by injecting rabbits with short, brominated double helices of alternating G and C nucleotides of Z-DNA. The rabbit made antibodies that were highly specific for Z-DNA and did not react at all with the B-form. Purified rabbit antibodies for Z-DNA were used to probe biological materials for the presence of Z-DNA. The polytene chromosome of *D. melanogaster* were treated with rabbit's anti z-antibody and incubated to enable the antibody to bind to any Z-DNA in the chromosomes.

The unbound antibodies were washed away. Further, the bound antibodies were made visible by adding a goat antibody specific for rabbit one. The goat antibody was conjugated to a fluorescent dye, so that it glowed under ultraviolet illumination. The anti z-antibody could be seen to bind to the chromosomes in a distinct and reproducible segmented pattern.

By comparing ultraviolet micrographs that reveal the bands and inter-bands of polytene chromosomes, it was established that the fluorescent segments are the inter-bands and not the bands. Therefore, there must be Z-DNA in the inter-bands.

It is conceivable that some Z-DNA is also present in the bands but is not visibly stained by the antibodies. Presently it appears that Z-DNA is either more abundant or more accessible in the inter bands than the bands.

C-Form DNA

The size of helix of C-form DNA is greater than A type of DNA but is smaller than B-DNA. It is about $31 \text{ \AA}$. There are 9.33 base pairs per turn of the helix. There is an axial rise of base pairs of $3.03 \text{ \AA}$ with a tilting of about 7.8° . C-form is found at 66% relative humidity: in presence of Li^+ (Lithium) ions.

D-Form and E-Form DNA

There are some other forms of DNA called D-form and E-form found rarely as extreme variants. In case of D-form there are 8 base pairs per turn of helix. An axial rise of base pairs is $3.03 \text{ \AA}$ with tilting of about 16.7° . In case of E-form, there are 7.5 base pairs per turn of helix.

Resemblances and Differences between B-DNA and Z-DNA

- Both B-DNA and Z-DNA are of double helical structure.
- In both DNAs two strands are antiparallel.
- G $\equiv$ C pairing is present in both B-DNA and Z-DNA.
- B-DNA is right-handed whereas Z-DNA is left-handed.
- The sugar-phosphate backbone in B-DNA is regular while in Z-DNA it follows zigzag course.
- One complete helix is $34 \text{ \AA}$ in B-DNA while it is $45 \text{ \AA}$ in Z-DNA.
- In Z-DNA, adjacent sugar residues exhibit $\rightarrow$ opposite orientations while in B-DNA they are in same orientation. This results in dinucleotide units in Z-DNA as against

mononucleotide units in B-DNA.

- In B-DNA there are ten base pairs at each turn while there are twelve base pairs at each turn of Z-DNA.
- The diameter of B-DNA is $20 \text{ \AA}$ whereas it is $18 \text{ \AA}$ in Z-DNA.
- Bases are closer to the axis in Z-DNA than B-DNA.
- The glycosidic torsion angle is anti in B- DNA and syn in B-DNA.
- Sugar pucker is C_2 endo for B-DNA and C'_3 endo for Z-DNA in deoxyguanosine.
- The angle of twist per repeating unit, i.e., dinucleotide is 60° as against 36° in B- DNA.
- Base pair tilt is 6° in B-DNA whereas it is 7° in Z-DNA.

Super-Twisting of DNA Molecules

Normally, in a double helix, DNA consists of two polynucleotide strands that twist or wind around each other. The additional term super-twisting is generally used to describe a special feature of DNA and it means that the helical coiling is over and above that of the twisting of the two strands of the DNA double helix. Super-twisting, supercoiling and superhelicity are all interchangeable terms to define the same feature.

Initially it was thought as a special type of small circular DNA such as viral DNA, bacterial plasmids and mitochondrial DNA. Sometimes it was believed to be an artifact of isolation.

At present, super-twisting or supercoiling is known to be an integral and extraordinarily important feature of nearly all chromosomes—whether a DNA may be circular or linear. The degree of super-twisting may be a crucial factor in stages of replication, transcription and recombination.

To understand the topology of super-twisting, suppose, two antiparallel twisted ribbons represent a DNA double helix. When the free ends of such twin ribbons are joined it will simply make a ring structure. When this circular structure lies flat on a planar surface without change in the ring geometry, it is called relaxed state.

However, if before joining the free ends of the ribbon one end of the rib-

bon is turned through one or more 360° revolutions in the direction of unwinding, while the other end is held fixed, the ring will no longer be relaxed but, instead, will be under-wound.

This means that DNA spontaneously winds to achieve the most stable conformations—one in which all bases are in a right-handed, double helical arrangement. However, a small segment of a linear DNA is unwound or untwisted and the free ends of the two strands are linked covalently to form a closed circle.

The joining of free ends makes the DNA molecule as a whole in an unwound state. When the unwound segments pair up again to assume the double helical conformation, the entire molecule must adjust its spatial shape by twisting, for example into a right-handed super-helical form.

An alternative deformation is for the molecule to assume a left-handed spiral. A supercoiled DNA that is under wound is called a negatively supercoiled DNA; the shape assumed is determined by the relative free energies of the various controlled forms. There are two types of negatively supercoils such as plectonemic and toroidal.

Plectonemic supercoils are the predominant form in negatively supercoiled DNA in solution, while toroidal coiling results when the DNA is wrapped around the surface of protein such as histone to form nucleosome.

Possibly due to their interaction with proteins (histone) DNA helices in cell are unwound about 5% to 8% with respect to the standard right-handed double helical conformation of DNA in solution. When DNA is removed from cells and stripped off its protein, it becomes negatively twisted.

If, instead, a linear DNA is twisted more tightly than normal before the free ends are joined together, an over-wound or positively supercoiled DNA is formed. A positively supercoiled DNA can assume the form of a left-handed superhelix or a right-handed spiral.

Some enzymes are also involved during supercoiling and relaxing state of DNA. Topoisomerases directly change the super helical state of DNA. Gyrase, an enzyme, induces negative supertwists and relaxing enzymes remove them.

DNA Bending

Long chains of DNA are more flexible than the short pieces of DNA (about 100 bp). Short chains of DNA are comparatively stiff and it is very difficult to bend them. However, certain sequences and structures present in DNA molecule generate an intrinsic bend and are known as bendable DNA segments.

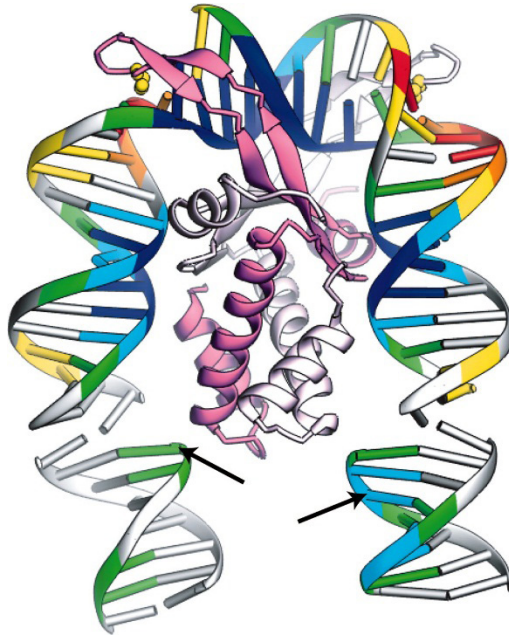
Such segments are inherently flexible. Bent segments are important features in regions of DNA that control replication, transcription and recombination.

Each mini circle of kinetoplast DNA (K- DNA) of Trypanosomes contains a small region of bendable DNA sequence. Polyacrylamide gel electrophoresis (PAGE) can be used as a technique for identifying DNA with a bend.

When DNA is cut into a number of small fragments by restriction digestion, then the fragments with the bend in the centre will move in the PAGE slowly while the fragments with the bend at the end move very fast. Therefore, on the basis of rate of mobility, the locus of bending can be defined by PAGE of DNA fragments that are more or less identical in length.

A bend, when it is a part of a longer molecule, creates a tight loop and is induced by adenine repeats. Although bends are caused by adenine repeats, thymine dimers, insertion of one or a few base pairs in one strand of the duplex, also bend DNA.

There are some non-intrinsic bends which are formed by the influence of a bound protein. Bending of DNA around a protein surface was first seen in a nucleosome in which at 145 bp segment is wrapped nearly twice around a histone octane and the segment bends more than 600° . Due to tight wrapping, the grooves of the DNA must be narrowed along the surface facing the protein and widened on the outside edges.



Many non-histone proteins also induce bending of DNA.

These are:

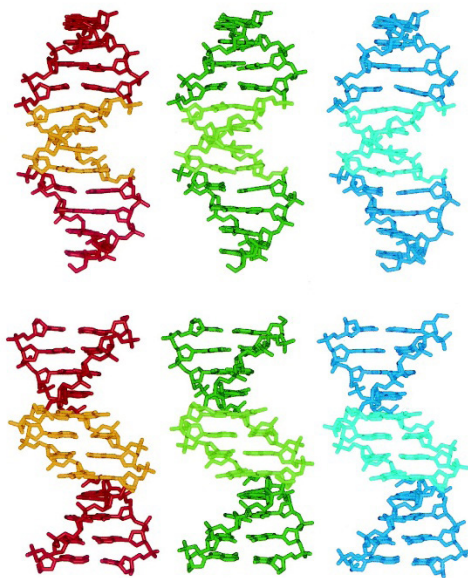
- Catabolic activator protein (CAP) and the galactose repressor
- RNA polymerase of *E. coli*
- Recombination enzymes such as Tn 3 resolvase and λ integrase
- Restriction enzymes Eco RI
- The small DNA binding proteins IHF, HU and Fis and (f) DNA gyrase.

The magnitude of protein induced bends is variable and ranges from 30° to 140° .

Functions of DNA Bending

The possible functions of DNA bending are:

- Condensation and packaging of DNA.
- Bringing together distant binding sites in linear DNA.
- Formation of special protein-DNA structure.
- Straining the DNA helix to stimulate its cleavage or melting.

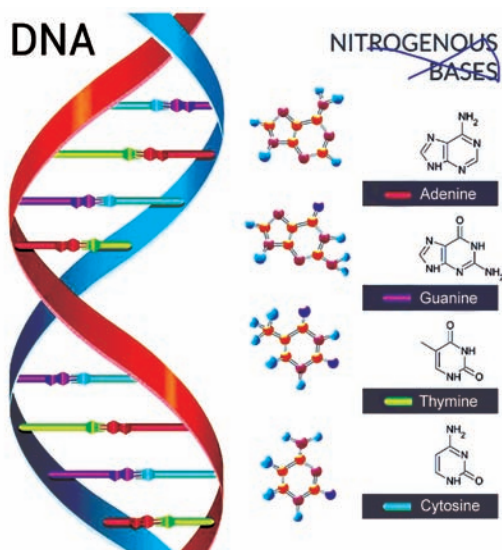


Functions of DNA

The base sequence of the DNA constitutes the informational signal called the genetic material. This nucleotide base sequence enables the DNA to function, store, express and transfer the genetic information. Hence it programs and controls all the activities of an organism directly or indirectly throughout its life cycle.

- (a) DNA stores the complete genetic information required to specify (form) the structure of all the proteins and RNA's of each organism.
- (b) DNA is the source of information for the synthesis of all cellular body proteins. Some of the proteins are structural proteins and some are enzymes. These enzymes arrange micro-molecules to form macromolecules. These macromolecules are arranged to form supra-molecular complexes or cell organelles which associate to form cells. These cells group to form tissues which in turn form different organs of a body, specifically peculiar to that organism during foetal development, growth and repair. Hence DNA programs in time and space the orderly biosynthesis of cells and tissue components.

- (c) It determines the activities of an organism throughout its life cycle, i.e., the period of gestation, birth, maturity, senescence and death.
- (d) It determines the individuality and identity of a given organism.
- (e) It duplicates (replicates to form two daughter DNA) itself and transfers one of the copy to the daughter cell during cell division, thus maintaining the genetic material from generation to generation.



The structure of DNA is highly rationalized as a design to fulfil several primary functions efficiently as follows:

- Primarily DNA acts as the cell's genetic blueprint which is mainly used for transcription of the encoded information and to serve as the template for replication.
- The linear array of four nucleotides, arranged in a defined order, allows the molecule to encode an essentially endless and diverse amount of information.
- The double helix structure of DNA maintains two copies of the information at all times. If lost by errors in replication or by DNA damage, the information is often retrievable from an undamaged complementary strand.

- Complementarity in the nucleotide sequence provides the basis for the transcription into the temporary but directly usable mRNA and for replication prior to cell division.
- Base complementarity, melting and annealing of the helix and its ability to accommodate at least transiently alien or new DNA molecule provide the foundations for recombination between DNA molecules.
- Recombination is essential for most pathways of damage repair, for replication of the ends of some linear molecules and for the initiation of replication.
- Recombination provides for the communication between DNA molecules in a population, across species and even across kingdom boundaries, affording a means for maintaining both genetic stability and diversity.

DNA must remain stable enough to allow on a modest level of change over many generations.

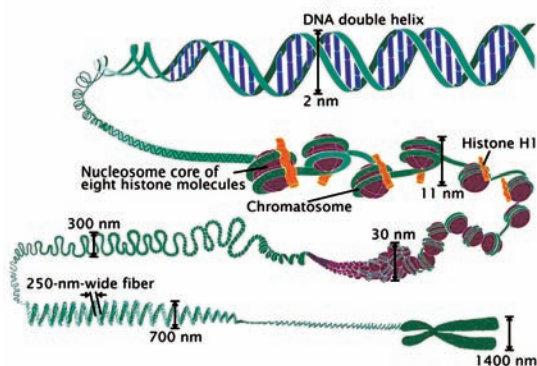
Finally, super twisting, bending and topological flexibility inherent in DNA play a key role in packing these long molecules into a cell that is only a tiny fraction of their length.

Packaging of DNA

Packaging of DNA within the Cells

The length of DNA molecules existing in a particular cell is much longer than the long dimensions of the cell or the organelle where they exist. The contour length (i.e., its helix length) of a double stranded DNA can be calculated from the molecular weight, presuming the average molecular weight of each nucleotide pair to be about 650 Da and there is one nucleotide pair for every 0.36 nm of the duplex.

DNA Packaging



CORE CONCEPT: THE PROPERTIES OF DNA

The use of DNA as a Nano scale construction material has come to be known as structural DNA nanotechnology – in order to

differentiate it from the field of biotechnology, in which DNA is used to encode genetic information for altering living cells, for example, by reprogramming them to produce novel or transplanted proteins. The primary goal of structural DNA nanotechnology is to exploit complementary base pairing in order to program the self-assembly of molecules into supramolecular complexes with desired properties. The importance of DNA for biological processes cannot be overstated. It is a biological polymer with a simple, yet robust information-encoding system. DNA's fundamental structure leads to efficient replication and transmission of encoded genetic information. DNA has also been recognized as a unique material for various nanotechnology applications (see section 'DNA as a Self-Assembling Construction Material' below). A fundamental understanding of DNA starts at the level of chemical structure and properties.

DNA has a number of special physical and chemical properties that are important to its structure and functioning. While the terms may sound quite foreign and perhaps even complicated, they are significant to our understanding of DNA and its many applications today.

Physical Properties

In living organisms such as humans, DNA exists as a pair of molecules rather than a single molecule. These strands are entwined in the shape of a double helix and the helix is kept stable by hydrogen bonds, which can be found between the bases attached to the two strands. A long polymer, DNA is made up of smaller units called nucleotides. In turn, each nucleotide consists of a phosphate group, a sugar and a nitrogenous base. To get a sense of just how long a DNA polymer is, consider that just one human chromosome is several hundred million base pairs long.

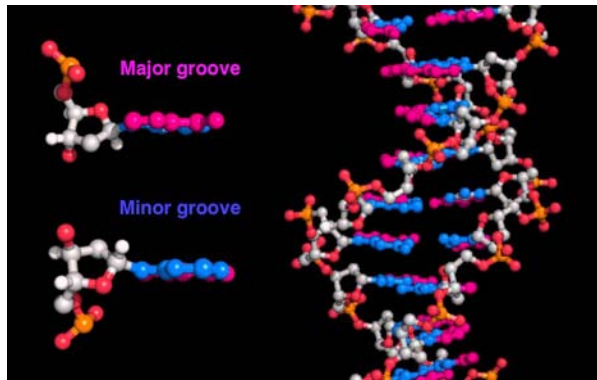
Base Pairing

Base pairing is a defining property of DNA and was particularly exciting when it was first discovered because it suggested the ‘copying’ mechanism for DNA. In DNA, bases are specific in that an adenine base, for example, only pairs with a thymine base. Following on that premise, a cytosine base will only bond to a guanine base. This base pairing is also known as complementary base pairing. The concept is quite simplistic but it is significant for DNA.

DNA Grooves

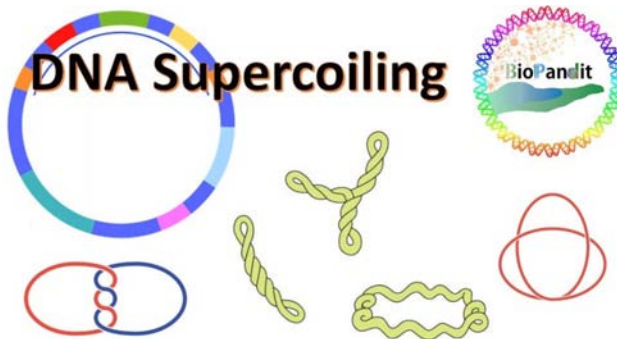
DNA has two kinds of grooves that play important roles in its functioning. Major and minor grooves are structures to allow for necessary proteins in your body to make contact with bases. You might wonder what makes it so important for proteins to attach to the bases. Some of these proteins are called transcription factors. While the word may well be a foreign one for those unfamiliar with DNA - which is most of the public - it is a key one. Transcription factors have numerous roles such as telling your cells how large they should become or suppressing tumour growth. They are also significant for cell development and communication. As such, these DNA grooves seen in the structure of DNA facilitate the binding of

proteins like transcription factors, which then serves to keep the cellular processes occurring effectively within your body.



DNA Supercoiling

If you try to picture a rope, you can get a visual image of DNA supercoiling. This coiling is a central property of DNA. DNA can be in a relaxed or coiled state and it is this coiling that allows our extremely long strands of DNA to fit or ‘pack’ into the comparatively much smaller cells in our bodies. This supercoiling property makes DNA more efficient by allowing it to make better use of space and fit neatly into a significantly smaller space relative to its relaxed, longer size.



DNA Conformations

DNA can exist in different conformations and these are important for a range of DNA mechanisms. These conformations interact with enzymes in your body and are also involved in aspects such as DNA repair.

DNA Sense and Antisense

To understand this aspect of DNA, you first need to be aware of RNA - ribonucleic acid. DNA holds the genetic 'codes' to make RNA. In turn, RNA is needed in your body because it holds the codes for the amino acids that make proteins. The antisense strand is the DNA strand that carries important information to make proteins by binding to the RNA. This antisense strand is the key for making proteins. In comparison, the sense strand is the one that does not code for RNA.

Despite the complex detail and challenges that come with understanding the properties of DNA, they are clearly important to its design and functioning in the human body. Since the discovery of DNA and our knowledge of its basic physical properties such as the double helix, we have progressed a great deal to learning about the chemical properties of DNA and interactions between DNA and its environment. Hopefully, we will learn even more detailed information about DNA structure and functioning in the future, which will have important applications in disease treatment and many other fields.

DNA Sequencing

DNA sequencing is an important concept although it can be a confusing one. For most of the public, understanding why DNA is important is only one step in learning about DNA and how it affects our lives. To fully get a sense of the relevance of DNA to each of us, however, it helps to know how sequencing works and why it is considered so important.



What exactly is DNA Sequencing?

The very basic unit of the human genome is a single DNA nucleotide. This nucleotide is extremely small and is made up of minuscule atoms, which creates a challenge for even an advanced microscope to be used for detection. Researchers still, however, need to be able to determine the sequence of bases in DNA that make up the human genome. As such, DNA sequencing has been developed but the process itself is a seemingly complex one.

DNA sequencing involves the determination of the order of DNA bases. You may be wondering what makes these bases so important. In a strand of DNA, there are some simple units known as nucleotides. These nucleotides have a ‘backbone’ that consists of sugars and a phosphate group. The DNA bases can be one of four kinds and they are attached to these sugars. These bases hold the important and unique genetic information for your body. These bases are:

- Adenine (A)
- Thymine (T)
- Cytosine (C)
- Guanine (G)

In the Beginning

The very first methods used for DNA sequencing were created in the 1970s. During this decade, researchers were only able to sequence a small number of base pairs, which was a far cry from the number needed to sequence just a single gene and an even more daunting and unlikely task to sequence an entire human genome. By 1990, things had improved somewhat but the number of laboratories able to sequence a hundred thousand bases was still few and far between. Not only was that, but the cost of sequencing itself extremely high and impractical. Fortunately, there have been vast improvements since then, particularly in terms of technological advancement. Better still, automation has made the process much faster and a great deal more practical. Now, individual genes are sequenced on a regular basis and can be done quickly and affordably for laboratories. In fact, some laboratories are sequencing more than a hundred million bases in a given year.

How is DNA Sequencing Performed?

DNA sequencing involves the process of figuring out the precise order of the four bases found in one piece of DNA. What this means is that the DNA is really just a template that is used to create a series of fragments. The fragments differ in length by one base and they are separated by size before the bases are identified, which then effectively recreates the original DNA sequence.

Each person has twenty-three pairs of chromosomes - one copy of the human genome. Because technology has limitations, we are limited in how many bases can be read at one time. Therefore, we can't just read each base from one end of a chromosome to the other. To make it feasible, the chromosome is cut down into smaller fragments.



Why Perform DNA Sequencing?

DNA sequencing is important to apply to the human genome. It allows scientists to sequence genes and genomes. Since there is a limit to how many bases can be sequenced in one experiment, larger DNA molecules - as mentioned - have to be 'broken' into smaller fragments before they can be sequenced and reassembled. To ensure that the sequencing is accurate, researchers also perform the sequencing several times. Clearly, finding just one single gene amongst the seemingly endless strands of DNA that constitute the human genome needs some very powerful equipment! With continued research, it is likely that better tools will be developed to make DNA sequencing much more rapid as well as cheaper

and more practical for researchers to complete. In doing so, we will have a better understanding of the base sequences that can tell us important genetic information in one specific segment of DNA. Once genes are identified and analyzed from sequence information, scientists can look for mutations that cause disease, thereby providing valuable medical information.



Importance of DNA

It has been argued that the discovery of DNA as well as our understanding of its structure and functioning may well be the most important discovery of the last century. The effect of the discovery of DNA on scientific and medical progress has been enormous, whether it involves the identification of our genes that trigger major diseases or the creation and manufacture of drugs to treat these devastating diseases. In fact, the identification of these genes and their subsequent analysis in terms of therapeutic treatment has ultimately influenced science and will continue to do so in the future.

Disease Diagnosis and Treatment

One important area of DNA research is that of genetics and medical research. Due to our discovery of DNA, our ability to actually diagnose diseases early on has been vastly improved. In addition, we have been able to better assess a person's genetic susceptibility to specific diseases. In doing so, we have also paved the pathway to formulate brand new drugs to treat these diseases. In fact, drugs can essentially be custom made to complement a person's personal biochemistry and genetic makeup.

For those diseases that were considered lethal and where treatment was either non-existent or largely unsuccessful, the discovery of DNA has essentially led to breakthrough drugs and treatments for patients with serious illnesses.

Paternity and Legal Impact

While the discovery of DNA has perhaps impacted medicine the most, its contribution to other areas is still similarly significant. Paternity cases have an enormous impact on families and children around the world. Through the assessment of DNA, the paternity of a child can be identified, which has a significant effect on the child's upbringing and his or her life.

Forensics and DNA

DNA has been notably important to the field of forensic science. The discovery of DNA has meant that the guilt or innocence of a person who is investigated for a crime can be determined. It also means that scarce evidence can still yield vital clues regarding the perpetrator of a crime. Also important is that the identification of victims can occur, particularly in cases where the victim's condition is unrecognizable to family or friends. DNA has been important in revolutionizing the entire field of forensic science.

Agriculture and DNA

The impact of DNA on agriculture has been a very important one because it has allowed breeders to facilitate the breeding of animals that have a better resistance to diseases. It also allows farmers to produce more nutritious produce, which has particularly important consequences in developing countries where the population subsists on a small range of staple foods that have little variety. This means that micronutrient deficiencies can be addressed in these countries.

Biological Significance or Properties of DNA

DNA has several biological significance particularly in the cellular environment and in the organism as a whole.

It shows the following properties:

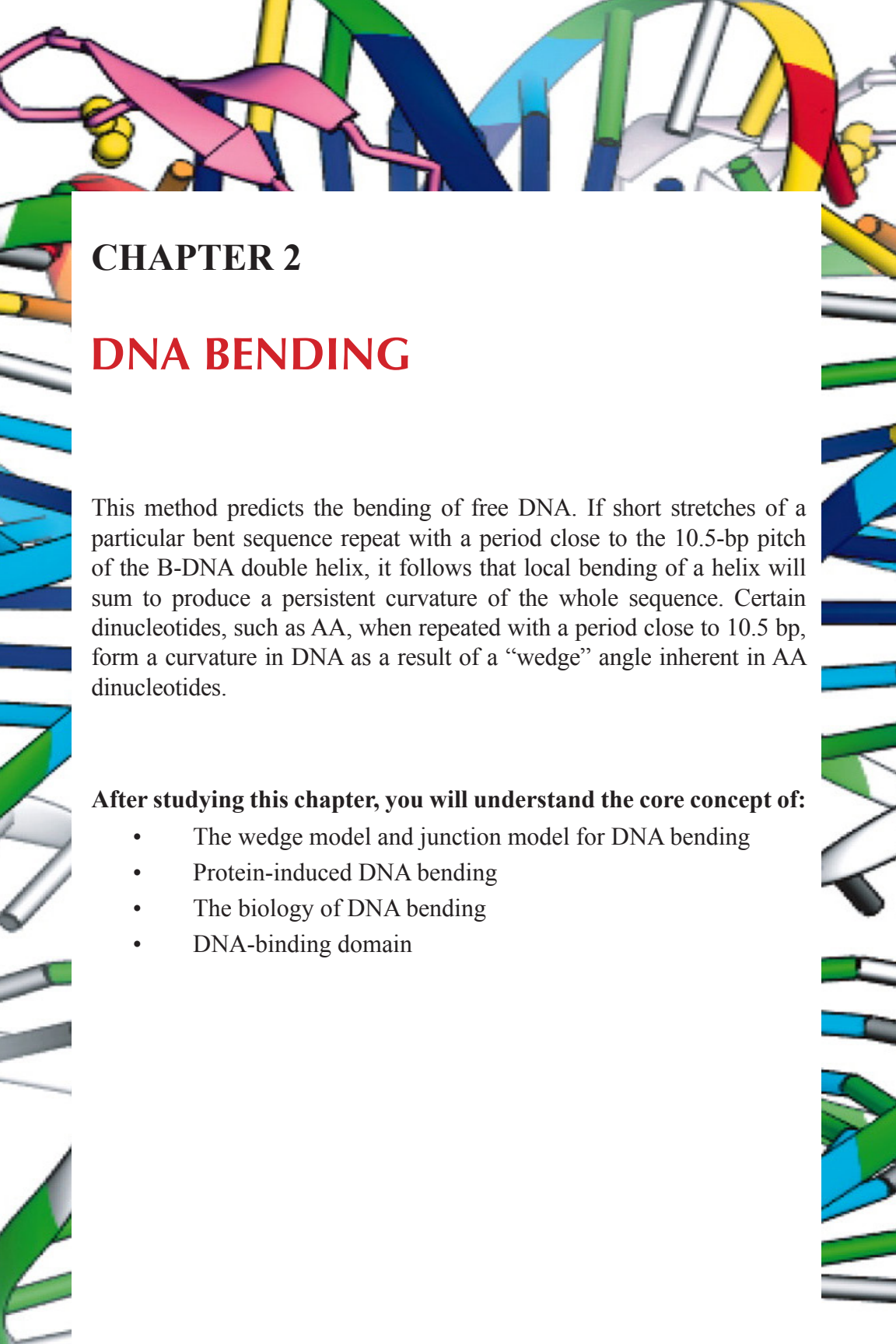
1. It replicates itself accurately during cell growth and duplication. This is known as auto synthesis.
2. It has the potentiality to carry all kinds of necessary biological information and is involved in gene action which—through a series of chemical reactions results in the ultimate expression of characteristics within the organism. The latter property is known as hetero-catalysis.
3. Its structure is fundamentally stable so that heritable changes (mutation) occur only very rarely.
4. It transmits all necessary biological information to the next cell generation or the next generation of the organism.
5. In most of the organisms DNA is the genetic material.
6. DNA must show a very wide diversity in form to account for the innumerable diversity in the characters of organisms occurring in nature.

REVIEW QUESTIONS

1. Focus on DNA.
2. What is DNA and how does it work?
3. Discuss about DNA replication.
4. Explain the DNA Mutation.
5. Understand the double helical structure of DNA.

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CHAPTER 2

DNA BENDING

This method predicts the bending of free DNA. If short stretches of a particular bent sequence repeat with a period close to the 10.5-bp pitch of the B-DNA double helix, it follows that local bending of a helix will sum to produce a persistent curvature of the whole sequence. Certain dinucleotides, such as AA, when repeated with a period close to 10.5 bp, form a curvature in DNA as a result of a “wedge” angle inherent in AA dinucleotides.

After studying this chapter, you will understand the core concept of:

- The wedge model and junction model for DNA bending
- Protein-induced DNA bending
- The biology of DNA bending
- DNA-binding domain

INTRODUCTION

Once a system or phenomenon is understood, future events related to the phenomenon are predictable. In some respects DNA is a simple molecule. A plasmid DNA of known sequence can be cut into a number of different-sized pieces by a restriction enzyme. Molecules of equal length should exhibit similar flexibility. Pieces of DNA shorter than the persistence length of 150- 200 bp will behave as rather stiff rods that cannot be easily bent into a circle (see Hagerman, 1988, for discussion of persistence length). Larger pieces adopt a “random coil” shape in solution. DNA molecules of defined size behave in a very predictable way when run on agarose or acrylamide gels. Shorter molecules migrate more rapidly than larger molecules. In agarose and acrylamide gels there is a linear relationship between the log of the distance migrated and the length of the DNA in base pairs.

A DNA band of known length does not run at its expected position on an acrylamide gel. One of the most striking examples of such anomalous migration was a 414 -bp piece of kinetoplast DNA from *Crithidia fasciculata*. This DNA migrated as if it were twice as long in an acrylamide gel but migrated at its proper position in an agarose gel. The anomalous migration of the kinetoplast DNA on polyacrylamide gels was attributed to the kinetoplast DNA being either stably bent or kinked Figure 1. Although the migration through acrylamide gels is not completely understood in physical terms, the migration is believed to depend on the ability of DNA to “snake” through the gel matrix. A relatively straight piece of DNA that is rather flexible can easily snake through the gel. On the other hand, DNA containing a permanent bend or kink is not as flexible and can get hung up in the gel matrix. (Consider a cooked piece of spaghetti that can snake its way through the gel more easily than a stiff piece that is bent or curved at the center. It will be much more difficult for the stiff piece to move through the gel matrix.)

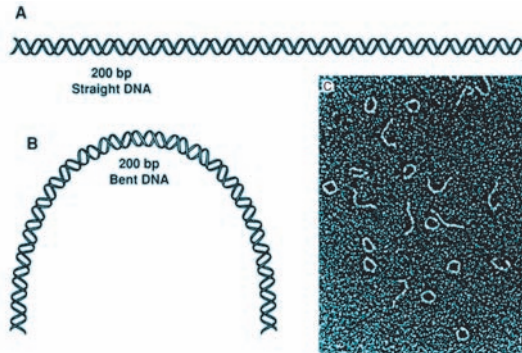
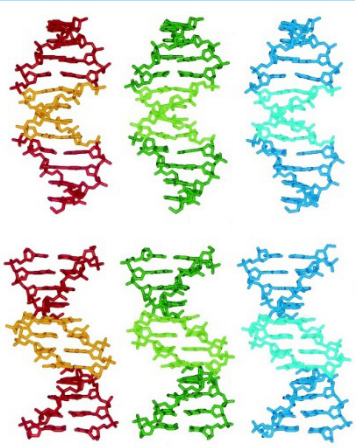


Figure 1. Straight DNA and bent DNA.



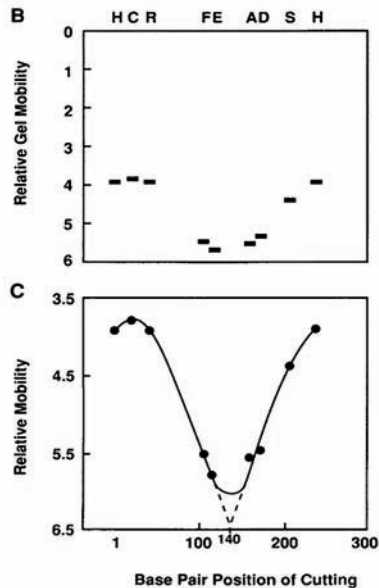
CORE CONCEPT: THE WEDGE MODEL AND JUNCTION MODEL FOR DNA BENDING

DNA-binding proteins recognize their DNA targets not only through the formation of specific contacts with the nucleotide bases but also through inherent properties of the DNA sequence, including increased bendability and rigidity.

Sequence Induced Bending of DNA

Wu and Crothers (1984) devised a clever way to map the site of bending in the kinetoplast DNA. They cloned the bent kinetoplast DNA fragment as a dimer of a 241-bp HindIII DNA fragment. By cutting the dimer with a series of different restriction enzymes that cut only once within each 241-bp fragment, different 241-bp DNA fragments were produced (Figure 2). The sequence organization of each fragment would be different

from that of the original fragment. Such a set of DNA molecules is called circularly permuted. Although all these isomers contain the same base sequence and, for the most part, the same bend in the DNA, they do not migrate at the same rate during electrophoresis in an acrylamide gel.



DNA double helix (Figure 4). The sum of wedges pointing in the same direction, a condition met by the 10-bp phasing, leads to the bending of DNA. As illustrated in Figure 5, the wedge angle can result from a wedge along the tilt axis or a wedge along the roll axis. Calladine and Drew (1992) discuss the physical basis for wedge angles for DNA.

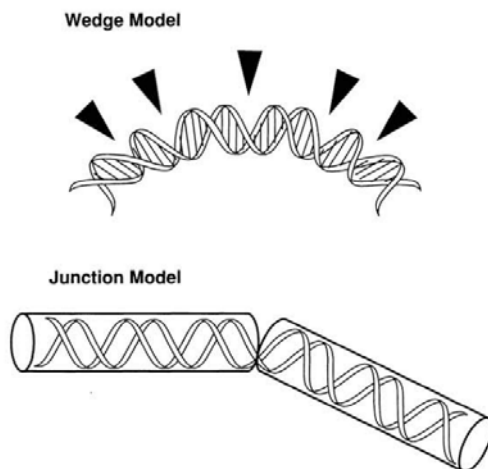


Figure 4. The wedge and junction model, for DNA bending.

Ulanovsky et al. used measurement of the efficiency of ligation of small DNA molecules into circles to calculate the wedge angle of an AA dinucleotide. As a short piece of DNA with a defined curvature is ligated together into increasingly long polymers, at some length the total angle of curvature will result in the formation of a circle of DNA. By determining the length at which the DNA forms a circle and by knowing the number of AA dinucleotides responsible for the 360° curvature, the individual AA wedge angle can be determined. To calculate the total wedge angle, Trifonov synthesized the following DNA sequence:

TCTCTAAAAAATATATAAAAA

TTTTTTATATATTTTATAGAGA

This sequence contains two runs of $(A)_5$ that are 10 bp apart. Since the average helix repeat is 10.5 bp, it is necessary to make a 21-bp fragment to ensure that the A tracts will remain in phase when multiple 21-mers are ligated to gerber. The optimal size for circle formation in polymers of this sequence was about 126 bp. This length is well below the persistence

length of DNA, which is 150-200 bp. A straight 126-bp piece of DNA will not circularize efficiently. Thus, efficient circularization is an indication of stable 360° curvature. Within the 126-bp molecule there are 66 AA dinucleotides; the sum of their individual wedge angles is responsible for the 360° curvature. Apparently making a few corrections, Trifonov estimated a total wedge angle (of both tilt and roll components) of 8.7° for each AA dinucleotide. This angle probably represents an upper limit of the wedge angle.

The Junction Model for DNA Bending

Wu and Crothers (1984) proposed the junction model for DNA bending (Figure 4) which suggests that there is a bend at the junction of B-form DNA and a non-B-DNA helix associated with A tracts. The suggestion that the A tracts might adopt a non-B-DNA helix was based, in part, on the knowledge that poly(dA) · poly(dT) can adopt a non-B-DNA helix called heteronomous DNA. In addition, modeling studies had suggested that DNA would bend at the junction between A-DNA and B-DNA. There are actually two junctions, a 5' and a 3' junction, relative to the A-tract helix. The site of bending could be associated with the 3' end, the 5' end, or both ends of the A-tract helix. Initially Crothers felt that bending was probably localized at the 3' end of the A tract. This conclusion was supported by experiments showing greater binding associated with a molecule in which the 3' ends of A tracts were in phase at 10 bp compared with a molecule in which the 5' ends of A tracts were in phase.

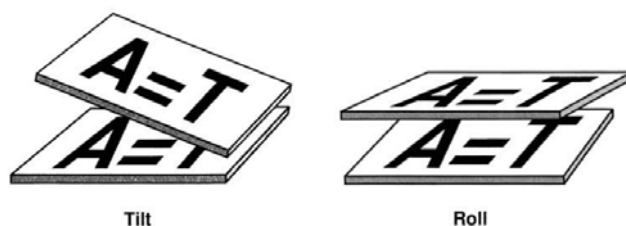


Figure 5. Tilt and roll wedges.

Tests of the Phasing Hypothesis

An important component of both the junction and the wedge model is 10-bp phasing. For DNA to contain a region of stable curvature and to produce

Electrophoresis is a technique commonly used in the lab to separate charged molecules, like DNA, according to size.

the anomalous gel mobility, the small individual bends must be oriented in the same direction or the same plane (if bends are not all directed in the same plane, the DNA will adopt a conformation that is effectively linear). If bends were placed every 1.5 turns of the DNA helix (15-16 bp), the DNA would be maintained in a zig-zag shape. Indeed, DNA with bends every 15 bp migrates slightly faster than a molecule with out-of-phase bends (or linear DNA) upon **electrophoresis** in an acrylamide gel. These patterns of curvature are shown in Figure 6.

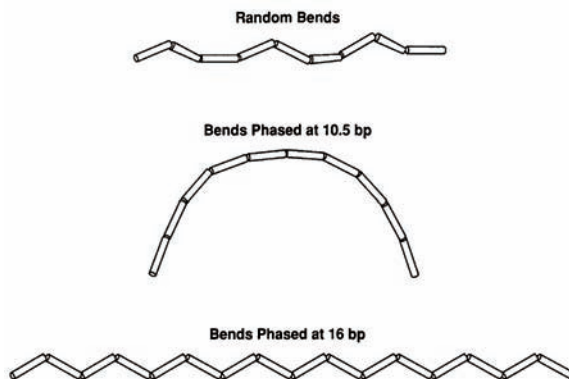


Figure 6. The significance of phasing in DNA bending. An A tract will introduce a small bend or deflection of the helix axis in DNA. However, the phenomenon of “bent DNA” or “stably curved DNA” that exhibits anomalous migration on polyacrylamide gel electrophoresis requires a number of small individual bends that are in phase. If bends are random, as shown in the top molecule, the DNA will migrate true to its length in an acrylamide gel. Only when bends are phased by 10.5 bp is the stable curvature shown in the middle structure observed. This DNA migrates anomalously slowly on electrophoresis in an acrylamide gel. If, as shown in the bottom structure, bends are phased by 16 bp, successive bends will be directed alternatively up and down. This creates a zigzag molecule, which is unusually straight and migrates slightly more rapidly than unbent DNA (or DNA containing random bends) in an acrylamide gel.

To test the phasing hypothesis, Hagerman (1985) synthesized oligonucleotides of sequence GA_3T_3C , $G_2A_3T_3C_2$, and $G_3A_3T_3C_3$. These oligonucleotides are self-complementary and can hybridize to form duplex DNA. These duplexes can then be ligated into longer polymers. By ligating the individual oligonucleotides, the A3 block in one DNA strand will be phased at 8-, 10-, and 12-bp intervals. If the 10-bp phasing were important, only the $G_2A_3T_3C_2$ polymer would exhibit anomalous migration on polyacrylamide gel electrophoresis. In Figure 7, mobility data are presented by dividing the apparent size in base pairs (determined by electrophoretic mobility) by the actual size of the DNA (determined by counting the ladders on the gel). The bP_{app}/bP_{seq} data points are plotted as a function of size of the DNA. The polymers with A tracts phased at 8 and 12 bp gave a horizontal line indicative of non bent DNA. The $G_2A_3T_3C_2$ polymer with A tracts phased at 10 bp exhibited an upward slope in the plot of bP_{app}/bP_{seq} until a plateau was reached (at $bP_{app}/bP_{seq} = 2.2$). This pattern of electrophoretic migration is diagnostic for curved or bent DNA, and demonstrates the importance of 10-bp phasing in DNA curvature.

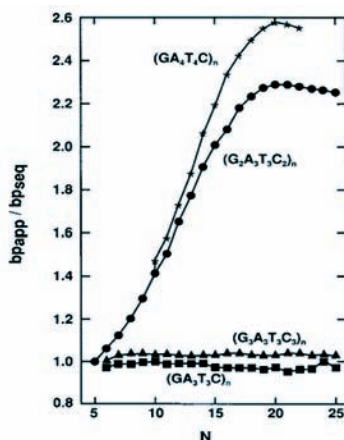


Figure 7. The analysis of DNA bending on polyacrylamide gels.

Sequence Requirements for DNA Bending

Koo, Wu, and Crothers synthesized a large number of oligonucleotides containing various lengths of A tracts that were phased at different lengths. These oligonucleotides were conceptually similar to those described by

Hagerman but were not symmetrical and thus had an A tract in only one strand of the DNA. Polymers with A_{4-9} were bent, with bending being optimal for A_6 . The Koo et al. polymer with A_3 phased at 10 bp was not significantly bent. (The Hagerman bent sequence with A_3T_3 contained an A_3 tract in both strands that must contribute to bending.) A continuous run of As is required for bending, since replacement of the central A in As with C, G, or T destroys the bending. There is no sequence requirement for a particular base 5' or 3' to an A tract for bending, although flanking sequences can influence curvature.

DNA sequences that do not contain runs of As can also be bent. The bends observed in DNA lacking phased A tracts are usually not as large as A-tract bends. These sequences have not been as well studied as A-tract-induced bends.

Analysis of $(GA_4T_4C)_n$ and $(GT_4A_4C)_n$ Leads to Development of a Refined Wedge Model

The wedge model is a “nearest neighbor model” which assumes that bending can be explained by wedge properties attributed solely to an AA dinucleotide. It assumes, in its simplest form, that there will be no influence on the AA dinucleotide wedge angle from flanking bases or the sequence contexts of DNA. Considering the polymers $(GA_4T_4C)_n$ and $(GT_4A_4C)_n$ one might expect them to migrate with identical rates on electrophoresis since they each contain A4 tracts that are phased at 10-bp intervals. However, the (GA_4T_4C) polymer is bent but the (GT_4A_4C) polymer is not bent. This result may seem to rule out the simple basic wedge model proposed by Trifonov. Ulanovsky and Trifonov realized that the relationships of phased A tracts in the two strands are actually quite different for (GA_4T_4C) and (GT_4A_4C) . Because GA_4T_4C and GT_4A_4C are symmetrical, A tracts in both DNA strands must be considered. Although both strands of a DNA duplex contain phased runs of As, the relationship between the A tracts in the opposite strands in GA_4T_4C compared to GT_4A_4C is different. The difference in A-tract phasing between these two molecules is illustrated in Figure 8. In GA_4T_4C , the 3' ends of the A

tracts in opposite strands occur at the same position in the helix axis. In GT_4A_4C , the 5' ends of the A tracts coincide at the same point along the helix axis.

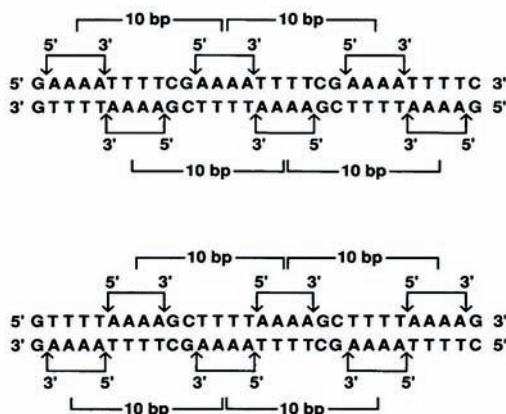


Figure 8. Asymmetry of GA_4T_4C and GT_4A_4C .

A key to understanding the difference in the electrophoretic mobility of $(GA_4T_4C)_n$ and $(GT_4A_4C)_n$ lies in the realization that the roll and tilt components of the wedge model are different symmetrically. For an AA dinucleotide, tilt can open either toward the As or toward the Ts. The tilt angle is believed to open toward the As as shown in Figures 5 and 9. In a DNA helix containing NNNAANN, the replacement of the As with Ts to form NNNTTNN results in a 180° reversal of the direction of the tilt angle. The roll angle, however, is unchanged when As are replaced by Ts because the roll angle opens toward the major groove and the positions of the grooves remain the same whether an AA or TT dinucleotide occurs in a helical region.

Assuming that bending is the result of a sum of all vectorial tilt and wedge components for AA dinucleotides in both strands of the DNA, Ulanovsky and Trifonov calculated that the AA dinucleotide has a tilt of 2.40° opening toward the AA and a roll angle of 8.40° opening toward the major groove (see Figure 9).

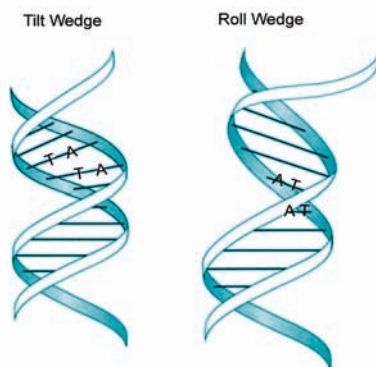


Figure 9. Tilt wedge and roll wedge.

A Refined Junction Model

The simple junction model first proposed by Crothers, in which bending was attributed predominantly to a deflection at the 3' end of the A tract, did not adequately explain experimental results. A refined wedge model was needed to explain the normal migration behavior of DNA that seemingly should be bent. As illustrated in Figure 10, Koo and Crothers proposed that bending was due to a tilt deflection at both the 3' and the 5' junction. At the 5' junction the tilt was toward the T, whereas at the 3' junction the tilt was toward the A. In addition, these investigators argued that there was a roll that opened the minor groove only at the 5' junction.

Differences in bending behavior of $(GA_4T_4C)_n$ and $(GT_4A_4C)_n$ are also predicted by the refined junction model. Since a roll component of bending is limited to the 5' junction of the A tract, this component of bending will be at the same point in the helix axis in GT_4A_4C and at different positions in GA_4T_4C (Figure 8). Although it is not intuitively obvious why the A_4T_4 sequence is bent whereas the T_4A_4 sequence is not bent, one should be able to appreciate the symmetry differences in the relationships of the tilt and roll components at the 3' and 5' junctions from analysis of Figure 8.

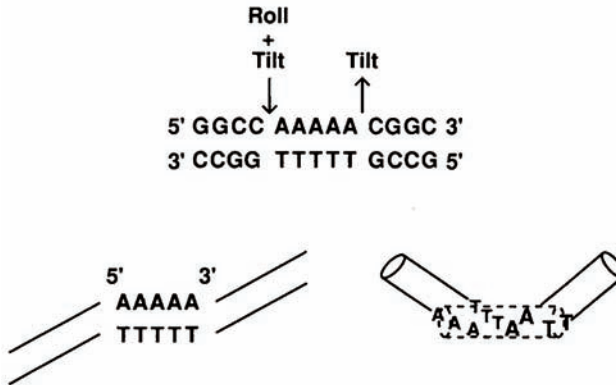
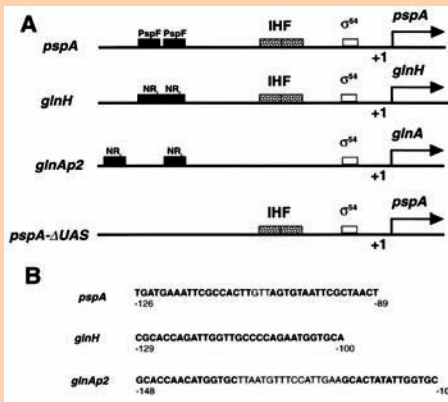


Figure 10. Refined junction model.



CORE CONCEPT: PROTEIN-INDUCED DNA BENDING

A significant number of the proteins which bind to specific sites in DNA bend that DNA. Bending proteins are involved in DNA replication, transcription, and recombination, and often participate in multicomponent complexes. Their

role in these complexes is probably to fix the geometry of DNA in such complexes so that all the components will interact optimally to effect the function of that complex.

There are many examples of DNA bending on association with protein. In some cases, the DNA is actually wrapped around a protein core. The wrapping of DNA around the enzyme DNA gyrase, around nucleosomes, and around the bacterial histone-like HU protein are extreme examples. In a less dramatic fashion, many other proteins bind

to DNA and introduce a slight bend that is more analogous to the bending introduced by a tracts in DN A. The protein-induced bending of DNA is shown in Figure 11.

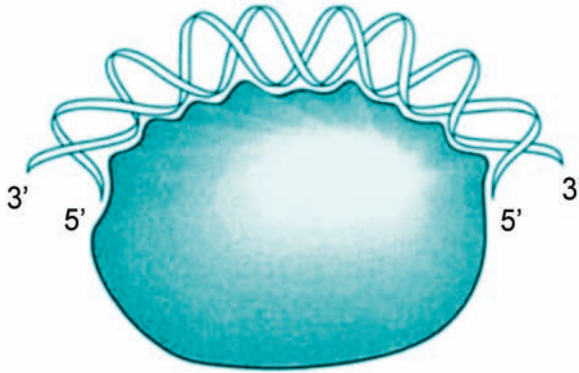
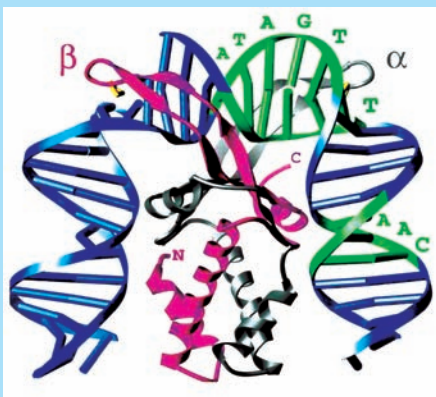


Figure 11. DNA bending around proteins.

The catabolite activator protein (CAP) and the lac repressor provide two examples of different effects of protein binding to DNA: CAP bends DNA whereas the lac repressor does not. In an experiment similar to that described for mapping the bend locus in kinetoplast DNA, Wu and Crothers (1984) demonstrated that CAP binding introduces a bend into DNA. Circularly permuted 203-bp DNA fragments containing the lac promoter region were constructed (the lac promoter region contains the CAP binding site and the lac operator sequence which binds the lac repressor). With no CAP bound, the circularly permuted DNAs had nearly identical mobility in the gel in the absence of CAP. CAP was bound to the fragments and the DNA protein complexes were analyzed by polyacrylamide gel electrophoresis. The mobility of DNA-protein complexes with the CAP binding site located at the center of the DNA was significantly reduced relative to when the CAP binding site was located near the end of the DNA fragment. The X-ray cocrystal structure of the CAP-DNA complex has subsequently shown a bent angle of 90°. To demonstrate that the difference in mobility was due to bending and not to the location of protein bound to DNA, the lac repressor was also bound to these DNA fragments. There was essentially no difference in the mobility of the circularly permuted DNAs bound with the lac repressor. In fact, lac repressor DNA complexes ran at nearly the same position as the DNA fragment with CAP bound near the end, despite the fact



CORE CONCEPT: THE BIOLOGY OF DNA BENDING

DNA-binding proteins play a pivotal role in the biology of the cell, being responsible for the transfer of biological information from genes to proteins, and have been estimated to constitute ~6–7% of all proteins expressed by eukaryotic genomes. A

large number of DNA-binding proteins and protein–DNA complexes are deposited in the protein data bank (PDB) and in the nucleic acid database.

Nature has found many uses for DNA bending. One is the control of access to promoters, the switch regions that turn genes on by bending or looping the DNA. Another is the control of initiation of DNA replication which, since it represents a major commitment for the cell, must be very carefully regulated. A third is site-specific integration of one DNA molecule into another. A fourth is **DNA repair**. Many chemicals that modify DNA and certain errors of DNA replication introduce bends into DNA. Finally, the organization and compaction of DNA into prokaryotic or eukaryotic cells involves the wrapping of DNA very tightly around DNA-packaging proteins (HU protein and histones). Although the organization of DNA with chromosomal packaging proteins does not involve sequence- or site-specific bends, the tight curvature of the DNA is required for this interaction. Our understanding of the significance of bending and the flexibility of DNA in biology continues to grow.

DNA repair is a collection of processes by which a cell identifies and corrects damage to the DNA molecules that encode its genome.

DNA Bending and Gene Regulation

In bacterial cells there are many examples in which DNA bending plays a critical role in the control of gene expression from a promoter. This control may involve the formation of a DNA loop in which the sequences that must be recognized by RNA polymerase are covered by repressor proteins that are bound to DNA stabilizing the loop. Alternatively, the critical RNA polymerase or other DNA binding sequences may be physically unavailable because they exist in a tightly curved loop of DNA that sterically is inaccessible to the protein.

The arabinose operon is one of the best understood examples of the role of bending or looping in gene expression. The arabinose operon encodes three proteins-AraB, AraA, and AraD-that are required for *Escherichia coli* to utilize the sugar arabinose. These genes are not normally expressed in *E. coli*. However, when arabinose is present in the environment, the three Ara proteins are produced. AraC mediates both negative regulation and positive regulation. The regulatory region of the arabinose operon is about 300 bp immediately upstream of the start of transcription (see Figure 12). In the arabinose operon, the dimeric repressor protein AraC binds to three operator sites on DNA: (1) $araO_2$ which is at position -265 to -294; (2) $araO_1$ which is at position -110 to 148; and (3) $araI$, which is at position -4 to -78. The $araI$ site actually contains two separate binding sites $araI_1$ (position -55 to -78) and $araI_2$ (position -35 to -4). During repression of the operon, a dimer of AraC binds to the $araO_2$ and $araI$ operator sites, which are 210 bp apart. For this simultaneous binding to occur, the DNA must form a loop between these two sites as illustrated in Figure 12. The formation of this repression loop requires that the DNA be supercoiled. Because the 210-bp length is close to the persistence length of DNA, there is little flexibility in the DNA between the two binding sites. In addition, the two binding sites occur on one “face” or side of the DNA double helix. Assuming a helical repeat of 10.5 bp per turn, binding sites separated by 210 bp are exactly 20 helical turns apart. Bending this DNA will bring the two sites on the same side of the DNA into physical contact. The introduction of 5 bp between $araO_2$ and $araI_1$ destroys repression by AraC because it puts the two sites out of phase on different sides of the double helix. Proteins bound to out-of-phase binding sites will not physically interact when the DNA bends.

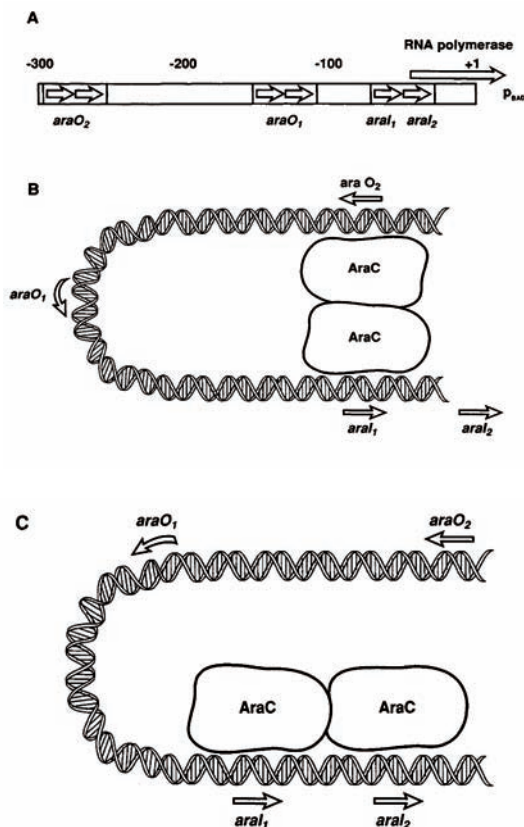


Figure 12. DNA bending in the arabinose operon. (A) The regulatory elements of the arabinose operon include the *araO₂* site at the left end of the operon, the *araO₁* site at about -120 bp, and an *araI₁*-*araI₂* site at around -30 bp. RNA polymerase begins transcription at bp +1. (B) A repression loop formed by an AraC dimer binding to both *araO₂* and *araI₁*. (C) A transcriptionally active configuration formed by an AraC dimer binding to *araI₁* and *araI₂*. The release by AraC of the *araO₂* site is mediated by the binding of the inducer arabinose.

To initiate transcription of the arabinose operon, the *araI*-*araO₂* repression loop must be broken. Arabinose, which is utilized by the products of this pathway, acts as an inducer by binding to AraC and inducing a conformational change that causes the release of one subunit of the AraC dimer from the *araO₂* site. This subunit of the AraC dimer then binds to the *araI₂* site (Figure 12). With an AraC dimer bound to *araI*, the DNA-protein complex recruits RNA polymerase to initiate transcription of the operon.

The CAP protein (cyclic AMP receptor protein), which bends DNA when bound, can also disrupt the repression loop. In the arabinose operon the CAP binding site is located between araO_I and araI_I . CAP binding may introduce a bend into the DNA that makes the initial repression loop energetically unfavorable (or may destabilize a preformed repression loop).

DNA Bending and DNA Replication

There are several examples in which natural bends in DNA exist at origins of replication. Specific bends can also be introduced by DNA-protein interactions at the origin of DNA replication. These phenomena have been found in both prokaryotes and eukaryotes.

Simian virus 40 (SV40) provides an example in which DNA bending is important in replication. The virus encodes a protein (large T antigen) which, among other things, is essential for DNA replication. Large T antigen binds to a specific site at the SV40 origin of replication. The recognition sequence for dimeric binding contains two 5-bp recognition sequences separated by an A tract. This binding site is naturally bent. If the A tract, which does not make contact with the protein, is replaced by “random” bases the origin region is no longer bent and the large T antigen does not bind as tightly. This result suggests that a slight bend is required for binding of the large T antigen.

Bacteriophage λ provides another example in which bending is required for DNA replication. Replication from the bacteriophage λ origin requires binding of the θ protein to four sites at the origin. Each of the four binding sites contains two G + C-rich recognition regions separated by a central region containing a run of 3 or 4 As. Moreover, the four binding sites are separated by λ tracts. As expected, these A tracts introduce a stable bend into the λ origin. Binding of θ protein increases the curvature of the origin region at two levels. First, binding of the θ protein increases the bending at the short λ tract between the two regions of the 19-bp θ binding site that make contact with the protein. Second, protein-protein contact between the θ proteins folds and organizes the entire origin region into a compact three-dimensional structure called an “O-some.”

DNA Bending and Site-Specific Integration of Bacteriophage λ

Bacteriophage λ , on infection into *E. coli*, can enter a lytic phase in which its DNA is replicated, new viruses are packaged, and the cell bursts releasing new virus particles. At other times, bacteriophage λ can enter a lysogenic phase in which the virus slips its genome into the *E. coli* chromosome. This integration into the chromosome is accomplished by site-specific recombination, which occurs through the interaction of two specific regions of DNA. These regions are called the attP site (the attachment site on the phage chromosome) and the attB site (the attachment site on the bacterial chromosome). Within these att sites, specific cuts are made by the bacteriophage AInt protein and the two molecules are joined together.

The integration host factor (IHF), an *E. coli* protein, is an important component in the integration of the bacteriophage λ chromosome. The binding of IHF to three sites-HI, H2, and H' on the attP sequence-results in the introduction of three 140° bends in the DNA. These bends result in the formation of a specific three-dimensional DNA-protein complex called an "intasome." The attB DNA then interacts with this intasome for the cutting and rejoining reactions required for site-specific integration.

DNA Bending and DNA Repair

When DNA is damaged, mutations can occur. Covalent modification of bases can result in a change in their electronic configuration. This will disrupt hydrogen bonding, effectively creating a mispair at the damaged site. If DNA replication occurred using a damaged base in its alternative electronic configuration, a mutation would result. For example, cytosine normally pairs with guanine in its keto configuration. If a chemical modification to guanine forced it into its enol configuration, G would hydrogen bond with T. On the next round of DNA replication the strand containing the T will have undergone a G · C to A · T transition. Thus, it is quite important that cells develop sensitive ways to detect DNA damage and prevent the introduction of mutations in DNA.

Enzymes involved in DNA repair have evolved to sense the shape of the DNA double helix. A bulge resulting from a natural mispair (G · A, for example) or a mispair resulting from DNA damage (C · 06-methyl G,

Pyrimidine dimers are molecular lesions formed from thymine or cytosine bases in DNA via photochemical reactions.

for example) is recognized by various repair enzymes. However, repair enzymes are probably not sensitive to subtle changes in twist or variations in the number of base pairs per turn. Smooth bends in the DNA introduced by phased A tracts or DNA looping by proteins do not seem to trigger attack by DNA repair enzymes.

Several types of damage to DNA introduce bends into DNA. An intrastrand cross-link, a covalent link between two bases in the same DNA strand, is one example. Ultraviolet light irradiation at 254 nm results in the formation of **pyrimidine dimers** in DNA, one type of intrastrand cross-link. The pyrimidine dimer introduces a bend of $7\text{--}30^\circ$ into the helical axis of DNA. Pyrimidine dimers are recognized by several different repair enzymes. The UvrABC exo-nuclease may recognize the sharply kinked or bent shape of intrastrand crosslinked DNA.

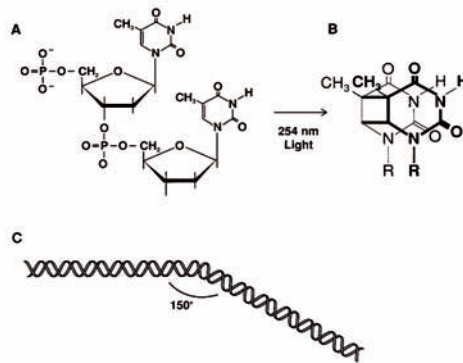


Figure 13. DNA bending introduced by pyrimidine dimers. (A) Adjacent thymines in one strand of DNA. Dimerization occurs when, on absorption of 254-nm light, the two 5,6 double bonds of adjacent thymines are covalently joined by a cyclobutane ring (B). This event will necessarily remove the 36° twist angle between the two adjacent base pairs and will introduce a kink of about 30° in the DNA (C).

Bulged nucleotides represent another helix anomaly that, if not corrected by DNA repair enzymes, can lead to a mutation. As shown in Figure 14, bulged nucleotides occur when there are extra nucleotides in one strand of DNA. During the process of DNA replication, slippage of the template can occur, especially in runs of a single base. Homopolymeric runs are often “hotspots” for naturally occurring addition and deletion mutations. The mechanism by which slipped misalignment can generate additions and deletions is described in Figure 15. A single extra nucleotide in one strand creates only a small perturbation of the structure of the double helix. In some cases, a single extra nucleotide remains “extrahelical” and bulged away from the double helix. In other cases, the extra nucleotide intercalates between two stacked base pairs (Figure 16). A single bulged nucleotide can kink the DNA by as much as 25° . Moreover, as the number of bulged nucleotides increases, the angle of the bend increases. The general feature of DNA kinking introduced by intrastrand cross-links or bulged nucleotides may be the structural feature that is recognized by repair enzymes such as UvrABC nuclease.

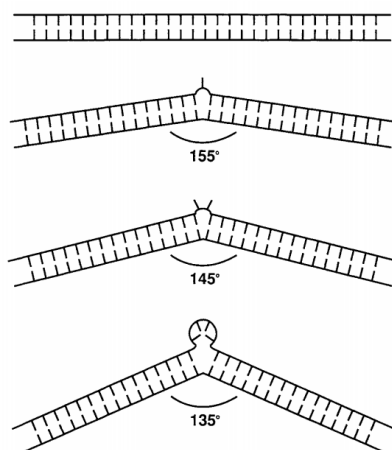


Figure 14. DNA bending by bulged nucleotides. A heteroduplex can be formed in which extra nucleotides exist in one strand. A perfectly paired molecule (top) is followed by heteroduplexes with 1, 2, and 4 extra nucleotides in the top strand. A single extra nucleotide can exist in an extrahelical structure (as shown) or can be intercalated between two stacked base pairs (as shown in Figure 16). A single bulged nucleotide can bend DNA by about 25° . When larger mismatches occur, a loop forms in one strand, and the magnitude of the bend increases.

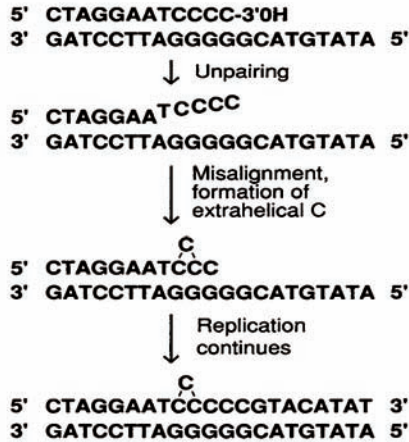


Figure 15. Slipped misalignment during DNA replication. A heteroduplex region, in which extra nucleotides occur in one strand, can result from an error of DNA replication in a process called slipped misalignment. As DNA polymerase replicates several Gs in the template strand, it can dissociate from the DNA allowing the 3' end of the newly synthesized strand to unpair or “breathe.” The 3' end progeny strand can reform hydrogen bonds but with the Cs and Gs in a different register so one C becomes extrahelical. The end of the progeny strand is said to have “slipped” or misaligned with respect to the template strand. As DNA polymerase rebinds this template and extends the progeny strand, there would be an extra C in the progeny strand. The slipped misalignment may also involve multiple base pairs. In addition, slipped misalignment can occur between direct repeats separated by hundreds of base pairs.

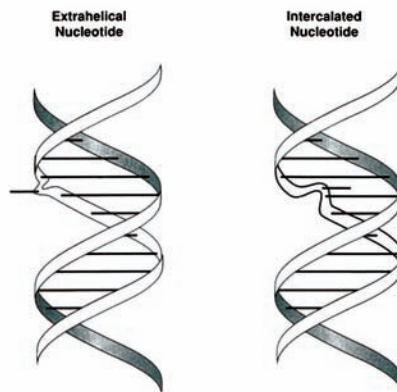


Figure 16. Extrahelical and intercalated extra nucleotides.

A-Tract Phasing and DNA Bending

To test the idea that DNA bending requires runs of As phased at 10-bp intervals, Hagerman synthesized three different 10-bp oligonucleotides: 5' GAAATTTTC 3', 5' GGAAATTTCC, and 5' GGGAAATTTCCC (alternatively written as $[G_iA_3T_3C_i]_n$, where $i = 1, 2, \text{ or } 3$). Since each single strand has inverted repeat symmetry, it is self-complementary. Thus, hybridization of each oligonucleotide (with itself) leads to double-stranded 8-, 10-, and 12-bp molecules:



In the presence of DNA ligase (and ATP and Mg^{2+}), the double-stranded oligonucleotides are joined together to form multimers. This creates a family of molecules that when analyzed by gel electrophoresis, shows a uniform repeating pattern referred to as a "ladder." The length of each member of the family increases by 8, 10, or 12 bp. For DNA molecules that do not contain a stable bend the migration through the gel is proportional to the log of the molecular weight. Hagerman used plasmid pBR322 DNA cut with the restriction enzyme HaeIII as molecular weight markers. Since the sequence of this plasmid has been determined, the exact sizes of the multiple fragments produced by the HaeIII digestion are known. These DNA fragments, for the most part, migrate true to size on electrophoresis in acrylamide gels. The sizes of these marker bands are indicated to the left of the lanes containing the three experimental samples, designated A, B, and C in Figure 17. By examining the bands migrating near the 80- and 89-bp marker bands, the 80-, 88-, 96-, 104-, ... bp bands for the $(GA_3T_3C)_n$ family (lane A) can be identified. Members of this family run true to size (as straight DNA in the gel). Likewise, the bands for the $(GGGA_3T_3CCC)_n$ family ran as expected for their lengths of 84, 96, 108, 120, ... bp (lane C). However, the spacing of the bands corresponding to $(GGA_3T_3CC)_n$ in lane B was not that expected for 10 bp. Assigning the band migrating equivalent to the 51-bp band in the marker lane as 50-bp, the next several molecules, which are physically 60, 70, 80, 90, ... bp in length, migrate as DNAs of about 65, 80, 100, 120, and 145 bp. Multimer number 20 (200 bp) migrates at about 2.3 times its apparent length or the same as a molecule of 460 bp.

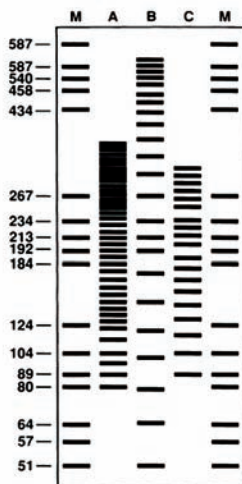


Figure 17. DNA bending analyzed by polyacrylamide gel electrophoresis.

The migration of these polymers is plotted in Figure 7 as $[\text{bP}_{\text{app}}/\text{bP}_{\text{seq}}]$ plotted as a function of length of the multimer. Each of these molecules contained the core sequence GAAATTC, which consists of bend-inducing A tracts (one in each of the complementary strands of DNA) and identical flanking bases. The distortion in helix structure at each A tract, which leads to the unusually slow migration for the 10-bp family, is present in the 8- and 12-bp families.

The only difference between the three molecules is the relative spacing of the A tracts, demonstrating that A tracts must be spaced at 10-bp intervals (not 8 or 12 bp) to observe stably bent DNA.

Phasing and DNA Looping

The formation of a DNA loop involves two sites on the DNA that are relatively close. If one were to loop a piece of DNA that is less than several hundred base pairs, the two surfaces at the ends of the DNA that would contact one another would constitute a defined area that is restricted by the helical phasing of the DNA. There is not enough flexibility in the rotation of the helix over short distances for all surfaces to have an equal probability of making contact. However, if looping occurs between sites

that are thousands of base pairs apart, the helical phasing between the two sites becomes less important because of the greater degree of flexibility inherent in longer pieces of DNA.

The helical phasing of protein binding sites on the DNA can be very important in gene regulation. Experimental data demonstrating the importance of helical phasing and looping is shown in Figure 18 for the arabinose system. Processes controlled by the interaction of two relatively closely spaced proteins on DNA should exhibit a 10.5-bp periodicity. When in phase, the process should occur readily. When out of phase, the process should not occur or should occur at a slower rate. Data from Lee and Schleif (1989) for transcription from P_{BAD} are shown in Figure 18. When the $araO_2$ and $araI$ sites are moved from 34 to 170 bp apart, a 10.5-bp helical phasing is observed. The addition of half a helical turn (5 bp) relative to optimally oriented sites reduces expression. A periodicity of gene expression due to changing the helical phase relationship of two control sites in the DNA of a promoter has been observed for several other systems including the lac operon.

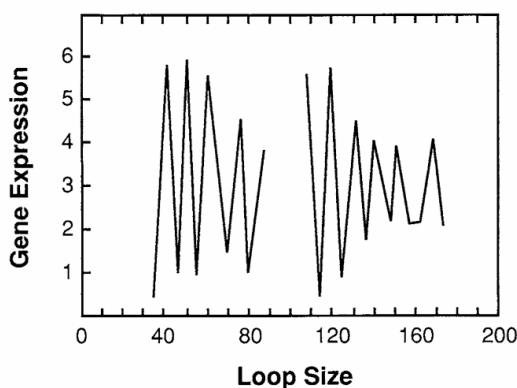


Figure 18. Phasing, looping, and gene expression.

Determining the Three-Dimensional Organization of the λ Site-Specific Recombination Complex

The organization of DNA binding sites for Int, IHF, and other proteins in the λ attP site is shown in Figure 19. Three IHF binding sites are bent by about 140° when IHF is bound. Since it would be difficult to interpret

changes in mobility when dealing with three bends in a DNA molecule, Snyder et al. analyzed changes in mobility of DNA containing pairs of IHF bend sites. Two molecules were constructed in which each contained about half the attP region. One molecule consisted of the left half of the attP site containing IHF sites H_1 and H_2 . The second consisted of the right half of the attP site containing H_2 and H' .



Figure 19. Sequence elements of the bacteriophage λ attP site.

The Crothers' laboratory demonstrated that the angle between two bends could be determined by the electrophoretic mobility of DNA. The rationale for this experiment is illustrated in Figure 20. Migration is a function of the end-to-end distance of a DNA molecule. In Figure 20A, the molecule with two bends on the opposite sides of the helix will migrate more rapidly on a polyacrylamide gel as shown in Figure 20B. The molecule with the bends on the same face or side of the DNA double helix will migrate most slowly. Molecules with intermediate angles of rotation at the second bend site will migrate with an intermediate mobility. Therefore, determining the spacing at which electrophoretic mobility is the slowest identifies a molecule in which two bends are in the same direction.

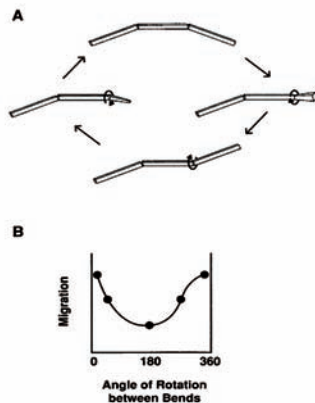


Figure 20. Effects of phasing two successive bends in DNA.

Snyder et al. (1989) inserted “spacers” between the H1 and H2 sites (in the Salt site) and between the H2 and H' sites (in the XhoI site) in each half of the attP DNA molecule. The length of the spacers varied in increments of 1 bp from 12 to 22 bp. Thus, these researchers varied the relative angle of rotation of the two IHF bend sites in increments of about 34° ($360^\circ \div 10.5$). Using the set of molecules in which the insert varied from 12 to 22 bp, the phasing of the two IHF binding sites moved through one complete helical turn of DNA. IHF was added to the family of half attP sites and the mobility of the DNA-protein complexes was analyzed by gel **electrophoresis**. Three bands were observed: “naked DNA” or unbound molecules, DNA with a single IHF bound, and DNA with two IHF molecules bound (one to each IHF binding site). The pattern of migration for the H1-H2 family is shown in Figure 21A. Several important conclusions can be drawn from these results. First, the attP DNA with H1 and H2 sites is naturally bent, as evident by the sinusoidal mobility of the unbound DNA as the spacer length was increased. The molecule with the 15-bp spacer migrated most slowly, indicating that this was the most bent molecule of the set. If the 15-bp spacer made the molecule maximally bent, then the natural attP sequence (without the 15-bp insert) would be relatively straight. The natural attP sequence must contain two natural bends flanking the Salt site that are out of phase by about 180° . Second, the majority of the DNA migrated as singly bound molecules. The migration of the DNA containing one bound IHF molecule was much slower than that of the unbound molecule because the protein bound to the DNA increases the molecular weight of the complex and because the geometry of the molecule changes. The singly bound species also showed a periodicity in gel mobility. The H1-H2 molecule with a 17-bp spacer migrated most slowly, indicating that this molecule

Electrophoresis

is the motion of dispersed particles relative to a fluid under the influence of a spatially uniform electric field.

was the most bent. Third, the amount of DNA migrating as the doubly bound species was reduced relative to the amount of DNA containing a single IHF molecule. In addition, the amount of doubly bound DNA varied as the spacer length changed. With a spacer length of 18 bp, very little doubly bound attP DNA was observed. The significance of this observation is discussed subsequently.

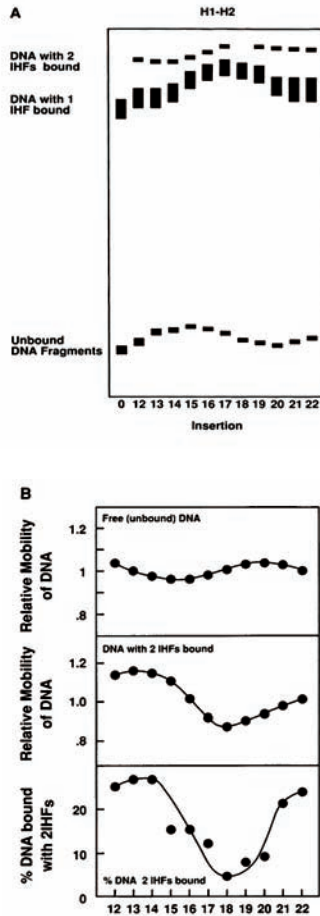


Figure 21. Determining the spatial organization of the λ attP-IHF complex.

From analysis of the mobility and the amounts of the various bands in this experiment (Figure 21A), conclusions can be drawn about

the directionality of the bends and, therefore, the three-dimensional organization of the attP DNA-protein complex. In Figure 21B, the relative mobility of unbound DNA, the relative mobility of doubly bound DNA, and the percentage of DNA with two IHFs bound are plotted as a function of the size of the spacer (or insertion). There is an inverse relationship between the extent of bending and the amount of the doubly bound species. The most bent molecule had the least doubly bound species. The doubly bound DNA molecule which was the most bent was the one with the 18-bp spacer, suggesting that the 18-bp spacer places the two 140 0 IHF-induced bends on the same face of the double helix. If the two bends are on the same side of the DNA helix, the two "arms" of DNA flanking both bend sites may be in physical contact with each other. This steric clash may be physically and energetically unfavorable. Moreover, this steric constraint may be responsible for the reduced amount of doubly bound species with the 18-bp spacer. Removal of the arms flanking the bend should relieve this steric constraint, allowing IHF to bind to both HI and H2 sites. The authors reported, as expected, that removal of 97 and 54 bp from the left and right arms, respectively, increased the amount of doubly bound molecules observed.

Using the information from the experiment shown in Figure 21 (and similar data for the H2-H' right half of the attP site) the authors calculated the angle between the three bend sites HI, H2, and H' in the natural sequence. In addition, they proposed the three-dimensional shape for the attP site shown in Figure 22. From the analysis of the doubly bound DNA molecules, it was evident that an 18-bp spacer placed the two IHF-induced bends on the same face of the helix. Knowing this, the relative angle between these bends in the natural attP sequence could be calculated by multiplying the average twist angle (34.3°) by the number of base pairs inserted between the IHF binding sites ($18 \times 34.3 = 617.4^\circ$). Since the helix repeat has a 360° per unit repeat interval (360° , 720° , 1080° , etc.), an introduced angle of 617.4° that restores the bends to a unit repeat interval (720°) means that the natural bends are offset by 102.6° or almost 3 bp worth of twist angle. This allows determination of the three-dimensional path of the DNA in space, as shown in Figure 22.

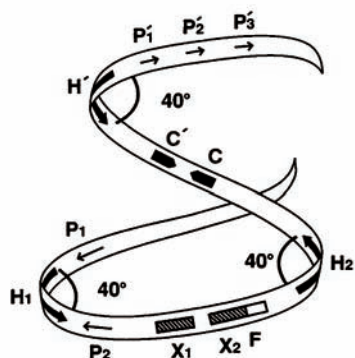
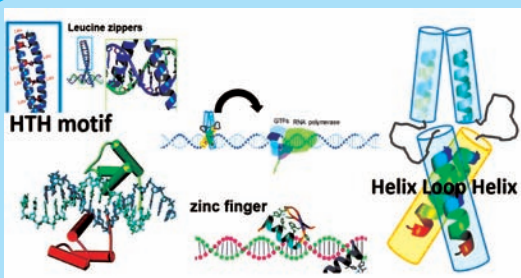


Figure 22. Three-dimensional organization of the λ attP region.



CORE CONCEPT: DNA-BINDING DOMAIN

DNA-binding domains adopt different structures, and the interaction of these domains

with DNA can be established through alpha helices, beta sheets, or disordered regions. The DNA-binding domain forms a module that can be separated from the rest of the transcription factor structure without losing activity. This facilitates structural studies of the isolated DNA-binding domains or their complexes with DNA using techniques that require low molecular weight proteins.

The DNA binding domain is located toward the carboxy terminus of the trans-activation domain. It consists of about 66–68 amino acids and is the most highly conserved region among the steroid receptor family. It comprises 20 invariant amino acids that form two zinc finger DNA binding motifs, providing the basis for the protein–DNA interactions in the regulator sequences of the target genes. Steroid receptors contain

nine conserved cysteine residues, of which eight interact in a coordinated fashion to form two separate tetrahedral metal-binding complexes, or fingers. The four cysteine residues in each finger bind one Zn^{2+} molecule, which permits interaction with DNA. Recent experiments have indicated that in addition to the cysteines, amino acids adjacent to the cysteines are critical in defining the specificity of binding the cognitive response elements. Elegant mutational analysis of the three amino acids at the base of the first zinc finger demonstrated that replacement of the amino acids Glu, Gly, and Ala found in the estrogen receptor (ER) with Gly, Ser, and Val, as in the **glucocorticoid receptor (GR)**, resulted in an ER that no longer recognized the estrogen response element (ERE), but instead recognized the glucocorticoid response element (GRE). The functional role of the second zinc finger is not clear, though it has been suggested that it aids in the stability of the binding of the protein to the DNA.

Glucocorticoid receptor (GR, or GCR) also known as NR3C1 (nuclear receptor subfamily 3, group C, member 1) is the receptor to which cortisol and other glucocorticoids bind. T

Function

One or more DNA-binding domains are often part of a larger protein consisting of further protein domains with differing function. The extra domains often regulate the activity of the DNA-binding domain. The function of DNA binding is either structural or involves transcription regulation, with the two roles sometimes overlapping.

DNA-binding domains with functions involving DNA structure have biological roles in DNA replication, repair, storage, and modification, such as methylation.

Many proteins involved in the regulation of gene expression contain DNA-binding domains. For example, proteins that regulate transcription by

binding DNA are called transcription factors. The final output of most cellular signaling cascades is gene regulation.

The hydrogen bonding pattern in the DNA major groove is less degenerate than that of the DNA minor groove, providing a more attractive site for sequence-specific DNA recognition.

The DBD interacts with the nucleotides of DNA in a DNA sequence-specific or non-sequence-specific manner, but even non-sequence-specific recognition involves some sort of molecular complementarity between protein and DNA. DNA recognition by the DBD can occur at the major or minor groove of DNA, or at the sugar-phosphate DNA backbone (see the structure of DNA). Each specific type of

DNA recognition is tailored to the protein's function. For example, the DNA-cutting enzyme DNase I cuts DNA almost randomly and so must bind to DNA in a non-sequence-specific manner. But, even so, DNase I recognizes a certain 3-D DNA structure, yielding a somewhat specific DNA cleavage pattern that can be useful for studying DNA recognition by a technique called DNA footprinting.

Many DNA-binding domains must recognize specific DNA sequences, such as DBDs of transcription factors that activate specific genes, or those of enzymes that modify DNA at specific sites, like restriction enzymes and telomerase.

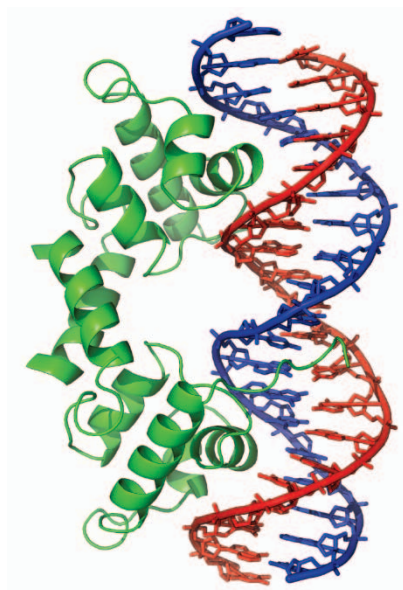
The specificity of DNA-binding proteins can be studied using many biochemical and biophysical techniques, such as gel electrophoresis, analytical ultracentrifugation, calorimetry, DNA mutation, protein structure mutation or modification, nuclear magnetic resonance, x-ray crystallography, surface plasmon resonance, electron paramagnetic resonance, cross-linking and microscale thermophoresis (MST).

Types

Helix-turn-helix

Originally discovered in bacteria, the helix-turn-helix motif is commonly found in repressor proteins and is about 20 amino acids long. In eukaryotes,

the homeodomain comprises 2 helices, one of which recognizes the DNA (aka recognition helix). They are common in proteins that regulate developmental processes (PROSITE HTH).



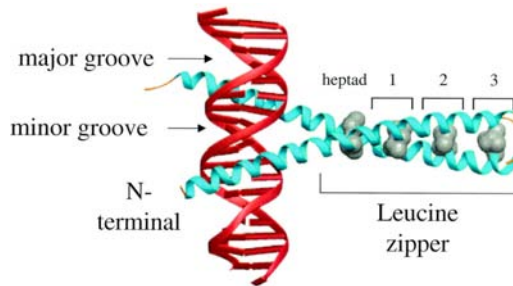
Zinc finger

The zinc finger domain is mostly found in eukaryotes, but some examples have been found in bacteria.^[5] The zinc finger domain is generally between 23 and 28 amino acids long and is stabilized by coordinating zinc ions with regularly spaced zinc-coordinating residues (either histidines or cysteines). The most common class of zinc finger (Cys2His2) coordinates a single zinc ion and consists of a recognition helix and a 2-strand beta-sheet.^[6] In transcription factors these domains are often found in arrays (usually separated by short linker sequences) and adjacent fingers are spaced at 3 basepair intervals when bound to DNA.

Leucine zipper

The basic leucine zipper (bZIP) domain is found mainly in eukaryotes and to a limited extent in bacteria. The bZIP domain contains an alpha helix with a leucine at every 7th amino acid. If two such helices find one another, the leucines can interact as the teeth in a zipper, allowing dimerization of two proteins. When binding to the DNA, basic amino

acid residues bind to the sugar-phosphate backbone while the helices sit in the major grooves. It regulates gene expression.



Winged helix

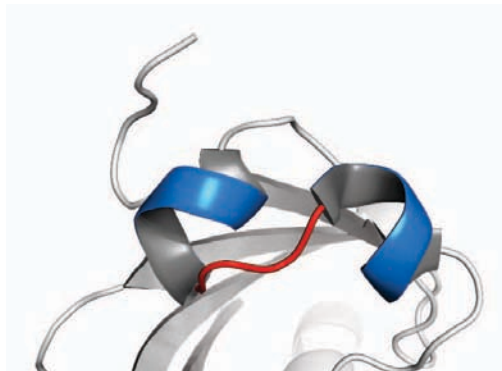
Consisting of about 110 amino acids, the winged helix (WH) domain has four helices and a two-strand beta-sheet.

Winged helix-turn-helix

The winged helix-turn-helix (wHTH) domain *SCOP* 46785 is typically 85-90 amino acids long. It is formed by a 3-helical bundle and a 4-strand beta-sheet (wing).

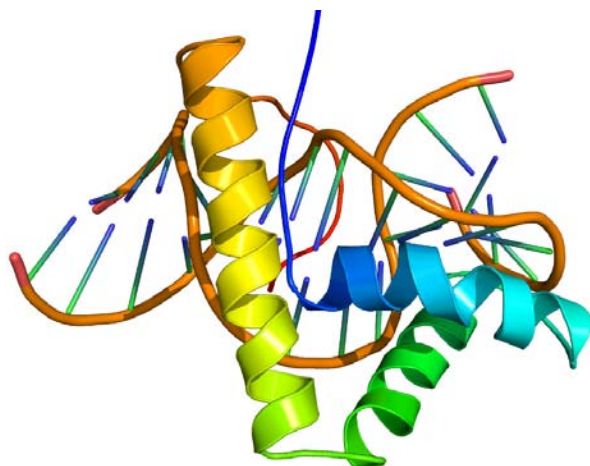
Helix-loop-helix

The basic helix-loop-helix (bHLH) domain is found in some transcription factors and is characterized by two alpha helices (α -helices) connected by a loop. One helix is typically smaller and due to the flexibility of the loop, allows dimerization by folding and packing against another helix. The larger helix typically contains the DNA-binding regions.



HMG-box

HMG-box domains are found in high mobility group proteins which are involved in a variety of DNA-dependent processes like replication and transcription. They also alter the flexibility of the DNA by inducing bends. The domain consists of three alpha helices separated by loops.



Wor3 domain

Wor3 domains, named after the White–Opaque Regulator 3 (Wor3) in *Candida albicans* arose more recently in evolutionary time than most previously described DNA-binding domains and are restricted to a small number of fungi.

OB-fold domain

The OB-fold is a small structural motif originally named for its oligonucleotide/oligosaccharide binding properties. OB-fold domains range between 70 and 150 amino acids in length. OB-folds bind single-stranded DNA, and hence are single-stranded binding proteins.

REVIEW QUESTIONS

1. What is wedge model for DNA bending?
2. Discuss about the junction model for DNA bending.
3. Describe the refined junction model.
4. Explain the protein-induced DNA bending.
5. What is DNA-binding domain?

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CHAPTER 3

DNA SUPERCOIL

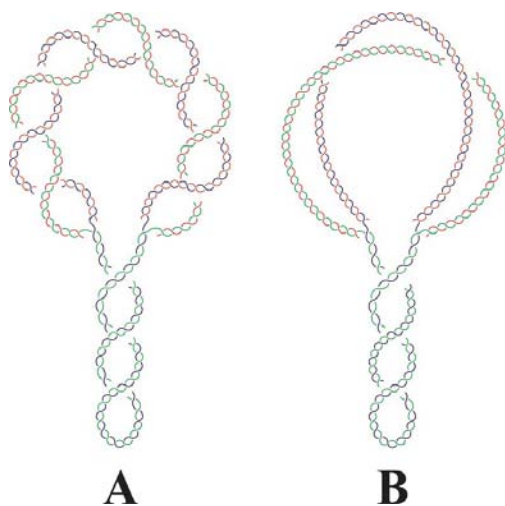
DNA supercoiling is a special property of circular, double-stranded DNA that is topological in origin. It confers new structural and energetic properties. When the ends of a linear DNA molecule are ligated to produce a covalently closed circle, the two strands become intertwined like the links of a chain, and will remain so unless one of the strands is broken. The number of times one strand is linked with the other is described by a fundamental property of DNA

After studying this chapter, you will understand the core concept of:

- Structure of DNA Supercoiling
- Supercoiling of Bacterial DNA
- Transcriptional Regulation by DNA Supercoiling

INTRODUCTION

DNA supercoiling (SC) has received considerable attention in recent years as a global and ancestral actor in genetic regulation. This is especially conspicuous in bacteria, where the chromosome is maintained at an out-of-equilibrium level of negative SC by a finely controlled balance of topoisomerase activity. And yet, in contrast to classical regulation based on transcription factors, quantitative models of the regulatory mechanisms by SC are essentially lacking. A possible explanation for this shortcoming is that SC affects transcription at several stages of the process, and can also be involved in various and complex interactions with regulatory proteins. As a result, virtually every investigated promoter exhibits a distinct SC response, making it difficult to dissect and model the underlying mechanisms. In this review, we wish to summarize existing evidence and models suggesting a widespread role of SC in bacterial genetic regulation, and more specifically in bacterial virulence. Here, we propose a complementary focus on proposed mechanistic and computational models of transcriptional regulation by SC as well as accumulating information obtained from transcriptomic data, which together underline the broad relevance of the investigated phenomenon in bacterial virulence and call for a combined experimental-theoretical research effort.





CORE CONCEPT: STRUCTURE OF DNA SUPERCOIL- ING

Supercoiling is introduced into DNA molecules when the double helix is twisted around its own axis in three-dimensional space. Generally, DNA molecules are negatively supercoiled inside cells, although the level of supercoiling is not equal throughout the genome and many supercoils may be constrained by bound proteins. Supercoiling increases the free energy of DNA and influences DNA metabolism by promoting or hindering specific enzymatic processes.

DNA supercoiling refers to the over- or under-winding of a DNA strand, and is an expression of the strain on that strand. Supercoiling is important in a number of biological processes, such as compacting DNA. Additionally, certain enzymes such as topoisomerases are able to change DNA topology to facilitate functions such as DNA replication or transcription. Mathematical expressions are used to describe supercoiling by comparing different coiled states to relaxed B-form DNA.

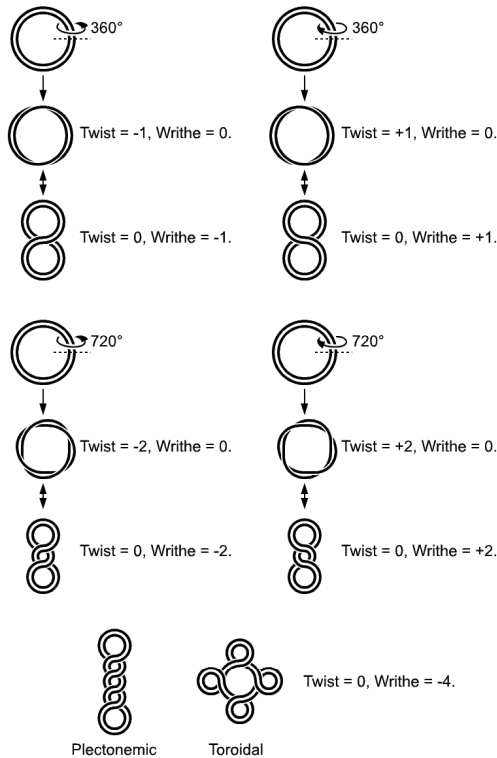


Figure 1: Supercoiled Structure of Circular DNA: This is a supercoiled structure of circular DNA molecules with low writhe.

As a general rule, the DNA of most organisms is negatively supercoiled. In a “relaxed” double-helical segment of B-DNA, the two strands twist around the helical axis once every 10.4 to 10.5 base pairs of sequence. Adding or subtracting twists, as some enzymes can do, imposes strain. If a DNA segment under twist strain were closed into a circle by joining its two ends and then allowed to move freely, the circular DNA would contort into a new shape, such as a simple figure. Such a contortion is a supercoil.

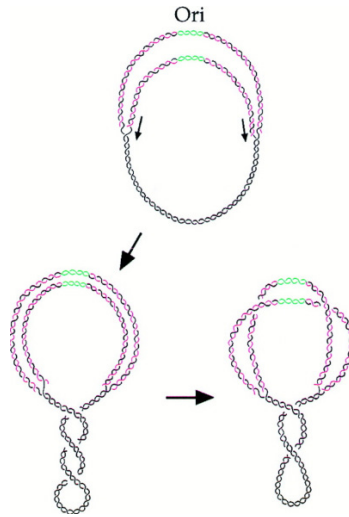
The simple figure is the simplest supercoil, and is the shape a circular DNA assumes to accommodate one too many or one too few helical

twists. The two lobes of the figure will appear rotated either clockwise or counterclockwise with respect to one another, depending on whether the helix is over or under wound. For each additional helical twist being accommodated, the lobes will show one more rotation about their axis.

The noun form “supercoil” is rarely used in the context of DNA topology. Instead, global contortions of a circular DNA, such as the rotation of the figure lobes above, are referred to as writhe. The above example illustrates that twist and writhe are interconvertible. “Supercoiling” is an abstract mathematical property representing the sum of twist and writhe. The twist is the number of helical turns in the DNA and the writhe is the number of times the double helix crosses over on itself (these are the supercoils).

Extra helical twists are positive and lead to positive supercoiling, while subtractive twisting causes negative supercoiling. Many topoisomerase enzymes sense supercoiling and either generate or dissipate it as they change DNA topology. In part because chromosomes may be very large, segments in the middle may act as if their ends are anchored. As a result, they may be unable to distribute excess twist to the rest of the chromosome or to absorb twist to recover from underwinding—the segments may become supercoiled, in other words. In response to supercoiling, they will assume an amount of writhe, just as if their ends were joined.

Supercoiled DNA forms two structures; a *plectoneme* or a *toroid*, or a combination of both. A negatively supercoiled DNA molecule will produce either a one-start left-handed helix, the toroid, or a two-start right-handed helix with terminal loops, the plectoneme. Plectonemes are typically more common in nature, and this is the shape most bacterial plasmids will take. For larger molecules, it is common for hybrid structures to form – a loop on a toroid can extend into a plectoneme. If all the loops on a toroid extend, it becomes a branch point in the plectonemic structure.



Energetics of Supercoiled DNA

cccDNA is a special DNA structure that arises during the propagation of some viruses in the cell nucleus and may remain permanently there.

All molecules, DNA will assume a configuration of minimum energy, and this is usually a helix of the B-form. Upon bending or twisting of the molecule, its energy is increased. For a **cccDNA** (covalently closed circular DNA) with a surplus or deficit in Lk , conformational modifications introduce specific changes to the free energy of the molecule. For example, to accommodate the same length of DNA in fewer helical turns, the double helix must be untwisted, leading to a substantial increase in the deformation energy of the molecule. By taking an appropriate writhed configuration, the cccDNA minimizes the amount by which it departs from the B configuration and reduces its deformation energy. On the other hand, writhing always introduces some curvature, and so it increases the bending contribution to the energy of the molecule. Since Wr and Tw are interconvertible, it is apparent that the underwound

DNA molecule will assume a configuration that optimizes twist while introducing the smallest possible amount of bending.

Experimental studies during the 1970s established that the free energy of a supercoiled DNA sample (DG_{sc}) has a quadratic dependence on DLk:

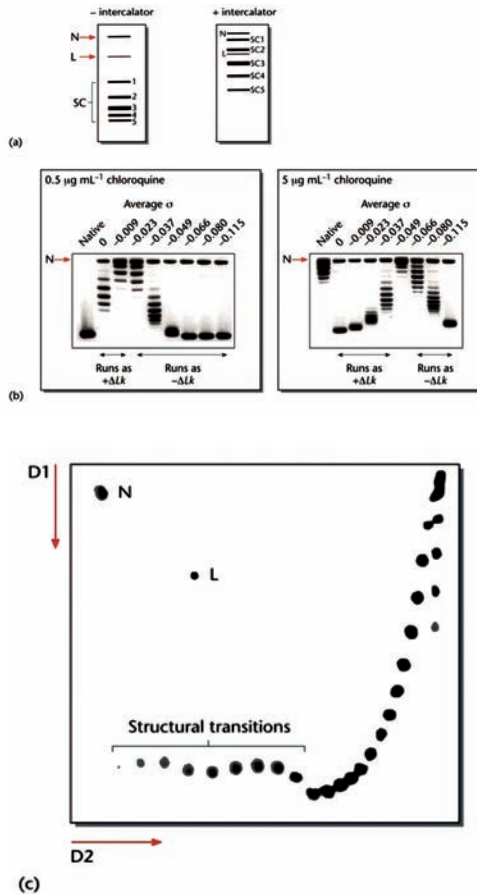
$$\Delta G_{sc} = K \cdot \Delta Lk^2 \quad (1)$$

where K is a DNA length-dependent constant. Thus, samples of cccDNA exist in a normal (Gaussian) distribution of topoisomers (i.e. molecules have a continuous, symmetrical distribution of Lk around the most intensely populated topoisomer – see Figure 2). Theoretical simulations suggest it is likely that DG_{sc} varies with ionic conditions and, in fact, it may not be a quadratic function of DLk under all conditions. The influence of environmental conditions on DNA supercoiling is due, at least in part, to the fact that ionic strength and temperature alter Tw of doublehelical DNA. Effects of ionic environment on the threedimensional structure of DNA are to be expected because DNA is a polyelectrolyte with a net negative charge at every nucleotide residue.

The free energy of supercoiling can be normalized to circle size in the same manner as described for DLk. Hence:

$$\Delta G_{sc}/N = NK(\Delta Lk/N)^2 \quad (2)$$

For DNA circles of 3–10 kb, it was shown that NK is independent of DNA circle size, and that $NK \approx 1100RT$, where R is the gas constant and T is the temperature. Since DLk/N is proportional to σ , the free energy of supercoiling per base pair is proportional to σ^2 , and is independent of circle size. Studies with small DNA molecules showed that NK increases gradually with decreasing DNA size, with a value of about 4000RT for a 300-bp circle. The interpretation of this finding was that, for small molecules, writhing of the DNA becomes increasingly unfavourable relative to twisting. Due to these energy considerations, it is thought that a higher proportion of supercoiling is partitioned into twisting rather than writhing for small DNA circles.



levels of supercoiling. Utilization of multiple gels with different concentrations of intercalator allows measurement of ΔL_k . For each sample, average superhelical density (σ). Note that in each gel, samples can have positively or negatively supercoiled topoisomers. The inclusion of intercalator in the running buffer alters the electrophoretic mobility of all topoisomers equivalently. Superhelical density can be measured for experimental samples ('native') by comparison with those of known σ . (c) Twodimensional agarose gel electrophoresis of topoisomers ranging from high negative σ to moderate positive σ . A DNA sample is loaded in a single well in a large agarose gel and electrophoresis is performed under specific conditions (usually without intercalator) in direction D1. After soaking of the gel in buffer containing intercalator, electrophoresis is continued in direction D2 (90° to D1). The gel shown contained 20 mg mL^{-1} chloroquine during the second electrophoresis, resulting in all topoisomers having positive σ . Deviation of **topoisomers** from a smooth curve indicates that structural transitions reduced their negative σ during the first direction of electrophoresis. Spots marked 'N' and 'L' indicate the position of migration of 'nicked' and 'linear' DNA molecules, respectively

Topoisomers are molecules with the same chemical formula and stereochemical bond connectivities but different topologies.

DNA Supercoiling and Stretching

DNA supercoiling have been carried out on small circular DNA molecules (plasmids). The circularity of these DNAs is useful in that it allows one to constrain the extremities of the molecule by a kind of boundary condition. Linear DNA molecules may also be torsionally constrained, by anchoring them (at their extremities or along their length) to an immobile surface or a massive object. The anchoring of chromatin to the nuclear scaffold, for example,

may generate topologically independent domains. The topology of such torsionally constrained molecules can then be described by a few simple quantities. The first is the twist (Tw) of the molecule, the number of times the two strands that make up the double helix twist around each other. For unconstrained B-DNA this is known to be one turn every 10.4 bp. Over winding increases the twist, under winding decreases it. The second topological quantity of interest is the writhe (Wr) of the molecule. On a twisted phone cord one often notices the formation of inter wound structures, called plectonemes which appear to be a way of releasing torsional stress. In these structures the number of times the cord loops over itself is its writhing number, and a positive or negative value may be assigned to it. The writhe represents another type of wrapping—that of the molecule's axis with itself.

If one constrains the ends of the DNA molecule, then the total number of times that the two strands of the helix cross each other (either by twist or by writhe) becomes a topological invariant of the system known as the linking number, Lk . For a circular plasmid the only way to change Lk is to break a strand and modify the number of times the DNA winds. In the case of linear anchored molecules, however, rotating one end of the molecule allows one to access and modify Lk without damaging the DNA. A mathematical theorem due to White states that

$$Lk = Tw + Wr \quad (1)$$

The linking number of a torsionally relaxed (linear or circular) DNA is written as $Lk_0 = Tw_0 + Wr_0$. Tw_0 is the number of helical repeats in B-DNA. $Wr_0 = 0$, assuming that DNA has no significant spontaneous curvature with which it could form coils or loops. The relative difference in linking number between the supercoiled and relaxed forms of DNA is called the excess linking number, σ :

$$\sigma = \frac{Lk - Lk_0}{Lk_0} \quad (3)$$

The molecule is overwound when σ is positive and under wound when it is negative.

As mentioned earlier, the first experiments concerning the force versus extension characteristics of single DNA molecules were carried

out on torsionally unconstrained DNA. A good fit over the full range of their data was achieved by treating the DNA as a worm-like chain. At low stretching forces such a system behaves like a random walk

$$F = \frac{3}{2} \frac{k_B T}{\xi} \frac{x}{l_0}$$

where F is the applied force, k_B is the Boltzmann constant, T is the temperature, ξ the walk's persistence length, and x/l_0 is the walk's relative extension. At high forces, the WLC's relative extension tends toward 1: $x/l_0 \approx 1 - 1/\sqrt{F}$ useful interpolation formula, correct to within 10%, was proposed to join these two regimes at intermediary forces:

$$\frac{F\xi}{k_B T} = \frac{1}{4} \left[\left(1 - \frac{x}{l_0} \right)^{-2} - 1 \right] + \frac{x}{l_0} \quad (4)$$

Where x is the polymer's extension and l_0 its contour length. In this paper we fit the WLC model's exact numerical solution (manuscript in preparation) to the experimental data to extract the contour length l_0 and the persistence length ξ of the system being stretched. One may thus determine a crucial parameter: the number of molecules being stretched. Indeed, a two-molecule system is twice as stiff as a single-molecule system. As a consequence, the effective persistence length of a two-molecule system is half that of a single molecule. In our experiments we can thus distinguish between DNA supercoiling (one DNA molecule) and DNA braiding (two DNA molecules).

The Importance of DNA Supercoiling

DNA supercoiling is important for DNA packaging within all cells. Because the length of DNA can be thousands of times that of a cell, packaging this genetic material into the cell or nucleus (in eukaryotes) is a difficult feat. Supercoiling of DNA reduces the space and allows for much more DNA to be packaged. In prokaryotes, plectonemic supercoils are predominant, because of the circular chromosome and relatively small amount of genetic material. In eukaryotes, DNA supercoiling exists on many levels of both plectonemic and solenoidal supercoils, with the solenoidal supercoiling proving the most effective in compacting the

DNA. Solenoidal supercoiling is achieved with histones to form a 10 nm fiber. This fiber is further coiled into a 30 nm fiber, and further coiled upon itself numerous times more.

DNA packaging is greatly increased during nuclear division events such as mitosis or meiosis, where DNA must be compacted and segregated to daughter cells. Condensins and cohesins are structural maintenance of chromosome (SMC) proteins that aid in the condensation of sister chromatids and the linkage of the centromere in sister chromatids. These SMC proteins induce positive supercoils.

Supercoiling is also required for DNA and RNA synthesis. Because DNA must be unwound for DNA and RNA polymerase action, supercoils will result. The region ahead of the polymerase complex will be unwound; this stress is compensated with positive supercoils ahead of the complex. Behind the complex, DNA is rewound and there will be compensatory negative supercoils. It is important to note that topoisomerases such as DNA gyrase (Type II Topoisomerase) play a role in relieving some of the stress during DNA and RNA synthesis.

Maintenance of DNA supercoils

Even living cells have their own tangled problems to solve. They involve knots that arise accidentally in DNA and that can compromise its functionality. Researchers now suggest that the DNA propensity to be supercoiled, just like telephone cords, prevents accidental knots. These coils can keep DNA knots locked in place long enough to be untied by specialised enzymes. The computational study, based on molecular dynamics simulations of bacterial DNA.

A tangled problem

DNA filaments in living cells is all crumpled and entangled, and to be functional, must be constantly unraveled, much like telephone cords that accumulate annoying curls or supercoils with careless use. Supercoils are present in DNA, too, and they are believed to be instrumental for keeping DNA compact and exposing or hiding genetic information to the cellular machinery that can process it, according to Lucia Coronel and Antonio Suma, young scientists and first authors of this study, which was coordinated by Cristian Micheletti.

At the same time, it has long been known that knots can accidentally form in DNA, with negative consequences for the cell. What had remained elusive until now was the dynamic interplay or coexistence on the same DNA filaments of complex knots and supercoiled regions. And this is what the researchers at SISSA set out to study using computer simulations, investigating the implications for DNA structure and its change in time, and whether such interplay might unlock new functional possibilities. They also sought to understand how the ensuing complex behavior fits with what is already known about the molecular maintenance of DNA.

For this study, the scientists used detailed and extensive molecular dynamics simulations. They first noticed that in supercoiled DNA rings, knots are likely found in one of two positions—at the tip of the supercoils or in a more central position. A closer inspection revealed a more surprising and unexpected effect. Coronel and Suma write, “We studied the DNA filament and noted that without supercoils, the knotted region would move relatively fast along the filament. Likewise, supercoiled regions can rapidly change in knot-free DNA. However, when knots and supercoiling are simultaneously present, then the crucial contact points in DNA knots become locked in place, persistently. And this unexpected effect is particularly interesting because it could be key to a specific and unexpected biological functionality.”

The authors write, “We know that living cells routinely deal with DNA knots, and we also know that these forms of entanglement are usually detrimental to biological functionality; for instance they can prevent the genetic information from being read and translated into protein products. Specific enzymes of the topoisomerase family are responsible for disentangling DNA. Their *modus operandi* recalls the efficient, if not drastic, way in which Alexander the Great untied the proverbial Gordian knot with a cut of his sword. Similarly, these enzymes untie DNA filaments by a sophisticated cutting and sealing action.»

There is still an ongoing debate on which molecular mechanisms can guide these enzymes, which are minute compared to the typical scale of DNA filaments, to intervene on the correct places where their cut-and-paste action can untie the knots. The authors conclude, “It has been suggested that the target places are recognized by specific geometric features, which is a plausible and elegant mechanism. However, it was unclear how these features could persist in spite of the incessant molecular

motion. Our study suggests that DNA supercoiling can favor the action of topoisomerases by keeping knots in a stable configuration for a time span that is much longer than other molecular rearrangements. In this way, the enzymes could have sufficient time to recognize the target sites and, in turn, their cut-and-paste action would be simpler, more reliable and efficient.

Influence of DNA supercoils *in vivo*

DNA supercoils have broad influence on DNA structure, and the identification of unconstrained supercoils in eukaryotes opens a huge field of regulatory potential. In recent years, a number of alternative DNA structures stabilized by under-wound DNA supercoils have been identified in higher eukaryotes *in vivo*, including R-loops, G-quadruplexes, cruciforms, Z-DNA and ssDNA. Proteins including transcription factors have been shown to specifically associate with such DNA structures, as well as more subtle differences in DNA structure such as the transition from a B-form to an A-form helix and the localized under-winding of DNA. In these cases, it has not been established whether the change in DNA structure causes or is a consequence of protein binding, although there is an increasing evidence supporting a role for DNA structure in this process. Furthermore, the direct association between changes in unconstrained DNA supercoiling and transcription factor binding has only been demonstrated for one example, the FUSE interacting protein and FIRs at the supercoil-sensitive FUSE DNA element. In future work, investigators must systematically test other supercoil-sensitive elements for (a) the formation of alternative DNA structures as a result of DNA supercoiling and (b) the specific binding of regulatory proteins to these alternative DNA structure and the specific regulation of transcription/replication as a result. It is noteworthy that DNA sequence motifs for alternative DNA structures are highly enriched and evolutionarily conserved at gene promoters and human replication origins, further supporting a potential functional relevance for supercoil-dependent DNA structural transitions. Identifying whether these structures are a general mechanism for real-time signalling of ongoing transcription, and function to enhance/suppress future transcription is an essential next step for the DNA supercoil field.

In addition to altering the helical structure of DNA, supercoiling can introduce a rotational torque into the DNA, which facilitates the formation of the pre-initiation complex and subsequent gene expression at specific eukaryotic genes *in vivo*. By this mechanism, supercoils generated by the transcription of one gene could dissipate through the DNA and influence transcription from the promoters of neighboring genes, and supercoils generated by abortive transcription could facilitate full-length gene expression by priming the DNA structure of a promoter region. Whether under-wound DNA supercoiling alters DNA structure directly, or provides the energy for proteins to do so, is unknown. Furthermore, the influence of under-wound DNA supercoiling on different eukaryotic promoters has not been widely tested, with a single *in vitro* study reporting increased transcription in two of three promoters. Recent work has shown that eukaryotic gene promoters are generally under-wound, particularly when active, and it is now important to establish the features of promoters *in vivo* that confer supercoil sensitivity. Only with this knowledge, can we begin to understand how domains of DNA supercoiling influence the expression properties of the gene promoters contained within them.

Effects of supercoiling on DNA

RNA polymerase generates about seven DNA supercoils per second. If these are not efficiently removed the residual energy may influence DNA or chromatin structure locally, or, if the energy can be propagated along the fiber, at more distant sites. The capacity of negative supercoiling to unwind DNA and facilitate processes such as transcription and replication and its ability to induce alternative DNA structures such as cruciform, G-quadruplexes and Z-DNA have been noted.

To address how transcription-generated force might directly alter DNA structure *in vivo*, Kouzine. used a tamoxifen-inducible Cre recombinase to excise a chromatin segment with its torsional stress trapped intact. As the segment, flanked by loxP sites, had been positioned on a plasmid between divergently transcribing promoters it was demonstrated that as transcription intensified the degree of negative supercoiling trapped within the excised segment increased. Using the c-myc FUSE element as a reporter they showed that supercoiling could propagate along the fibre, melt the FUSE element and promote the binding of ssDNA binding proteins (Figure 3a).

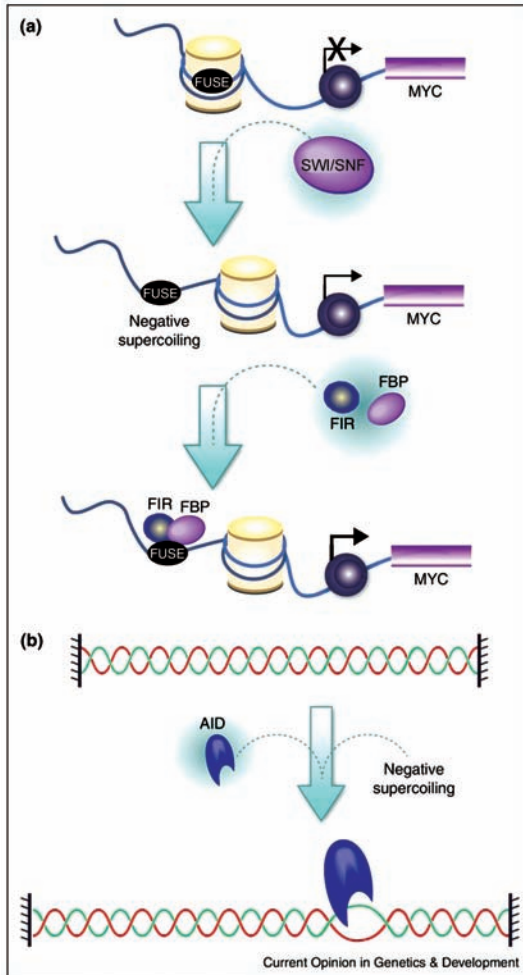


Figure 3: (a) Transcriptional activation of MYC is accompanied by a SWI/SNF activity revealing the FUSE element. A buildup of negative supercoiling propagates along the fibre, melting the FUSE element promoting the binding of the ssDNA binding proteins, FBP and FIR. (b) Negative supercoiling promotes ssDNA patch formation allowing the mutagenic activity of activation-induced cytidine deaminase (AID).

Although negative supercoiling promotes transcription initiation, supercoiling can also hinder polymerase elongation. To investigate how polymerase responds to different supercoiling environments Ma in a single-molecule approach, used an angular optical trap. RNA polymerase was immobilized on a slide whilst its DNA template, attached to a

quartz cylinder, was held in the trap. Rotation and torque could be applied to and measured from the DNA by manipulation of the quartz bead whilst its height provided a measure of displacement. Upon transcription into a negatively supercoiled template, the polymerase initially relaxed the DNA and then introduced positive supercoiling. As positive supercoiling accumulated ahead of the polymerase, it stalled. Thus, resisting torque slows RNA polymerase and increases its pause frequency.

In addition to facilitating the binding of polymerases or transcription factors, negative supercoiling can generate DNA substrates for more complex activities. In yeast, topoisomerase I inhibition promotes the formation of large ssDNA bubbles in highly expressed **rRNA** genes, which can be visualised by Miller spreads. Parsa have shown that substantially smaller ssDNA patches located within active genes (Figure 3b) may be substrates for the mutagenic activity of activation-induced cytidine deaminase (AID). Employing an inducible gene in a hyper-negatively supercoiled *E. coli* strain they demonstrated that negative supercoiling increased ssDNA patch density compared to wild type and promoted a higher mutation rate. It will be interesting to know whether a similar effect is observed in eukaryotic cells where the DNA is packaged into chromatin and levels of supercoiling are probably buffered.

rRNA is the RNA component of the ribosome, which is essential for protein synthesis in all living organisms. rRNA is the predominant RNA in most cells, composing around 80% of cellular RNA.



CORE CONCEPT: SUPERCOILING OF BACTERIAL DNA

DNA extracted from bacterial cells is negatively supercoiled, a feature that is mediated by the action of DNA topoisomerases. Negative supercoiling arises when DNA has a linking deficit, i.e. when a topologically closed DNA molecule has fewer duplex turns than would be found in a nicked or linear DNA of the same length.

DNA molecules exist in two states, relaxed and supercoiled. A relaxed state of DNA molecule is one in which the exact number of turns of the helix can be predicted by calculating the number of base pairs, whereas when the relaxed DNA molecules is highly twisted to accommodate itself in a little space the state is called supercoiled.

The example of *Escherichia coli* makes clear why relaxed DNA molecule enters into the supercoiled state. The DNA of *E. coli* in its relaxed state becomes more than 1 mm in length, about 400 times longer than the *E. coli* cell itself. How can so much DNA be packed into such a little space of *E. coli* cell? Nature has solved this problem by imposing a kind of 'high order' structure on the DNA molecule.

In this structure the double-stranded helical DNA is further highly twisted employing the process called supercoiling, and is easily packed within *E. coli* cell. The supercoiled DNA molecule remains under torsion almost similar to the added tension to a rubber band that occurs when it is twisted.

DNA molecule supercoils in either a positive or negative direction; it is the negative supercoiling that predominantly operates in all non-hyperthermophilic organisms in nature. In negative supercoiling, the DNA molecule is twisted about its axis in the opposite direction from that of the right-handed double helix.

DNA molecule can be alternately supercoiled and relaxed. It is because the supercoiling is necessary for packaging the DNA into the bacterial cell and relaxing is necessary for the replication of DNA. Figure 4 shows how supercoiling takes place in a relaxed covalently closed circular dsDNA molecule, and it also shows how a supercoiled DNA molecule becomes relaxed one.

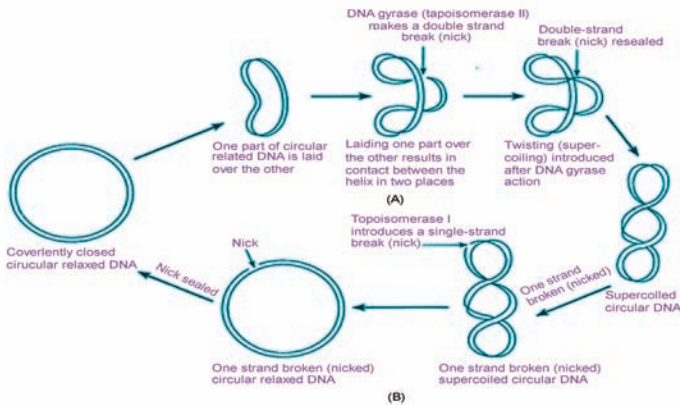


Figure 4. Interconversion of relaxed and supercoiled DNA in bacteria (A) Introduction of supercoiling into a circular DNA by the activity of DNA gyrase (topoisomerase II). DNA gyrase makes double-strand breaks (nicks). (B) Conversion of supercoiled circular DNA into relaxed circular DNA by the activity of topoisomerase I.

Supercoiling in bacterial and most archaeal DNA is carried out with the help of an enzyme called DNA gyrase, which is a type of enzyme called a topoisomerase, particularly a topoisomerase II.

The process completes via several steps

- (i) One strand of circular relaxed DNA is laid over the other,
- (ii) Laid one part over the other results in contact between the helix in two places, and DNA gyrase (topoisomerase II) makes a double-strand break (nick) in the region of contact at one place, and (3) the broken (nicked) double-strand is resealed on the opposite side of the intact strand resulting in supercoiling in DNA molecule.

Supercoiling in DNA molecule removed by the action of another enzyme called **topoisomerase I**. This enzyme introduces a single-strand break (nick) in the supercoiled DNA and causes the rotation of one

Topoisomerases

are enzymes that participate in the overwinding or underwinding of DNA.

single- strand of the double-strand DNA around the other resulting in the return of supercoiled state of DNA to the relaxed state.

However to prevent the entire bacterial DNA from becoming relaxed every time a break (nick) is made, the DNA contains supercoiled domains. Actually, the double-stranded bacterial DNA is arranged not in one supercoil but in several supercoiled domains.

In *E. coli* over 50 supercoiled domains are reported to exist, each of which is thought to be stabilized by binding to specific proteins. However, a nick in one strand of double-stranded DNA in one of these domains does not allow DNA to relax in other domains.

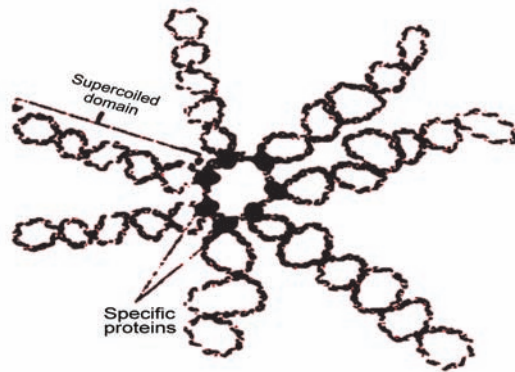


Figure 5: Supercoiled domains thought to be stabilized by binding to specific proteins



CORE CONCEPT: TRANSCRIPTIONAL REGULATION BY DNA SUPERCOILING

The role of DNA supercoiling (SC) in transcriptional regulation has attracted considerable attention in recent years. Due to the helical nature of DNA, mechanical torsion affects transcription at both initiation and elongation steps, and can thereby be considered as a non-conventional transcriptional regulator in eukaryotes as well as bacteria. In the latter, fast changes in DNA topology play a central role in the global transcriptional response to environmental stress. Inheritable changes in DNA topology are also under positive selection during evolution experiments with bacteria, in which SC-modifying mutations can provide a substantial fitness gain.

The abovementioned transcriptional responses induced by SC variations differ qualitatively from those induced by classical transcriptional factors (TFs). The latter recognize, bind and regulate a specific subset of the genome defined by a well-defined (although often degenerate) target sequence motif, and their action can be modelled using classical thermodynamic models of activation or repression. SC affects the transcriptome of all investigated species globally, without any identified promoter sequence determinant. And yet strikingly, regulatory models comparable to those involving TFs are essentially lacking. There are two reasons for this: first, experimentally, SC regulates gene activity in a continuous “more or less” manner as opposed to the stronger “on or off” mode of regulation by TFs; second, SC can modulate transcription in a variety of ways, making them difficult to decipher. In the following, we discuss such mechanisms, most of which will be illustrated on the promoter of *pele*, one of the major virulence genes of the phytopathogen *Dickeya dadantii* where SC-mediated regulation was studied extensively.

Basal Regulation of the RNA Polymerase-DNA Interaction

RNA polymerase is an enzyme that synthesizes RNA from a DNA template.

The ancestral and global mode of regulation by SC results, in the first instance, from it affecting the interaction between DNA and **RNA Polymerase** (RNAP) itself, independently from any additional regulatory protein. But this basal regulation already involves distinct mechanisms occurring at successive steps of the complex process of bacterial transcription: closed-complex formation, open-complex formation, promoter clearance, and elongation.

The most clearly identified – and possibly strongest – effect of SC on transcription initiation results from the requirement for RNAP to open the DNA strands and stabilize a “transcription bubble”, in order to gain access to the DNA bases in the template strand. In torsionally unconstrained DNA, this melting transition represents a substantial free energy cost of around $10 k_B T$ (6 kcal/mol), which in eukaryotes is provided through ATP hydrolysis by the basal transcription factor TFIID. Crucially, this cost reduces drastically when the double helix is destabilized by negative torsion at SC levels physiologically relevant in bacteria (Figure 6A, upper panel), thus providing the physical basis for the bacterial transcription process that does not require any external energy consumption. Interestingly, the same can become true of eukaryotic RNA Polymerase II when operating on a comparably negatively supercoiled template, whereas temperature may replace SC as the source of melting energy in thermophilic archaea relying on reverse gyrase. This strong regulatory effect of SC can be observed in well-controlled in vitro transcription assays (Figure 6A, lower panel), where plasmids carrying a model promoter are prepared at different supercoiling levels, resulting in a drastic variation

of expression strength without any modification of the DNA sequence or regulator concentration. This activation curve is quantitatively reproduced using thermodynamic models of DNA opening relying on knowledge-based enthalpic and entropic parameters for base-pairing and stacking interactions of all base sequences.

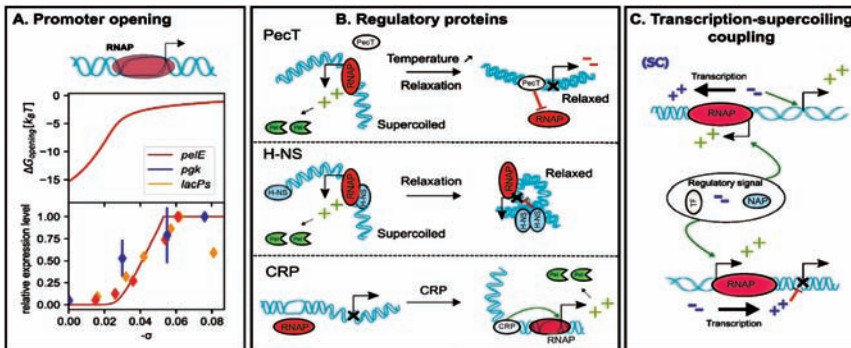


Figure 6. Mechanisms of transcriptional regulation by DNA supercoiling.

(A) Opening free energy of the *pelE* promoter of *D. dadantii* (upper panel) and expression level of three promoters from different species (lower panel, solid line for the model prediction); (B) Various mechanisms involving SC and NAPs/TFs at the *pelE* promoter; (C) Local SC-mediated regulatory interaction: transcription from a given gene (left) can activate (top) or repress (bottom) its neighbor depending on their relative orientation.

How may this mechanism lead to transcriptional regulation, i.e., the selective activation of a subset of promoters by global SC variations? The opening curve is strongly dependent on the promoter base sequence (mostly though its GC content) in the -10 region and that immediately downstream referred to as the “discriminator”. Remarkably, although the latter region does not harbour any consensus binding signal for RNAP, mutation studies showed that it plays a predominant role in the SC-sensitivity of promoters: GC-rich discriminators are typically more activated by negative SC than AT-rich ones, the canonical example being those of stable RNAs strongly induced during exponential growth. And yet, a systematic model of this regulation mechanism is still lacking. A possible obstacle is that the thermodynamic description used above

might be insufficient, if the expression level of investigated promoters is limited not by the rate of promoter opening but of promoter escape. As an example, some promoters might attract RNAP and easily form an open-complex, but still exhibit low expression levels if the latter is too stable, resulting in abortive rather than processive transcription. Since the influence of SC on these subtle kinetic steps was not dissected in detail, little can be predicted from such a scenario that may explain why promoters respond differently to SC variations, and some even in opposite ways. Yet we still note that many *in vitro* investigated promoters do follow the behavior expected from thermodynamic modelling (Figure 6 A, lower panel), which might thus account for at least a significant fraction of transcriptomic responses to SC variations.

A regulatory action of SC was proposed at least at two other transcription steps. RNAP binds promoters by recognizing the -10 and -35 elements, which are separated by a spacer of variable size (15–20 nt, with an optimum at 17 nt). Because of the helical structure of DNA, these variations are associated to different relative angular positions of the recognition sites around the helical axis, possibly placing them out-of-phase for RNAP binding and closed-complex formation. A suitable (spacer length-dependent) level of torsional stress could then modulate this binding rate by untwisting the spacer DNA toward a favorable orientation. This scenario was invoked to explain the response of several promoters to SC variations, including ribosomal RNAs which usually exhibit a suboptimal spacer length of 16 nt.

SC may also affect transcription at the elongation stage, where positive torsion hinders the progress of RNAP, as demonstrated from single-molecule experiments. Such positive SC levels are not observed *in vivo* by usual techniques involving plasmids, but might be transiently generated locally downstream of transcribed genes. However, most *E. coli* genes were found to be transcribed at a comparable speed, suggesting that positive supercoils are readily eliminated by topoisomerases and do not play any regulatory role during elongation of these moderately expressed genes, whereas mechanical stalling of RNAP might occur at the most strongly expressed genes such as those of ribosomal RNAs.

Specific Regulation Involving DNA Regulatory Proteins

SC affects transcription via RNAP itself, but also via promoter-specific regulatory proteins (TFs). In classical thermodynamic regulation models, the latter act (as activators or repressors, depending on their action on RNAP activity) at distinct binding sites characterized by their affinity, the latter being inscribed in the genomic sequence and usually described by sequence motifs. In this simplistic view, the cell has limited regulatory freedom, since the only tunable parameter is the TF concentration. In the classical example of the global regulator CRP which binds (and usually activates) hundreds of promoters in *E. coli*, increasing its concentration (or rather that of its cofactor cAMP) may activate many of these target promoters, but in this model, the *relative* amount of CRP bound at these promoters, and thus also their relative activities, is out of the control of the cell. This view is increasingly challenged since additional layers of complexity were identified, including epigenetic modifications of the regulators or promoter DNA and post-transcriptional regulation. However, we note that many regulatory proteins recognize not only the base sequence, but also the DNA shape, a mechanism often referred to as indirect readout. In that case, the activity of classical regulators is intrinsically modulated, not only by chemical (epigenetic) modifications requiring dedicated enzymes, but also by the ubiquitous mechanical deformations. A computational estimation of this recognition mode suggests that, in the case of CRP, the latter may in fact be the strongest determinant of its loose sequence selectivity. Crucially, in contrast to the classical “static” sequence-motif model, this selectivity is now dependent on the mechanical state of chromosomal DNA, and thus subjected to cellular control through SC. Considering that the SC distribution itself is non-uniform and locally affected by NAPs, this additional mechanical dimension is probably a key contributor in the complexity of the binding selectivity of regulatory proteins.

This mechanism has not been investigated experimentally in many systems. One example is the thermoregulation of *peleE* by the repressor PecT, which is achieved not by a change in regulator concentration but rather by an increase of its binding affinity for the promoter resulting (at least partly) from temperature-induced chromosomal DNA relaxation (Figure 6B, top). PecT belongs to the LysR-like family of TFs relying on indirect sequence readout, which is the largest TF family in enterobacteria,

and similar mechanisms are used by other families. Altogether, the role of DNA mechanics in TF binding selectivity is thus likely much more important than generally considered, although the precise mechanisms remain to be identified and modelled. Because most dimeric TFs are smaller than RNAP, they are probably less sensitive to the untwisting of bound DNA than the latter, but this sensitivity might be augmented substantially in the case of several binding sites arranged in helical phase. For example, H-NS was found to repress *pele* on relaxed DNA (as most of its targets), but it is not the case on negatively supercoiled DNA. A proposed explanation is that H-NS traps RNAP in a small loop at this promoter by bridging its two binding sites, which in the latter condition is prevented by their unfavourable helical phasing (Figure 6B, middle). On the other hand, regulatory proteins also exert twist deformations themselves; a spectacular example is MerR, which in the presence of Hg_{2+} binds and untwists the 19-bp spacer of the mercury resistance operon of Tn501 (a transposable element isolated from *Pseudomonas aeruginosa*), thus enabling RNAP binding.

In our opinion, the interplay between regulatory proteins and local SC may mostly rely on 3D modifications of the promoter conformation, i.e., writhe rather than twist. On the one hand, writhe could facilitate the formation of loops required for many regulatory interactions. These loops are favoured by the distinct mechanical properties of promoter DNA sequences, and were already included in regulatory models based on the transfer matrix formalism. SC-dependent reduction of looping free energies can thus strongly modify the binding landscape of regulatory proteins, as already described. Conversely, many regulatory proteins induce strong bends in DNA, e.g., CRP ($\sim 90^\circ$), LexA ($\sim 35^\circ$), as well as the NAPs FIS ($\sim 45^\circ$) and IHF ($\sim 180^\circ$). Since such deformations drastically reduce the energetic cost of DNA loops, they are expected to displace the twist-writhe equilibrium within the bound region in favour of the latter, and may thus induce local topological changes similar to those induced by SC. In summary, just like the NAPs are involved in a complex double-sided interaction with SC that shapes the global structure of the **chromosome**, a comparable effect probably occurs with many more regulatory proteins at the more local scale of gene promoters, with direct consequences for local transcription.

A final important ingredient is the widespread occurrence of structural transitions in genomic DNA. The latter can switch from double-stranded B-DNA to, among others, denaturated, Z-DNA, G-quadruplex or cruciform states. The rates of these different transitions can be computed by thermodynamic modelling, and depend not only on the DNA sequence, but also on torsional stress that destabilises B-DNA and can be accommodated more favourably in alternate states. It was also shown that these transitions occur predominantly at bacterial gene promoters which they regulate according to various mechanisms, some involving TFs. Denaturated AT-rich regions located 50–200 bp upstream of the TSS can act as “sinks” for negative SC and impede the proper opening of the promoter by thermodynamic competition. This was shown to occur for *peIE*, which in vitro is not expressed in absence of CRP due to such upstream strand opening; when present, CRP not only favours the correct binding of RNAP, but also “closes” the upstream AT-rich tract, possibly by bending DNA (Figure 6B, bottom). Since denaturation bubbles are extremely flexible, they may also strongly facilitate the formation of loops required by TFs. Additionally, some regulatory proteins may selectively bind non B-DNA regions, as occurs at the mammalian oncogene *cMYC* where negative SC triggering DNA melting is provided by adjacent transcription. Finally, since rho-independent termination of transcription involves RNA hairpin structures, SC might also favour structural transitions in the DNA template itself at the transcription termination site, which could then modulate the termination rate, as already observed for the B-Z transition. Since transcriptional read-through was highlighted as a widespread feature in bacterial genomes in recent years, an additional underestimated layer of regulation might thus also occur at this later stage of the transcription process.

A **chromosome** is a deoxyribonucleic acid (DNA) molecule with part or all of the genetic material (genome) of an organism.

Spatial Heterogeneities of DNA Supercoiling: The Transcription-Supercoiling Coupling

The intimate relationship between SC and transcription is not single-sided. In the elongation step, the helical structure of DNA imposes a fast rotation of the bulky RNAP relative to it (around two turns per second), but this movement is strongly hindered by the viscosity and crowding of the surrounding medium, resulting in an asymmetric accumulation of torsional stress from back to front, as recognised more than 30 years ago. This phenomenon thus leads to an intrinsic dynamical coupling between SC and transcription highly dependent on gene orientations (Figure 6C). This coupling regained significant interest in recent years since it was shown to underpin transcriptional bursting in bacteria, i.e., the nonlinear auto-induction of a promoter that can typically give rise to phenotypical heterogeneities among isogenic populations of cells. Several theoretical models were proposed, most of them focusing on biophysical properties of transcription. A strong obstacle for their application in genomic regulation is the lack of experimental knowledge of the distribution of SC along a bacterial chromosome. A promising method involving the intercalating agent psoralen was developed in eukaryotes but did not yet provide high-resolution data in bacteria. Recently, indirect information was provided from binding distributions of topoisomerases obtained by ChIP-Seq at different resolutions. These data, together with a systematic analysis of bacterial transcriptomes, confirmed that the distribution of SC along the chromosome is highly heterogeneous and strongly affected by gene orientations, leading to a fine-tuned and ancestral regulation of promoters depending on their genomic context. In summary, the *local* level of SC experienced by a given promoter can strongly differ from the *global* (average) level of the chromosome, and depends on the orientation and activity of adjacent genes, providing a strong mechanistic basis for the co-regulation of co-localised operons. Direct evidence for these effects was obtained from experiments involving supercoiling-sensitive promoters inserted on the chromosome in different artificial configurations. Comparable evidence is more difficult to establish for native promoters, but was highlighted in at least two examples of divergently organised operons: the *ilv* promoters of *E. coli* and the *leu-500* promoter of *S. typhimurium*. Interestingly, in these two examples, the local nature of SC (generated by the divergent

genes) is combined with a complex regulatory mechanism involving DNA binding proteins. In the first example, the activation of *ilvP_G* is prevented by the denaturation of an upstream AT-rich tract, except when the NAP IHF closes that region and favors the opening of the promoter (like CRP at the *peIE* promoter, Figure 6B). The pattern is similar for *leu-500*, although the repression is here achieved by H-NS binding at an AT-rich tract, and relieved by the TF LeuO in presence of locally generated negative SC. Since divergent genes involved in the same function and simultaneously expressed are commonly found in bacteria, including among those involved in pathogenicity, these examples may only be the first of a large unexplored class.

The high density of bacterial genomes implies that the interaction between neighboring genes could in fact give rise to a collective behavior along larger distances, forming “topological domains”. Indeed, when promoters were displaced over the chromosome, their expression and supercoiling sensitivity were found to change depending on their location and neighbouring activity. These domains, shaped by transcription and architectural proteins, remain a poorly defined notion in bacteria. Proposed lengthscales vary from 10 to 20 kb, to 50 kb and up to hundreds of kilobases; while the former may underpin an extension of the notion of operons, the latter probably reflect a higher order folding of bacterial chromatin involving different actors, and this hierarchical organisation remains to be characterised.

REVIEW QUESTIONS

1. Discuss the energetics of supercoiled DNA.
2. Define the DNA supercoiling and stretching.
3. Explain the maintenance of DNA supercoils.
4. Discuss the basal regulation of the RNA polymerase-DNA interaction.
5. Discuss the spatial heterogeneities of DNA supercoiling.

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CHAPTER 4

CRUCIFORM STRUCTURE IN DNA

Cruciform structures at origins of DNA replication in mammalian cells have been shown to recruit cruciform-binding proteins that function during the initiation of DNA synthesis.

Cruciform DNA structures are stabilized through supercoiling and their formation alleviates stress generated from DNA supercoiling. Cruciform structures block the recognition of the tet promoter in pX by RNA polymerase. The cruciform structures can also disrupt a step in the kinetic pathway, shown when gyrase is inhibited by novobiocin. Cruciform structures regulate transcription initiation such as the suppression of pX transcription. DNA replication can then be inhibited by cruciform containing tertiary structures of DNA formed during recombination, which can be studied to help treat malignancy.

After studying this chapter, you will understand the core concept of:

- Overview of cruciform structures in DNA
- Formation and presence of cruciform structures in the genome

INTRODUCTION

Cruciform DNA is a form of non-B DNA that requires at least a six nucleotide sequence of inverted repeats to form a structure consisting of a stem, branch point and loop in the shape of a cruciform, stabilized by negative DNA supercoiling. Two classes of cruciform DNA have been described; folded and unfolded. Folded cruciform structures are characterized by the formation of acute angles between adjacent arms and main strand DNA. Unfolded cruciform structures have square planar geometry and 4-fold symmetry in which the two arms of the cruciform are perpendicular to each other. The formation of cruciform structures in linear DNA is thermodynamically unfavourable due to the possibility of base unstacking at junction points and open regions at loops. Cruciform DNA is found in both prokaryotes and eukaryotes and has a role in the transcription of DNA, double strand repair, DNA translocation and recombination. Cruciform structures can increase genomic instability and are involved in the formation of various diseases, such as cancer.



CORE CONCEPT: OVERVIEW OF CRUCIFORM STRUCTURES IN DNA

Cruciform DNA structure appears as several hairpin loops, creating a crucifix-like structure composed of DNA. DNA structure is

formed by incomplete exchange of the strands between the double-stranded helices. Cruciform DNA Eukaryotic cells contain DNA-binding protein that can specifically recognize cruciform DNA. Interactions with ubiquitous protein plays a crucial role for the conformation of cruciform DNA.

The genetic information of all organisms contains unique sequence information necessary to encode proteins and RNA molecules. Within the noncoding regions, and on occasion within coding regions, there are defined ordered DNA sequences (dosDNA) that contain various symmetry elements. The symmetry elements inverted repeats, mirror repeats, and direct repeats are summarized in Figure 1. A number of non-B-DNA conformations can form within dos DNA sequences including cruciform, intermolecular triplex DNA, and slipped mis-paired structures.

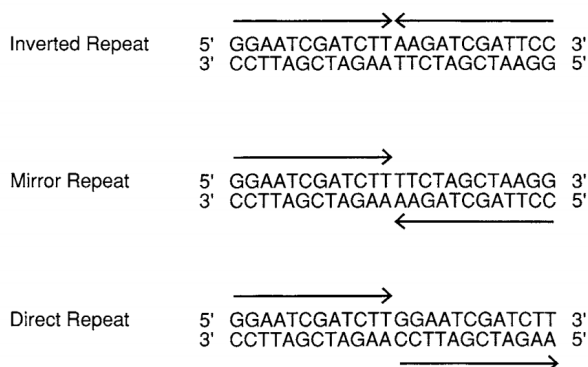


Figure 1. Symmetry elements. This figure shows the organization of three different types of symmetry elements in DNA: inverted repeats, mirror repeats, and direct repeats.

Before the advent of DNA sequencing, hybridization analysis of DNA revealed snap-back sequences that form a double-stranded region after denaturation and rapid renaturation. On rapid renaturation, double-stranded DNA does not form. However, if an inverted repeat is present, a snap-back region or hairpin can form as shown in Figure 2. The inverted repeat orientation has also been called a palindrome. Linguistically a **palindrome** is a sentence that reads the same in either direction. For example, “mad am, I’m Adam” is a palindrome. Since the DNA has polarity and is “read” by RNA polymerase in the

Palindrome is a word, number, phrase, or other sequence of characters which reads the same backward as forward, such as “taco cat” or madam or racecar or the number 10801.

5' to 3' direction, the term palindrome can be used for perfect inverted repeats. Inverted repeat regions that are not completely symmetrical or that have a centre region that is not an inverted repeat should not be called palindromes (imperfect palindromes or quasi-palindromes are acceptable terms).

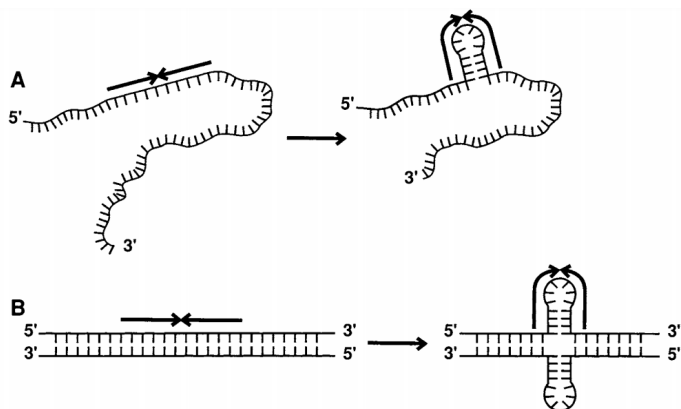


Figure 2. Snap backs and cruciforms. (A) An inverted repeat in a single-stranded region of DNA is shown forming a hairpin or snap-back region.

DNA sequencing has uncovered many palindromic regions of DNA. Often inverted repeats occur near putative control regions of genes or at origins of DNA replication. This localization has made it tempting to speculate that the ability of DNA to exist in an alternative conformation could provide a molecular switch for controlling transcription or replication.

The idea for cruciform structures (Figure 2B) was first proposed by Platt in 1955 and Gierer in 1966. In the early 1970s, Lebowitz proposed that cruciform structures would explain their chemical modification data. To form a cruciform the inter-strand hydrogen bonds in the inverted repeat must melt and intrastrand hydrogen bonds must form between complementary bases in each single strand (Figure 3). On cruciform formation, there will necessarily be a loop of 3-4 unpaired bases at the tip of the hairpin, resulting in a loss of hydrogen bonding and base stacking interactions that provide stability to the linear double helix. From thermodynamic and energetic considerations, cruciform initially seemed an unlikely structure and the suggestion of their existence was met with some degree of skepticism. It was nearly 10 years before the insightful

suggestion of Lebowitz was finally accepted. In part, this acceptance paralleled a growing appreciation of the energetics and dynamics of negatively supercoiled DNA.

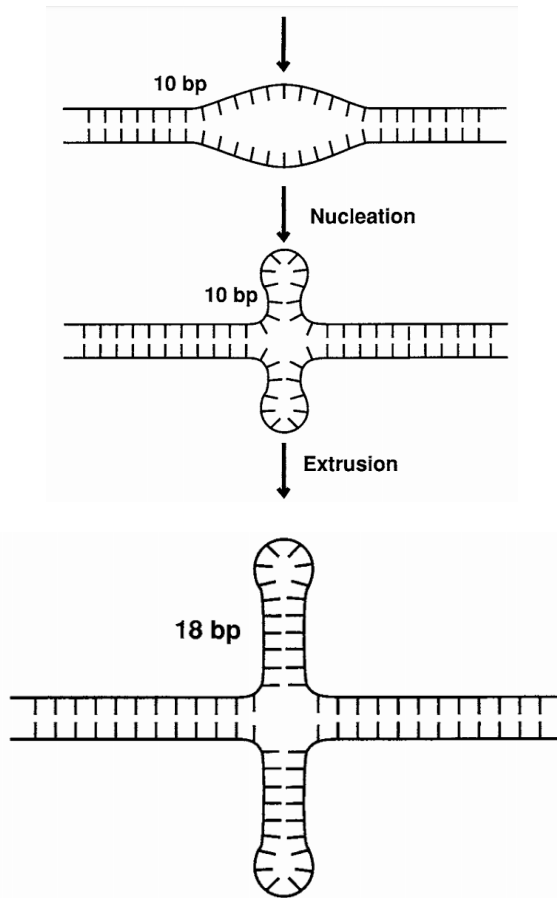


Figure 3. Mechanism of cruciform formation

Formation and Stability of Cruciform

Cruciform structures in DNA are of considerable interest, both as extreme examples of sequence-dependent structural heterogeneity and as models for four-way junctions such as the Holliday junction³ of homologous genetic recombination. Cruciforms are of lower thermodynamic stability than regular duplex DNA, and have been observed only in negatively

supercoiled molecules, where the unfavourable free energy of formation is offset by the topological relaxation of the torsion-ally stressed molecule.

Super-helical Energy Requirement for Cruciform Formation and Cruciform Stability

Energy is required to melt the centre of the inverted repeat to allow intrastrand nucleation and to stabilize extrusion into the cruciform form. This energy comes from negative DNA supercoiling. Cruciforms form in supercoiled DNA but not in relaxed or linear DNA. The energy required to form a cruciform has been shown to be critical. Analysis of cruciform-containing DNA using two-dimensional agarose gel electrophoresis has shown that a very specific minimal level of energy is required for cruciform formation. Within a family of topoisomerase, the formation of cruciforms is an all-or-none phenomenon. The addition of one more negative supercoil results in the formation of cruciforms in all molecules of that (and more negative) super-helical density. The critical super-helical density for cruciform formation.

A probable mechanism for cruciform formation about 10bp, must unwind at the centre of symmetry to allow cruciform formation. The super-helical density dependence for cruciform formation in plasmid populations at different average levels of DNA supercoiling. Since DNA supercoiling promotes breathing in the double helix, it is expected that more cruciforms would form in DNA samples with higher levels of negative DNA supercoiling.

Cruciforms are stable and thermodynamically favoured in negatively supercoiled DNA because, when cruciforms are formed, negative supercoils are relaxed. To form a cruciform, helical turns between the two strands of the DNA (inter-strand turns) are unwound, reducing the value of T. Intrastrand helical turns form in the cruciform arms but, at a first approximation, these do not enter the supercoiling equation

$$L = T + w$$

The formation of cruciforms results in the relaxation of one negative supercoil for every 10.5bp of the inverted repeat that participates in cruciform formation. The stability of cruciforms results from a loss of the free energy ($\sim G$) inherent in supercoiled DNA. Thus the stability of cruciforms is proportional to their length.

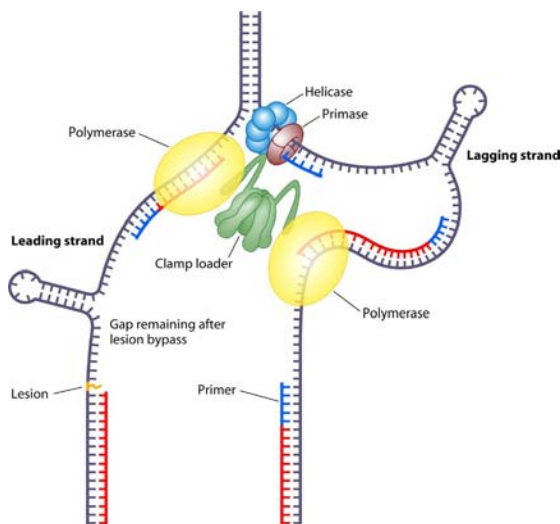
Once formed, cruciforms are quite stable even in relaxed DNA because removal of a cruciform in relaxed DNA requires the introduction of one negative supercoil for each 10.5bp of the cruciform arm converted back into a linear duplex. Thus, removal of a cruciform from relaxed DNA would increase the free energy of the molecule.

The removal of a cruciform from relaxed DNA would increase the free energy of the molecule.

Biology of Cruciform

Biochemical, crystallographic and computer simulation analyses of the A-DNA structure and protein-DNA complexes indicate that an A-like DNA conformation may either form upon binding of certain proteins to DNA, or be an important intermediate step in forming the strongly distorted DNA conformation observed within at least some complexes with proteins.

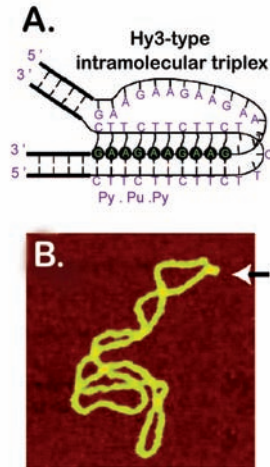
While DNA mostly has a seemingly random distribution of nucleobases in the sequence, defined order sequences may rather frequently occur. These include inverted repeats that can form cruciforms, mirror repeats that may adopt intramolecular triplex DNA conformations, and direct repeats, that can form slipped mispaired structures, and $(GC)_n$ and $(GT)_n$ tracts that can form Z-DNA.



Cruciform Structure

An inverted repeat or a palindrome is a DNA sequence that reads the same from the 5' to 3' in either strand. For example, many type II restriction enzyme sites are palindromic. To form a cruciform the *interstrand* hydrogen bonds in the inverted repeat must be broken and *intrastrand* hydrogen bonds then established between complementary bases in each single strand, thus forming two hairpin-like arms with 3-4 unpaired bases at their tips image in (A). As a whole, the cruciform consists of two rather long duplex DNA arms, and two comparatively short hairpin arms which form a four-way junction. The structure of the four-way junction is such that the nucleobases in and around the junction are fully involved in base pairing. Cruciforms can form in topologically closed molecules where they use energy from DNA supercoiling to melt the center of the inverted repeat, allowing the intrastrand hairpin nucleation. The thermodynamic stability of the cruciform comes from relaxation of one negative supercoil per 10.5 bp of DNA sequence that converts into the cruciform.

The propensity for cruciform formation increases in longer inverted repeats that relax more supercoils than shorter ones. It also depends on temperature and the base composition of the inverted repeat, most importantly, in its center, in accordance with a requirement of partial DNA melting before the hairpin base pairing. Although schematically the cruciform is usually shown as having a cross shape as in the schematic representation, such an extended structure is favored only under the low-salt conditions, where electrostatic repulsion between phosphates pushes all four cruciform arms apart image in (B). Under physiologically relevant salt conditions, where the phosphates are partially shielded and repulsion is reduced, the cruciform adopts an X-type structure with unequal inter-arm angles as seen in the AFM image in (C). The extended cruciform is rather stiff, as judged from little fluctuation of the inter-arm angles, whereas the X-type cruciform has a pronounced mobility of the hairpin arms observed by atomic force microscopy in liquid. The distribution of inverted repeats in eukaryotic DNA is nonrandom and they are clustered at or near genetic regulatory regions, which suggests that they are important biologically.

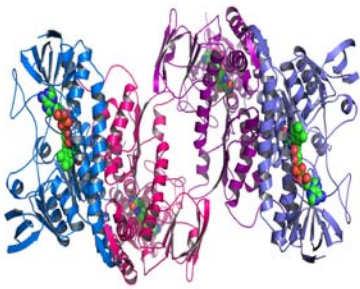


Z-DNA Structure

Left-handed Z-DNA has been mostly found in alternating purine-pyrimidine sequences $(CG)_n$ and $(TG)_n$. Z-DNA is thinner (18 Å) than B-DNA (20 Å), the bases are shifted to the periphery of the helix, and there is only one deep, narrow groove equivalent to the minor groove in B-DNA. In contrast to B-DNA where a repeating unit is 1 base pair, in Z-DNA the repeating unit is 2 bp. For Z-DNA in $(CG)_n$ sequences the twist angle for a CpG step is 9° , whereas it is 51° for the GpC step, totaling 60° in the 2 bp repeating unit. The helix repeat in Z-DNA is 12 bp/turn and an average rise is 3.7 Å/bp, compared with 10.5 bp/turn and 3.4 Å/bp in B-DNA. The backbone follows a zigzag path as opposed to a smooth path in B-DNA. The sugar and glycosidic bond conformations alternate: C2' endo in *anti* dC or dT and C3' endo in *syn* dG or dA. Electrostatic interactions play a crucial role in Z-DNA formation. Because of the zigzag backbone path, some phosphate groups are closer and **electrostatic** repulsion between them is greater than in B-DNA.

Electrostatics is a branch of physics that studies electric charges at rest.

Therefore, Z-DNA is stabilized by high salt concentrations or polyvalent cations that shield interphosphate repulsion better than monovalent cations. Other factors also contribute to Z-DNA stability. If an alternating purine-pyrimidine sequence occurs in a circular DNA molecule, DNA supercoiling is a major driving force for Z-DNA formation. Z-DNA formation unwinds DNA about two supercoils per 12 bp of DNA. The junctions between the B- and Z-DNA in supercoiled DNA span several base pairs in which nucleobases behave as if they were unpaired. In particular, they are partially reactive to single-strand specific chemicals. A computer analysis of over one million base pairs of human DNA, containing 137 complete genes identified 329 potential Z-DNA-forming sequences. Like inverted repeats, potential Z-DNA-forming sequences have a distinctly nonrandom distribution with a strong bias toward locations near the site of transcription initiation.



CORE CONCEPT: FORMATION AND PRESENCE OF CRUCIFORM STRUCTURES IN THE GENOME

Genome sequencing projects have inundated us with information regarding the genetic basis of life. DNA cruciforms play an important role in the regulation of natural processes involving DNA. These structures are formed by inverted repeats, and their stability is enhanced by DNA supercoiling. Cruciform structures are fundamentally important for a wide range of biological processes, including replication, regulation of gene expression, nucleosome structure and recombination.

Cruciform structures are important regulators of biological processes. Both stem-loops and cruciform are capable of forming from inverted repeats. Cruciform structures consist of a branch point, a stem and a loop, where the size of the loop is dependent on the length of the gap between

inverted repeats (Figure 4). Direct inverted repeats lead to formation of a cruciform with a minimal single-stranded loop. The formation of cruciform from indirect inverted repeats containing gaps is dependent not only on the length of the gap, but also on the sequence in the gap. In general, the AT-rich gap sequences increase the probability of cruciform formation.

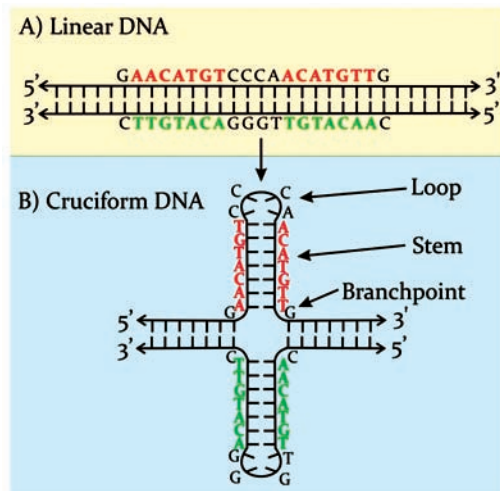


Figure 4. Changes associated with transition from the linear to cruciform state in the p53 target sequence from the p21 promoter. The promoter sequence contains a 20bp p53 target sequence with 7bp long inverted repeat (red), (A) as linear DNA and (B) as an inverted repeat as a cruciform structure. In the cruciform structure, the p53 target sequence is presented as stems and loops.

It is also possible that the gap sequence can form an alternative DNA structure. The formation of DNA cruciforms has a strong influence on DNA geometry whereupon sequences that are normally distal from one another can be brought into close proximity. The structure of cruciforms has been studied by atomic force microscopy. These studies have identified two distinct classes of cruciforms. One class of cruciforms, denoted as unfolded, have a square planar conformation characterized by a 4-fold symmetry in which adjacent arms are nearly perpendicular to one another. The second class comprises a folded (or stacked) conformation where the adjacent arms form an acute angle with the main DNA strands (Figure 5). Two of the three structural motifs inherent to cruciforms, the branch point and stem, are also found in Holliday junctions. Holliday junctions

Genomics is an interdisciplinary field of biology focusing on the structure, function, evolution, mapping, and editing of genomes.

are formed during recombination, double-strand break repair, and fork reversal during replication. Resolving Holliday junctions is a critical process for maintaining **genomic** stability. These junctions are resolved by a class of structure-specific nucleases: the junction-resolving enzymes.

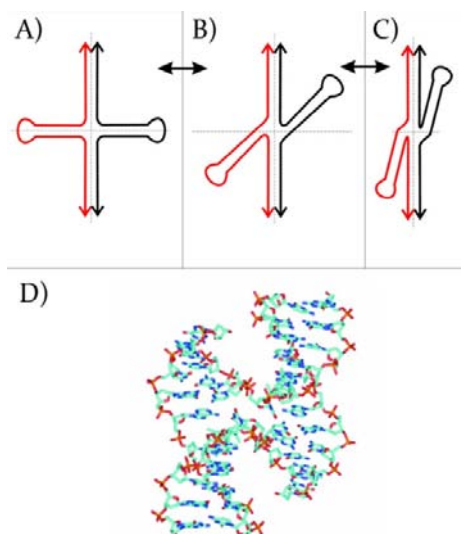


Figure 5. Conformations of a cruciform structure. Conformations of a cruciform can vary from (A) “unfolded” with 4-fold symmetry to (B) bent, and to (C) “stacked” with 4 chains of DNA in close vicinity. D) Topology of a Holliday junction stabilized by a psoralen cross-linking agent (PDBID 467D). Here, the junction takes the form of an anti-parallel stacked x-structure.

Cruciforms are not thermodynamically stable in naked linear DNA due to branch migration. Cruciform structure formation *in vivo* has been shown in both prokaryotes and eukaryotes using several methodological approaches. The presence of the cruciform structure was first described in circular plasmid DNA where the negative superhelix density can stabilize cruciform formation. Plasmids with native superhelical density usually contain cruciform structures *in vitro* and *in vivo*. For example, higher order structure in the pT181 plasmid was shown to

exist *in vivo* using bromoacetaldehyde treatment. Deletion of the sequence which forms this structure at the ori site leads either to a reduction or failure in replication. Similarly, deletion of the cruciform binding domain in 14-3-3 proteins results in reduced origin binding which affects the initiation of DNA replication in budding yeast. Monoclonal antibodies against cruciform structures have also been used successfully to isolate cruciform-containing segments of genomic DNA.

Furthermore, these sequences were able to replicate autonomously when transfected into HeLa cells. Stabilization of the cruciform structures by monoclonal antibodies 2D3 and 4B4, with anti-cruciform DNA specificity, resulted in a 2- to 6-fold enhancement of replication *in vivo*. 14-3-3 sigma was found to associate *in vivo* with the monkey origins of DNA replication *ors8* and *ors12* in a cell cycle-dependent manner, as assayed by a chromatin immunoprecipitation (ChIP) assay that involved formaldehyde cross-linking, followed by immunoprecipitation with anti-14-3-3 sigma antibody and quantitative PCR. Similarly, the 14-3-3 protein homologs from *Saccharomyces cerevisiae*, Bmh1p and Bmh2p, have cruciform DNA-binding activity and associate *in vivo* with ARS307. Several studies show that transcription is regulated directly by the presence of cruciform structure *in vivo*. Another example includes the ability of the d(AT)_n-d(AT)_n insert to spontaneously adopt a cruciform state in *E. coli*, resulting in a block of protein synthesis. Using site-directed mutational analysis and P1 nuclease mapping, it was demonstrated that the formation of a cruciform structure is required for the repression of enhancer function in transient transfection assays and that Alu elements may contribute to regulation of the CD8 alpha gene enhancer through the formation of secondary structure that disrupts enhancer function. Transcriptionally driven negative supercoiling also mediates cruciform formation *in vivo* and enhanced cruciform formation correlates with an elevation in promoter activity. It was also shown that the secondary DNA structures of the ATF/CREB element play a vital role in protein-DNA interactions and its cognate transcription factors play a predominant role in the promoter activity of the RNMTL1 gene. Hypo-methylation of inverted repeats by the Dam methylase show that these sequences are consistent with an unusual secondary structure, such as DNA cruciform or hairpin *in vivo*. The *in vivo* effects of cruciform formation during transcription. Interestingly hairpin-capped linear DNA

Chromatin is a complex of DNA and protein found in eukaryotic cells.

(in which the replication of hairpin-capped DNA and cruciform formation and resolution play central roles) was stably maintained for months in a human cancer cell line as numerous extra-chromosomal episomes. Long palindromes can also induce DNA breaks after assuming a cruciform structure. Palindromes in *S. cerevisiae* are resolved, *in vivo*, by structure-specific enzymes. *In vivo* resolution requires either the Mus81 endonuclease or, as a substitute, the bacterial HJ resolvase RusA. These findings provide confirmation of cruciform extrusion and resolution in the context of eukaryotic **chromatin**. Taken together, these studies show that cruciform have been detected *in vivo* using a variety of independent techniques and that they are an intriguing and integral phenomenon of DNA biology and biochemistry.

Proteins involved in Interactions with Cruciform Structures

Numerous proteins have been shown to interact with cruciforms, recognizing features such as DNA crossovers, four-way junctions, and curved or bent DNA. Structural transitions in chromatin occur concomitantly with DNA replication or transcription and in processes that involve a local separation of DNA strands.

Junction-resolving enzymes

There are a large number of proteins that recognize cruciforms and, of these, the junction-resolving enzymes have been studied extensively. These proteins have been identified in many organisms from bacteria (and their phages) to yeast, archaea and mammals. The majority of the junction-resolving enzymes can be divided into one of two super-families. Those in the first class target specific DNA

sequences for enzymatic activity, although they will bind equally well to junctions of any sequence. This superfamily includes *E. coli* RuvC, the yeast integrases, Cce1, Ydc2, and RnaseH. The second group includes the phage T7, endonuclease RecU, the Hjc and Hje resolving enzymes, the MutH protein family and related restriction enzymes. The x-ray structures of the junction-resolving enzymes in complex with 4-way junctions highlight the flexibility inherent to DNA (Figure 6) in that these enzymes recognize and distort the junction. This enables them to carry out such key roles as the cleavage of allogene DNAs and maintenance of genomic stability to name but a few. The recognition of non-B-DNA structure by junction-resolving enzymes has been the subject of several reviews.

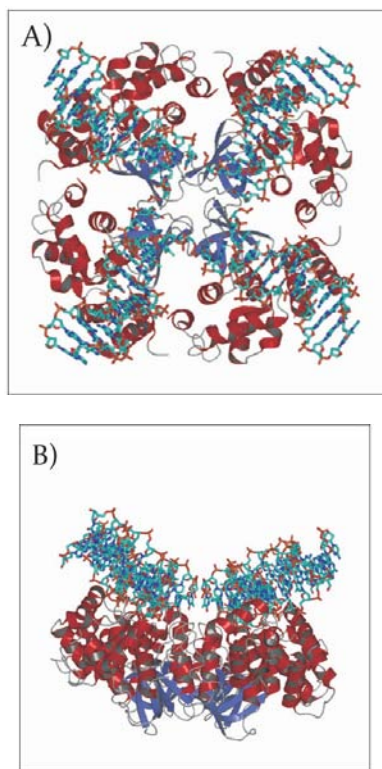


Figure 6. Crystal structure of the *E. coli* RuvA tetramer in complex with a Holliday junction (PDBID 1C7Y). A) The Holliday junction is depressed at the centre where it makes close contacts with RuvA. Each of the arms outside of the junction centre takes on a standard beta DNA conformation B) Rotation of A) by 90°.

Proteins Involved in Transcription and DNA repair

The maintenance of a cell's genomic stability is achieved through several independent mechanisms. Arguably, the most important of these mechanisms is DNA repair. Protein binding to damaged DNA and to the local alternative DNA structures is therefore a key function of these processes. The promoter regions of genes are often characterized by presence of inverted repeats that are capable of forming cruciforms *in vivo*. A number of DNA-binding proteins, such as those of the HMGB-box family [47], Rad54 [48], BRCA1 protein [49, 50], as well as PARP-1 (poly(ADP-ribose) polymerase-1) [51], display only a weak sequence preference but bind preferentially to cruciform structures. Moreover, some proteins can induce the formation of cruciform structures upon DNA binding [51, 52]. Among the DNA repair proteins which bind to cruciforms are the junction-resolving enzymes Ruv and RuvB [53, 54], DNA helicases [55], XPG protein [56], and multifunctional proteins like HMG-box proteins [57] BRCA1, 14-3-3 protein family including homolog's Bmh1 and Bmh2 from *S. cerevisiae*, and GF14 from plants. Footprinting analysis of the gonadotropin-releasing hormone gene promoter region indicated the human estrogen receptor (ER) to be another potential cruciform binding protein. In this case, extrusion of the cruciform structure allowed the estrogen response elements motifs to be accessed by the ER protein. Formation and presence of cruciform structures in the genome

Cruciform structures are important regulators of biological processes [3,5]. Both stem-loops and cruciforms are capable of forming from inverted repeats. Cruciform structures consist of a branch point, a stem and a loop, where the size of the loop is dependent on the length of the gap between inverted repeats (Figure 1). Direct inverted repeats lead to formation of a cruciform with a minimal single-stranded loop. The formation of cruciforms from indirect inverted repeats containing gaps is dependent not only on the length of the gap, but also on the sequence in the gap. In general, the AT-rich gap sequences increase the probability of cruciform. The maintenance of a cell's genomic stability is achieved through several independent mechanisms. Arguably, the most important of these mechanisms is DNA repair. Protein binding to damaged DNA and to the local alternative DNA structures is therefore a key function of these

processes. The promoter regions of genes are often characterized by presence of inverted repeats that are capable of forming cruciforms *in vivo*. A number of DNA-binding proteins, such as those of the HMGB-box family, Rad54, BRCA1 protein, as well as PARP-1 (poly (ADP-ribose) polymerase-1), display only a weak sequence preference but bind preferentially to cruciform structures. Moreover, some proteins can induce the formation of cruciform structures upon DNA binding. Among the DNA repair proteins which bind to cruciforms are the junction-resolving enzymes Ruv and RuvB, DNA helicases, XPG protein, and multifunctional proteins like HMG-box proteins BRCA1, 14-3-3 protein family including homolog's Bmh1 and Bmh2 from *S. cerevisiae*, and GF14 from plants. Foot-printing analysis of the gonadotropin-releasing hormone gene promoter region indicated the human **estrogen receptor (ER)** to be another potential cruciform binding protein. Extrusion of the cruciform structure allowed the estrogen response elements motifs to be accessed by the ER protein

Estrogen receptors are a group of proteins found inside cells. They are receptors that are activated by the hormone estrogen.

PARP-1

PARP-1 is an abundant, nuclear, zinc-finger protein present in ~1 enzyme per 50 nucleosomes. It has a high affinity for damaged DNA and becomes catalytically active upon binding to DNA breaks. In the absence of DNA damage, the presence of PARP-1 leads to the perturbation of histone-DNA contacts allowing DNA to be accessible to regulatory factors. PARP-1 activity is also linked to the coordination of chromatin structure and gene expression in *Drosophila*. It was reported that PARP can bind to the DNA hairpins in heteroduplex DNA and that the auto-modification of PARP in the presence of NAD⁺ inhibited its hairpin binding activity. Atomic force microscopy studies revealed that, *in vitro*, PARP protein has a preference

for the promoter region of the PARP gene in superhelical DNA where the dyad symmetry elements form hairpins (Figure 7). PARP-1 recognizes distortions in the DNA backbone allowing it to bind to three- and four-way junctions. Kinetic analysis has revealed that the structural features of non-B form DNA are important for PARP-1 catalysis activated by undamaged DNA.

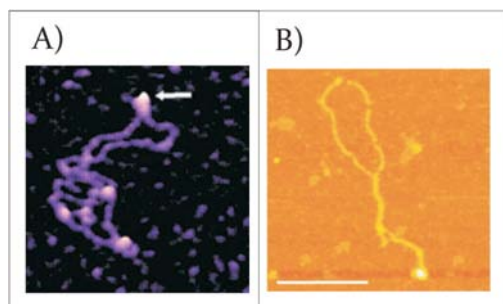


Figure 7. AFM and SFM images of proteins binding to a cruciform structure.

The order of PARP-1's substrate preference has been shown to be: cruciforms > loops > linear DNA. These results suggest a link between PARP-1 binding to cruciforms structures in the genome and its function in the modulation of chromatin structure in cellular processes. Moreover, it was shown that the binding of PARP-1 to DNA can induce changes in DNA topology as was demonstrated using plasmid DNA targets

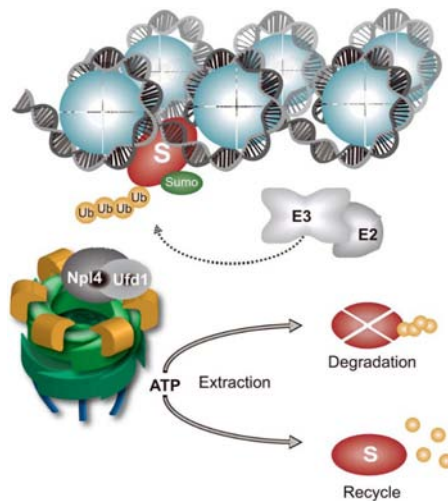
P53

P53 is arguably one of the most intensively studied tumor suppressor genes. More than 50% of all human tumors contain p53 mutations and the inactivation of this gene plays a critical role in the induction of malignant transformation. Sequence-specific DNA binding is crucial for p53 function. P53 target sequences, which consist of two copies of the sequence 5'-RRRC (A/T)(T/A)GYYY-3', often form inverted repeats. It was reported that p53 binding is temperature sensitive and dependent on DNA fragment length. Moreover, it was demonstrated, *in vivo*, that p53 binding to its target sequence is highly dependent on the presence of an inverted repeat at the target site. Preferential binding of p53 to superhelical DNA has also been described. Non-canonical DNA structures such as mismatched duplexes, cruciform structures, bent DNA, structurally flexible chromatin DNA, hemicatenated DNA, DNA bulges, three- and

four-way junctions, or telomeric t-loops can all be bound selectively by p53. There is a strong correlation between the cruciform-forming targets and an enhancement of p53 DNA binding. Target sequences capable of forming cruciform structures in topologically constrained DNA bound p53 with a remarkably higher affinity than did the internally asymmetrical target site. These results implicate DNA topology as having an important role in the complex, with possible implications in modulation of the p53 regulon.

Chromatin-associated Proteins

The chromatin-associated proteins cover a broad spectrum of the proteins localized in the cell nucleus. They are partly involved in modulating chromatin structure, but are also implicated in a range of processes associated with DNA function. They fine-tune transcriptional events (DEK, BRCA1) and are involved in both DNA repair and replication (HMG proteins, Rad51, Rad51ap, topoisomerases). Another family of enzymes deemed important in these processes is that of topoisomerases. These enzymes occur in all known organisms and play crucial roles in the remodeling of DNA topology. Topoisomerase I binds to Holliday junctions, and topoisomerase II recognizes and cleaves cruciform structures and interacts with the HMGB1 protein.



These processes are particularly important for maintaining genomic stability due to their ability to diffuse the stresses that are levied upon

The recognition of branched structures during homologous recombination is a critical step in this process.

a DNA molecule during transcription, replication and the resolving of long cruciform that would otherwise hinder DNA chain separation. The Rad54 protein plays an important role during homologous recombination in eukaryotes. Yeast and human Rad54 bind specifically to Holliday junctions and promote branch migration. The binding preference for the open conformation of the X-junction appears to be common for many proteins that bind to Holliday junctions. Human Rad54 binds preferentially to the open conformation of branched DNA as opposed to the stacked conformation. Similarly, RAD51AP1, the RAD51 accessory protein, specifically stimulates joint molecule formation through the combination of structure-specific DNA binding and by interacting with RAD51. RAD51AP1 has a particular affinity for branched-DNA structures that are obligatory intermediates during joint molecule formation. The recognition of branched structures during homologous recombination is a critical step in this process.

Mixed lineage leukemia (MLL) protein

Chromosome translocation is a phenomenon that results in unusual rearrangement of chromosomes.

The MLL gene encodes a putative transcription factor with regions of homology to several other proteins including the zinc fingers and the so-called “AT-hook” DNA-binding motif of high mobility group proteins. The 11q23 **chromosomal translocation**, found in both acute lymphoid and myeloid leukemias, results in disruption of the MLL gene. Leukemogenesis is often correlated with alternations in chromatin structure brought about by either a gain or loss in function of the regulatory factors due to their being disrupted by chromosomal translocations. The MLL gene, a target of such translocation events, forms a chimeric fusion product with a variety of partner genes.

The MLL AT-hook domain binds cruciform DNA, recognizing the structure rather than the sequence of the target DNA. This interaction can be antagonized both by Hoechst 33258 dye and distamycin. In a nitrocellulose protein-DNA binding assay, the MLL AT-hook domain was shown to bind to AT-rich SARs, but not to non-SAR DNA fragments. MLL appears to be involved in chromatin-mediated gene regulation. In translocations involving MLL, the loss of the activation domain combined with the retention of a repression domain alters the expression of downstream target genes, thus suggesting a potential mechanism of action for MLL in leukemia. AF10 translocations to the vicinity of genes other than MLL also result in myeloid leukemia. A biochemical analysis of the MLL partner gene AF10 showed that its AT-hook motif is able to bind to cruciform DNA, but not to double-stranded DNA, and that it forms a homo-tetramer *in vitro*.

Cruciform Binding Proteins and Disease

The recognition of DNA junctions and cruciform structures is critical for genomic stability and for the regulation of basic cellular processes. The resolution of Holliday junctions and long cruciforms is necessary for genomic stability where the dysregulation of these proteins can lead to DNA translocations, deletions, loss of genomic stability and carcinogenesis. The large numbers of proteins which bind to these DNA structures work together to keep the genome intact. We believe that the formation of cruciform structures serves as a marker for the proper timing and initiation of some very basic biological processes. The mutations and epigenetic modifications that alter the propensity for cruciform formation can have drastic consequences for cellular processes. Thus, it is unsurprising that the dysregulation of cruciform binding proteins is often associated with the pathology of disease.

As stated above, the cruciform binding proteins including p53, BRCA1, WRN and the proto-oncogenes DEK, MLL and HMG are also associated with cancer development and/or progression. Some of these proteins play such important roles that their mutation and/or inactivation result in severe genomic instability and sometimes lethality. For example, *Brca1* $-/-$ mouse embryonic stem cells show spontaneous chromosome

breakage, profound genomic instability and hypersensitivity to a variety of damaging agents (e.g. γ radiation) all of which suggests a defect in DNA repair. The connection between the BRCA1 mutation and breast cancer is well known. P53's transcriptional regulation is fine-tuned by its timely binding to promoter elements. The formation of a cruciform structure in p53 recognition elements may be an important determinant of p53 transcription activity.

REVIEW QUESTION

1. Discuss the formation and stability of cruciform.
2. What do you understand by biology of cruciform?
3. Define the Z-DNA Structure.
4. Focus on proteins involved in interactions with cruciform structures.
5. Explain the chromatin-associated proteins.

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CHAPTER 5

LEFT –HANDED Z-DNA

Left-handed Z-DNA is favored by nucleotide sequences with alternating purines and pyrimidines. The structural basis of this preference is described, as well as the factors which stabilize Z-DNA. The most important biological factors stabilizing Z-DNA are negative supercoiling and Z-DNA binding proteins.

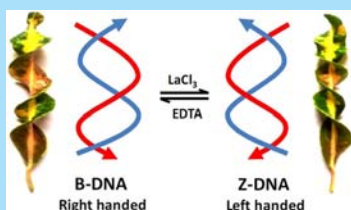
Z-DNA is one of the many possible double helical structures of DNA. It is a left-handed double helical structure in which the helix winds to the left in a zigzag pattern, instead of to the right, like the more common B-DNA form. Z-DNA is thought to be one of three biologically active double-helical structures along with A- and B-DNA.

After studying this chapter, you will understand the core concept of:

- Z – DNA or Left-Handed DNA
- Formation and Stability of Z-DNA

INTRODUCTION

Z-DNA is a high energy conformer of B-DNA that forms *in vivo* during transcription as a result of torsional strain generated by a moving polymerase. An understanding of the biological role of Z-DNA has advanced with the discovery that the RNA editing enzyme double-stranded RNA adenosine deaminase type I (ADAR1) has motifs specific for the Z-DNA conformation. Editing by ADAR1 requires a double-stranded RNA substrate. In the cases known, the substrate is formed by folding an intron back onto the exon that is targeted for modification. The use of introns to direct processing of exons requires that editing occurs before splicing. Recognition of Z-DNA by ADAR1 may allow editing of nascent transcripts to be initiated immediately after transcription, ensuring that editing and splicing are performed in the correct sequence. Structural characterization of the Z-DNA binding domain indicates that it belongs to the winged helix-turn-helix class of proteins and is similar to the globular domain of histone-H5.



CORE CONCEPTS: Z – DNA OR LEFT-HANDED DNA

Z-DNA is one of the many possible double helical structures of DNA. It is a left-handed double helical structure in which the helix winds to the left in a zigzag pattern, instead of to the right, like the more common B-DNA form. Z-DNA is thought to be one of three biologically active double-helical structures along with A- and B-DNA.

Z-DNA is a new family of DNA structure which is left-handed, i.e., the strands of DNA are twisted towards left-side instead of the usual right-handed twisting. B-form is twisted clockwise whereas Z-form is twisted anti-clockwise. Therefore two possible helical forms of DNA are mirror images of each other.

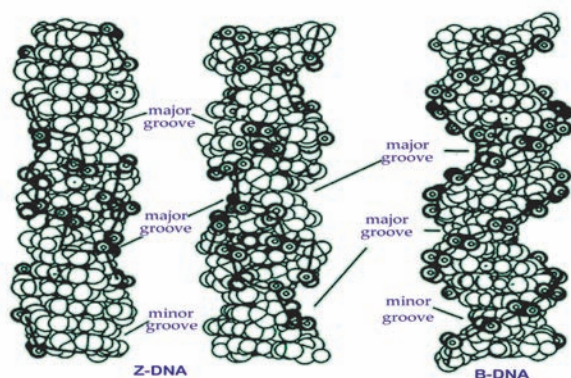


Figure 1. Side views of Z-DNA and B-DNA.

Alexander Rich and his colleagues of USA have discovered the Z-form of DNA. Soon the left-handed DNA was observed in other laboratories, but only under certain conditions. Factors that promote the Z-structure include methylation, bromination, specific DNA binding **protein** and sufficient torsional stress as in negatively supercoiled DNA. The Z-DNA was so-named because of the zigzag course of the backbones.

Z-DNA differs from the A and B forms not only in its left-handedness, but also in the orientation of the glycosyl bonds. The syn and anti-orientations alternate, in contrast with the all-anti-conformation in A- and B- DNA.

The pyrimidine nucleotides are in the standard anti-conformation with a C'_2 -endo sugar pucker, while the purine residues are syn and contain a C'_3 -endo conformation. The Z-form is more likely to occur in alternating purine-pyrimidine sequences because steric repulsion makes it less-favorable for a pyrimidine to adopt the syn conformation.

The crystal structures have a strong dinucleotide repeat unit due to the large alternation in helix twist

Protein is a nutrient needed by the human body for growth and maintenance.

between the C and G in a CpG step which is about 15° while that between the G and C in the subsequent GpC is close to 45° .

Both B-DNA and A-DNA have two types of grooves—one major and one minor. But Z-DNA has the single groove which is quite deep, extending to the axis of the helix. The diameter of Z-DNA is about $18 \text{ \AA}$ and the length of pith is $45 \text{ \AA}$.

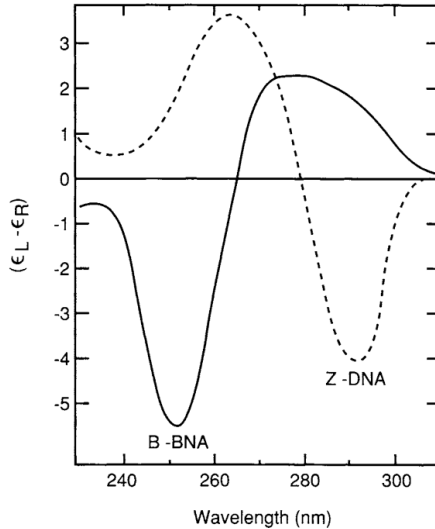


Figure 2. Circular dichroism spectra of B-DNA and Z-DNA.

The Left-Handed Z-DNA Helix

The structure of Z-DNA is shown in Figure 3 (B-DNA is shown for comparison). There are only a few similarities between B-DNA and Z-DNA, including the double-stranded, antiparallel nature of the two strands and Watson-Crick hydrogen bonding. Apart from these similarities, there are a number of differences between B-DNA and Z-DNA. First, there is a zigzag phosphate backbone in Z-DNA compared with a smooth backbone in B-DNA. Second, B-DNA has a distinct major and minor groove. In Z-DNA, the major groove has all but disappeared into a nearly flat surface. The one visible groove is deep and narrow. This groove is structurally analogous to the minor groove in B-DNA. Third, the Z-DNA helix, at $18 \text{ \AA}$ in diameter, is narrower than B-DNA ($20 \text{ \AA}$ in diameter). Fourth, the helix repeat in Z-DNA is 12 bp per turn

compared with 10.5 bp per turn for right-handed B-DNA. Finally, in B-form DNA the helix pitch is, on average, 34 Å with an average rise of 3.24 Å.

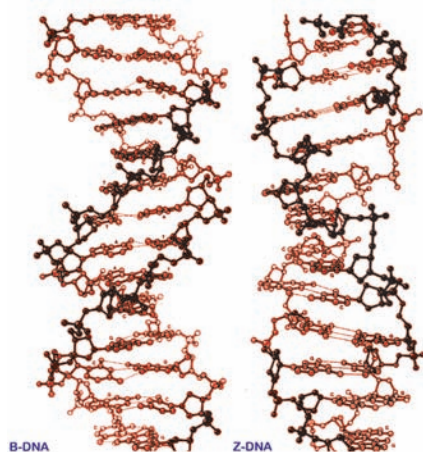


Figure 3. The B-DNA and Z-DNA helices.

The basic helix geometry of Z-DNA is:

- The length of one pitch is 45 Å.
- Number of base pairs/pitch is 12.
- The axial rise per base is 3.7 Å.
- Tilting of base pairs from the axis of helix is 7°.
- Diameter of the helix is 18 Å.

Z-DNA is a matter of intense current interest. It also occurs in nature. Its presence has also been established in the chromosome of the fruit-fly *Drosophila melanogaster* and also in the chromosomes of some other organisms.

There is growing evidence that certain DNA regions rich in guanine and **cytosine** within long molecules do adopt a Z-like conformation. This peculiar form of DNA is presumed to have a role in the regulation of gene activity.

Cytosine is one of the four main bases found in DNA and RNA, along with adenine, guanine, and thymine.

It could be a recognition signal for some important function of DNA. To understand the function of Z-DNA in *Drosophila melanogaster* chromosomes, research on synthetic DNA was conducted.

It has been demonstrated that purine-pyrimidine sequences can undergo transitions between right-handed B-conformation depending on the salt concentration or on chemical modification of the bases. The reversibility suggests that Z-DNA has some regulatory role.

For example, some nucleotide sequences can be established in the Z-form by being methylated. Methylation and de-methylation are very important in controlling the activity of genes.

There could be many possibilities by which the Z-DNA controls the gene regulations such as when certain control sites are established in the Z-form by methylation—regulatory protein binds to the site and keeps gene turned off; de-methylation might switch the site to the B-conformation, causing the regulatory molecule to let go.

Identification of Z-DNA

Rich et al could raise antibodies by injecting rabbits with short, brominated double helices of alternating G and C nucleotides of Z-DNA. The rabbit made antibodies that were highly specific for Z-DNA and did not react at all with the B-form.

Purified rabbit antibodies for Z-DNA were used to probe biological materials for the presence of Z-DNA. The polytene chromosome of *D. melanogaster* were treated with rabbit's anti z-antibody and incubated to enable the antibody to bind to any Z-DNA in the chromosomes.

The unbound antibodies were washed away. Further, the bound antibodies were made visible by adding a goat antibody specific for rabbit one. The goat antibody was conjugated to a fluorescent dye, so that it glowed under ultraviolet illumination. The anti z-antibody could be seen to bind to the chromosomes in a distinct and reproducible segmented pattern (Figure 4).

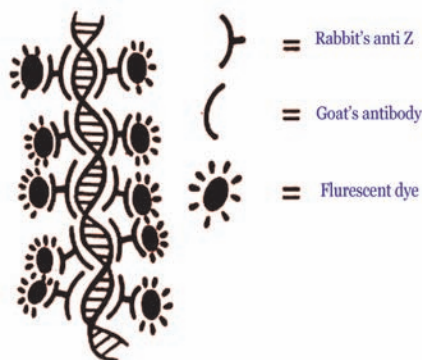


Figure 4. Identification of Z-DNA.

By comparing ultraviolet micrographs that reveal the bands and inter-bands of polytene chromosomes, it was established that the fluorescent segments are the inter-bands and not the bands. Therefore, there must be Z-DNA in the inter-bands.

It is conceivable that some Z-DNA is also present in the bands but is not visibly stained by the antibodies. Presently it appears that Z-DNA is either more abundant or more accessible in the inter bands than the bands.

Sugar Pucker and Glycosidic Bond Rotation in Z-DNA

Two major structural differences between B-DNA and Z-DNA are the sugar pucker and the configuration of the glycosidic bond in dG. In B-form DNA all sugar residues exist in the C2' endo configuration. In Z-DNA the sugar pucker for dC remains C2' endo but the pucker changes to C3' endo in dG residues. The structures of C2' endo, anti dG and C3' endo, syn dG are shown in Figure 5. The zig-zag phosphate backbone is the result of the alternating C2' endo conformation in pyrimidines and C3' endo conformation in purines. One major consequence of the change in sugar pucker is the effect this has on the distance between the phosphate groups attached to the C5' and C3' ribose positions (Figure 5). Changing the sugar pucker from C2' endo to C3' endo reduces this distance dramatically. This shortening has the result of pulling the base away from the center of a B-DNA helix and closer to the phosphate backbone, which is on the outside of the DNA helix. The other significant change

occurring in dG in Z-DNA is a 180° rotation around the glycosidic bond. The configuration goes from the anti-configuration found in B-form (and A-form) DNA to the syn configuration in Z-DNA. Some of these changes can be seen in the representation of Z-DNA shown in Figure 6.

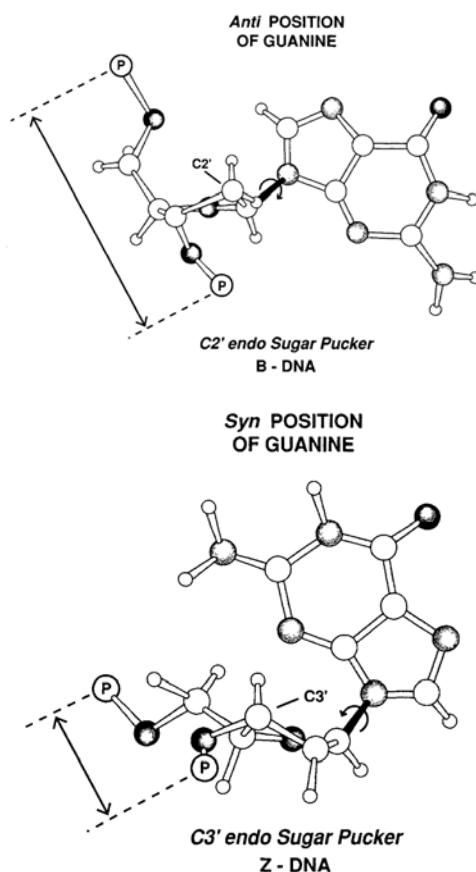


Figure 5. The structure of guanosine in B-DNA and I-DNA.

Bases Do a Backflip to Form Z-DNA

A major structural rearrangement required for the B-DNA-to-Z-DNA transition is a flipping or inversion of the bases relative to the helix axis. Within a region of Z-DNA the bases flip over 180° (Figure 6). This flipping is accompanied by the rotation of the glycosidic bond of the

purines from anti to syn and the sugar pucker change. For the pyrimidine nucleotides in Z-DNA, the sugar accompanies the base in its 180° rotation.

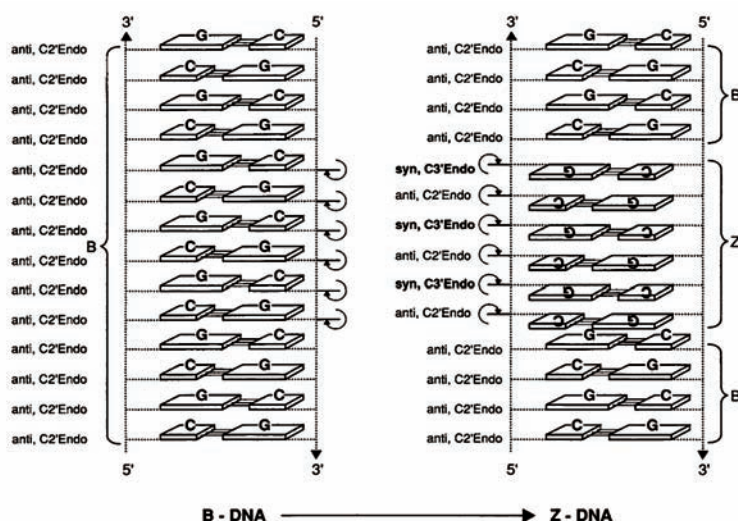


Figure 6. Topological changes in the positions of the base pairs on Z-DNA formation.

The Dinucleotide Repeat in Z-DNA and Base Stacking

In B-form DNA each base pair is oriented in the helix similarly with respect to its adjacent base pairs, so all dinucleotide base pairs are structurally similar. The dinucleotides CpG and GpC have twist angles of 29.8° and 40.0° , respectively which are not similar to the average helical repeat of 34.3° . Another prominent feature of B-DNA is that the base pairs are stacked regularly on top of one another. As shown in Figure 7, there is reasonable overlap between the bases in the GpC dinucleotide. There is somewhat less physical overlap between the bases in the CpG stack. In B-form DNA the bases are hydrogen bonded in a position almost perfectly centered within the double helix (the dot in Figure 7 denotes the center of the DNA helix). As illustrated in Figure 8, the bases in B-form DNA stack into a cylinder within the center of the double helix, with the sugars and phosphates on the outside of the helix. Because all base pairs are positioned similarly with respect to other base pairs, B-form DNA has a base pair repeating unit of one.

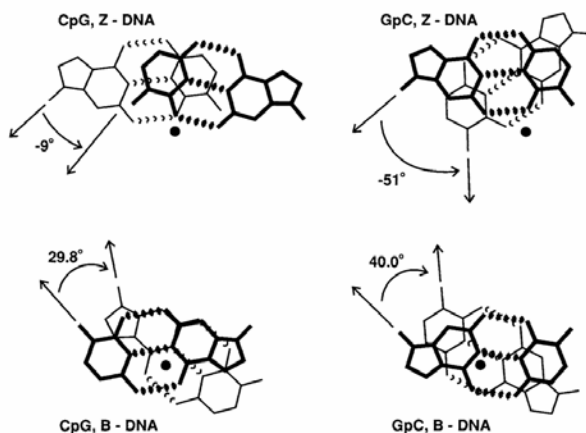


Figure 7. Base stacking in I-DNA and B-DNA.

In Z-DNA the repeating base pair unit is two. The CpG dinucleotide is very different from the GpC dinucleotide. There are 12 bp per turn of ZDNA which should produce an average helix twist of -30° (the value is negative to denote the left-handed rotation in Z-DNA compared with a positive rotation found in right-handed helices). However, the twist angle for CpG is only -9° , whereas the twist angle for GpC is -51° (Figure 7). Consequently, in contrast to B-DNA, each base pair in Z-DNA is not oriented in the helix in a similar fashion with respect to the adjacent base pairs. However, every two base pairs are oriented within the helix in a fashion identical to the next two adjacent base pairs. The twist angle for two base pairs relative to the next two base pairs is -60° , producing a helix repeating unit of two.

Helix is a type of smooth space curve, i.e. a curve in three-dimensional space.

Another major difference between B-DNA and Z-DNA is that in ZDNA the base pairs are sheared with respect to their stacking positions. The orderly stacking in the center of the **helix** found in B-DNA is nonexistent in Z-DNA. Analysis of the CpG dinucleotide in Figure 7 shows that the two cytosines are stacked but the guanines are positioned under

and over the ribose sugars. Thus, in the CpG dinucleotide the bases are sheared by being effectively pulled away from each other with respect to the center of the helix. In the GpC dinucleotide in Z-DNA, the bases are relatively well stacked.

The position of bases within the helix also distinguishes DNA from BDNA. In B-DNA the bases form a cylinder within the double helix, with the hydrogen bonds at the center of the helix. However, as shown in Figures 7 and 8, the bases in Z-DNA are positioned toward the outside of the Z-DNA helix so that neither the bases nor the hydrogen bonds overlap the center of the helix. In B-DNA, the bases are protected from the solvent by their location at the center of the helix. In Z-DNA, certain ring positions are much more chemically reactive than in B-DNA. For example, the N7 and C8 positions of guanine are exposed to the solvent in Z-DNA (Figure 8).

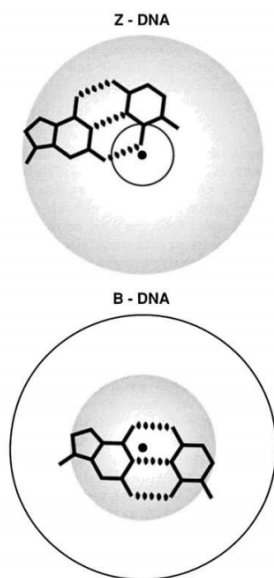


Figure 8. Position of the base pairs within the helix.

The B-Z Junction

When a region of Z-DNA exists within a larger B-form DNA molecule, there must be a junction between the right- and left-handed helices. The best estimation of the physical structure of a B-Z junction is that

it consists of a region of probably 3-4 unpaired or weakly paired bases. There is likely not a sharp transition from the left- to the right-handed helix but a region of a few base pairs that is partially unwound. The selective or differential reactivity of many chemicals at B-Z junctions has provided sensitive assays for Z-DNA.

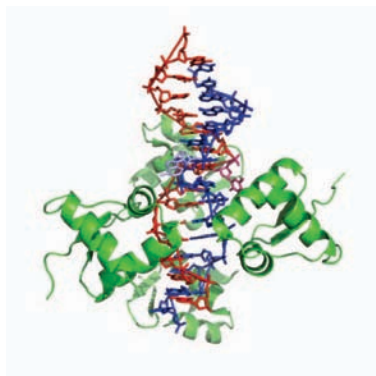


Figure 9. B–Z junction bound to a Z-DNA binding domain.



CORE CONCEPTS: FORMATION AND STABILITY OF Z-DNA

DNA encodes biological information in at least two different ways. First, through the linear order of nucleotides, it specifies the composition of proteins. Second, through its shape, DNA can provide information that is used directly or indirectly by a variety of macromolecules to regulate the assembly of cellular machines. DNA is capable of assuming many shapes. One of the most dramatic changes in shape is that which occurs in going from the familiar right-handed B-DNA double helix to the slightly thinner and elongated left-handed Z-DNA conformation.

DNA is capable of assuming many different conformations other than the familiar right-handed B-DNA double helix. One of the most dramatic examples is the Z-DNA conformer, which is left-handed. Like B-DNA, the two strands of Z-DNA are antiparallel and joined by Watson–Crick base-pairing. In contrast to B-DNA, which has all its bases in the anti-conformation, the bases in the Z-DNA helix alternate between the anti-conformation and the unusual syn-conformation. This dinucleotide repeat causes the backbone to follow a zigzag path, giving rise to the name Z-DNA. In Z-DNA, there is only a single narrow groove that corresponds to the minor groove of B-DNA. No major groove exists. Instead, the ‘information’ rich residues that allow sequence-specific recognition of B-DNA lie exposed on the convex outer surface of Z-DNA (Figure 10). This transition from B- to Z-DNA occurs most readily in sequences, with alternations of purines and pyrimidines, especially alternating deoxycytosine and deoxyguanine residues.

The biological role of Z-DNA is an area of active study. The aim of these investigations is to determine whether this alternate DNA conformation exists *in vivo*, how its formation is regulated, and what information it conveys. Here we will review recent studies that bear on the role of Z-DNA in biological systems.

The Sequence Requirement for Z-DNA

Z-DNA can form in regions of alternating purine-pyrimidine sequence; $(GC)_n$ sequences form Z-DNA most easily. $(GT)_n$ sequences also form Z-DNA but they require a greater stabilization energy for formation than $(GC)_n$. $(AT)_n$ generally does not form Z-DNA since $(AT)_n$ easily forms cruciforms. $(AT)_n$ can form Z-DNA under two special conditions. First, up to 10 alternating A · T base pairs embedded within a $(GC)_n$ or $(GT)_n$ region will form Z-DNA. Second, conditions of high negative supercoiling, high salt, and the presence of $NiCl_2$ cause $(AT)_n$ to form left-handed DNA.

Formation of Z-DNA in Vitro

The existence of Z-DNA was first suggested by optical studies demonstrating that a polymer of alternating deoxyguanosine and deoxycytidine residues ($d(CG)_n$) produced a nearly inverted circular dichroism spectrum in a high salt solution. The physical reason for this finding remained a mystery until an atomic resolution crystallographic study of $d(CG)_3$ revealed a left-handed double helix, which maintained Watson-Crick base pairing(2). The Z-DNA helix is built from a dinucleotide repeat with the deoxycytidines in the *anti*-conformation while deoxyguanosines are in the unusual *syn* form. In Z-DNA, there is a single narrow groove that corresponds to the minor groove of B-DNA. There is no major groove. Instead, the “information”-rich residues that allow sequence-specific recognition of B-DNA lie exposed on the convex outer surface of Z-DNA (Fig. 10). The transition from B-DNA to Z-DNA involves “flipping” the base pairs upside down. During this process, deoxycytidine remains in the *anti*-conformation because both the sugar and base rotate, while only the base of deoxyguanosine inverts, moving it into the *syn* conformation. As a consequence, the backbone follows a zigzag path, giving rise to the name Z-DNA. Z-DNA can form from B-DNA under physiological salt conditions when **deoxycytidine** is 5-methylated. The demonstration that Z-DNA formed under conditions of negative superhelical stress raised considerable excitement as this brought the left-handed conformation within the realm of biology.

Deoxycytidine is a deoxyribonucleoside, a component of deoxyribonucleic acid.

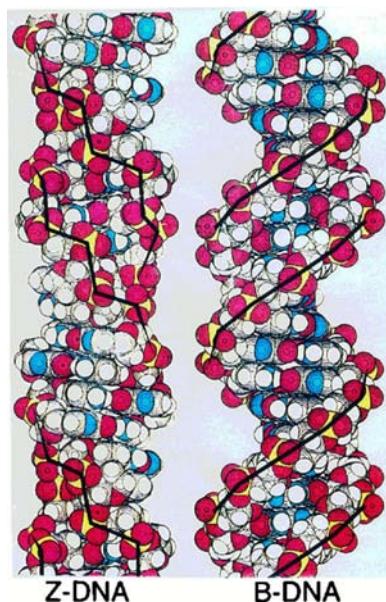


Figure 10. The “information-rich” residues that allow sequence-specific recognition of the major groove of B-DNA lie on the convex surface of left-handed Z-DNA helix. The two DNA strands of each duplex are highlighted by *solid black lines*. The “zigzag” nature of the Z-DNA backbone is clearly seen.

Stabilization of Z-DNA by negative supercoiling illustrates a number of features. First, Z-DNA is a higher energy conformation than B-DNA and will only form when plasmids are torsionally stressed. The energy necessary to stabilize Z-DNA can be determined by measuring the plasmid superhelical density at which Z-DNA formation occurs, and it is proportional to the square of the number of negative supercoils. Second, sequences other than alternating purines and pyrimidines can form Z-DNA. The ease with which this occurs depends on the sequence; $d(CG)_n$ is best, $d(TG)_n$ is next, and a $d(GGGC)_n$ repeat is better than $d(TA)_n$. Third, formation of B-Z DNA junctions, each of which has a free energy ΔG near +4 kcal/mol, is a significant energetic barrier to Z-DNA formation.

Based on these empirical findings, computer models have been developed to rank the Z-DNA-forming potential of naturally occurring sequences. One analysis of 137 fully sequenced human genes demonstrated that sequences which could form Z-DNA easily were present in 98 and that they were distributed non-randomly throughout

the gene; sequences were 10 times more frequent in 5' than in 3' regions. This fits with the expectation that the energy necessary to form Z-DNA *in vivo* is generated by transcription. Negative supercoils arise behind a moving RNA polymerase as it ploughs through the DNA double helix. The torsional strain generated by passage of RNA polymerases then becomes a potent source of energy to stabilize Z-DNA.

Formation of Z-DNA in Vivo

Due to the requirement for energy, formation of ZDNA *in vivo* is an active process. One source of available energy is provided by transcription. As pointed out by Liu and Wang, negative supercoils arise behind a moving RNA polymerase as it ploughs through the DNA double helix, providing one mechanism for the initiation of Z-DNA formation *in vivo*. Computer models are consistent with this prediction. One analysis of 137 fully sequenced human genes demonstrated that sequences which could form Z-DNA easily were present in 98 genes. These sequences were distributed nonrandomly throughout a gene – sequences were ten times more frequent in 50 than in 30 regions. They lie precisely in the regions of a gene where negative supercoiling is highest during transcription.

Experimental Demonstration of Z-DNA Formation in Vivo

A number of experiments in prokaryotes have been used to demonstrate that Z-DNA forms *in vivo*, and that this occurs as a result of transcription. One approach is to detect Z-DNA using chemical modification of DNA. Through use of either osmium tetroxide or potassium permanganate, the formation within *E. coli* of Z-DNA in plasmids with a $d(CG)_n$ insert can be demonstrated. UV crosslinking of bacteria treated with psoralens have confirmed these results, and made possible a precise measurement of the amount of unrestrained supercoiling present within *E. coli* necessary to initiate formation of Z-DNA. Another approach has used a construct in which an EcoRI site is embedded in a Z-DNA forming sequence. In the bacterial cell, this fragment can be methylated when it is in the B-DNA conformation but it becomes resistant to methylation while in the ZDNA conformation. Susceptibility to methylation of the EcoRI site thus provides an *in vivo* measure of ZDNA formation. In *E. coli*,

Z-DNA is formed in the absence of external perturbation and is increased by transcription, an effect that is enhanced by mutations inactivating topoisomerase I. In *Morganella*, *Klebsiella*, or *Enterobacter* formation of Z-DNA was not observed.

It has been difficult to directly demonstrate the existence of Z-DNA in eukaryotic systems due to their increased complexity. A number of early observations clearly suggested its existence. Unlike B-DNA, Z-DNA is highly immunogenic, and polyclonal as well as monoclonal antibodies can be made that recognize this conformation. One natural source rich in anti-Z-DNA antibodies is the sera obtained from patients with autoimmune diseases, especially lupus erythematosus. These antibodies are produced during the exacerbations of the disease, along with antibodies to many other nuclear components. The high specificity of these antibodies strongly suggest that Z-DNA is the cognate antigen and, by implication, that Z-DNA exists *in vivo*.

Antibodies raised in rabbits and sheep were used in staining experiments with both fixed and unfixed polytene chromosomes of *Drosophila*. These antibodies produced an unusual staining pattern of interband regions but did not stain bands. Staining was especially intense in the puffs which are associated with high levels of transcriptional activity. Antibodies were also used in staining ciliated protozoa which have both a macronucleus and a micronucleus. The micronucleus is used for genetic reproduction, but the macronucleus is the site of all transcriptional activity. Here, again, the macronucleus stained exclusively with no staining in the micronucleus. Both of these early experiments suggested, somewhat indirectly, a link between transcriptional activity and the presence of Z-DNA.

Analysis of intact mammalian systems has been more complicated. There are a number of limitations in these experiments. As yet, no phenotype has been associated with presence or absence of Z-DNA forming sequences, thus excluding the use of genetic approaches. In order to model Z-DNA formation *in vivo* a number of experiments have been carried out using metabolically active permeabilized mammalian nuclei which were formed by embedding intact cells in agarose microbeads using the method of Jackson and Cook. Here, low concentrations of detergent are used to lyse the cytoplasmic membrane and permeabilize the nuclear membrane. These nuclei have been shown to replicate DNA at

85% of the rate observed in the intact cell, and they are transcriptionally competent. In these experiments the amount of Z-DNA present in the gene is measured by diffusing biotin-labeled anti-Z-DNA monoclonal antibodies into the beads. The amount of Z-DNA present can be measured by quantitating how much radioactive streptavidin binds within the nucleus. Such experiments show that, at low concentrations of antibody, the amount of Z-DNA measured was independent of the antibody added over a 100-fold change in antibody concentration, suggesting that the Z-DNA is present *de novo* in these preparations rather than being induced by antibody. Furthermore, the amount of Z-DNA present increased dramatically during active transcription, consistent with the model of Liu and Wang, but was largely unaffected by DNA replication.

In further experiments, it was found that individual genes could be assayed by cross linking the antibody to DNA using a 10-ns exposure of a laser at 266 nm. Release of DNA fragments with attached antibody was accomplished by diffusing in restriction endonucleases and performing an *in situ* DNA digest. Following isolation of biotin-labeled antibody-DNA complexes with streptavidin magnetobeads, free DNA was obtained by proteolysis. These experiments made it possible to determine the site of Z-DNA formation in particular genes. Using hybridization or PCR techniques, the *c-myc* gene was studied in murine U937 cells. Three transcription-dependent ZDNA forming segments were identified in the 50 region of the gene with two of them near promoters. Retinoic acid, which induces the cells to differentiate into macrophages, was then used to down regulate expression of *c-myc*. Loss of *c-myc* expression was accompanied by a rapid reduction in the amount of Z-DNA present in these three regions. In contrast, Z-DNA formation in the beta actin gene, which is not down regulated with differentiation, was detected under all the conditions tested.

In other studies with a primary liver cell line, induction of Z-DNA was measured in the corticotropin hormone-releasing gene. Z-DNA formation increased when the gene was up-regulated and decreased when it was down regulated. This finding suggests that physiological events are being measured in these systems.

A major conclusion from these studies is that ZDNA forms largely, if not exclusively, behind a moving RNA polymerase and is stabilized by the negative supercoiling generated by DNA transcription.

Functional Consequences of Z-DNA Formation

The role of Z-DNA in biological processes is currently unknown. In principle, Z-DNA formation could have a functional role that need not involve its recognition by proteins. For example, *E. coli* RNA polymerase does not transcribe through Z-DNA raising the possibility that the formation of Z-DNA behind (5') to a moving polymerase may block a trailing RNA polymerase

Processing of an RNA would then be physically and temporally removed from that of subsequent transcripts, perhaps minimizing non-functional trans-splicing in eukaryotes.

from transcribing through that region of a gene until the torsional strain stabilizing the Z-DNA is relieved by topoisomerases. This mechanism might ensure spatial separation between successive polymerases.

Alternatively, formation of Z-DNA may relieve topological strain that arises when intact duplexes are intertwined as occurs during recombination events involving Holliday junction intermediates. For example, the Z-forming $d(\text{CA/GT})_n$ sequence has been shown to be recombinogenic in yeast, but is found to be less efficient than $d(\text{CG})_n$ in human cells. Furthermore, several reports have correlated chromosomal breakpoints in human tumors to potential Z-DNA forming sequences, although no causal relationship has yet been established. In addition, Z-DNA formation could affect the placement of nucleosomes as well as the organization of chromosomal domains by providing regions from which histones or other architectural proteins are excluded. Lastly, Z-DNA may perform unexpected roles in organisms such as the primitive eukaryote dinoflagellate *Prorocentrum micans*, which lack histones and nucleosomes but forms immunologically detectable Z-DNA at the nuclear periphery and at the segregation fork of dividing chromosomes.

There have been many attempts to find proteins that bind to Z-DNA in the hope that they would indicate indirectly the presence of Z-DNA *in vivo*, and help establish a biological role for this conformation. Early studies were unfruitful and caused widespread skepticism that Z-DNA would be associated with any biological function. Many of the apparently positive results reported in these studies may have been due either to

artefacts or misinterpretation of data. However, absence of proof was confused with absence of existence.

Identification of a High Affinity Z-DNA Binding Protein with Enzymatic Activity

Our work has recently shown that one type of doublestranded RNA adenosine deaminase (ADAR) called ADAR1 binds Z-DNA in vitro with high affinity. The dissociation constant of the Z-DNA binding domain is nanomolar, making it likely that this interaction is functional. The binding of ADAR1–Z-DNA was identified initially in bandshift assays, using competition with high concentrations of unlabeled polynucleotides to indirectly confirm specificity of binding. Mapping studies showed the presence in ADAR1 of two Z-DNA binding motifs, called $Z\alpha$ and $Z\beta$ (Figure 11). $Z\alpha$ alone is able bind to Z-DNA with high affinity, but can interact with $Z\beta$ to form a domain with slightly different binding properties. The specificity of recombinant $Z\alpha$ for Z-DNA has now been directly confirmed using biophysical techniques such as circular dichroism and Raman spectroscopy. NMR studies have confirmed structure predictions that $Z\alpha$ belongs to the winged-helix-turn-helix family of proteins (Figure 12). The fold is similar to that found in the globular domain of histone H5 and the transcription factor HNF γ 3. The domain consists of a helix-turn-helix motif (incorporating $\alpha 2$ and $\alpha 3$ shown in Figure 12) and a C-terminal β -sheet that constrains the fold through contacts with residues lying between $\alpha 1$ and $\alpha 2$. Mutagenesis studies confirm that $\alpha 3$ has the properties of a recognition helix and also show that residues in the C-terminal β -sheet are also involved in binding to ZDNA. Further structural studies are necessary to resolve how the winged helix-turn-helix fold can be used to recognize both right- and left-handed DNA.

Domain Structure of ADAR1

ADAR1 has a complex structure (Figure 11). It has two copies of a ZDNA binding motif that are absent in ADAR2, the only other known

member of this enzyme family. In addition, ADAR1 has three copies of a double-stranded RNA binding motif (DRBM) as well as a catalytic domain related to that of *E. coli* cytidine deaminase.

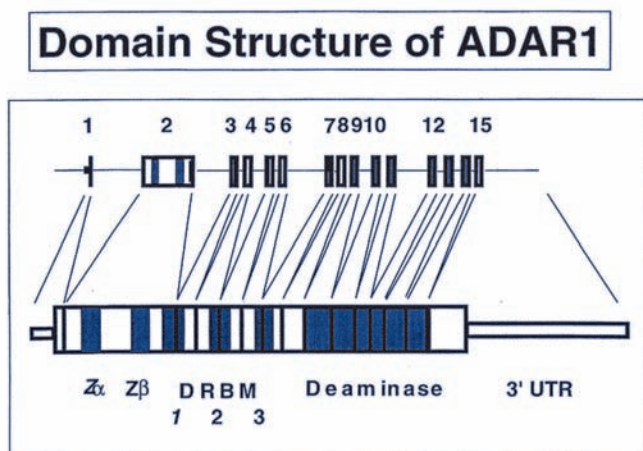


Figure 11. The domain structure of ADAR1. ADAR1 has two Z-DNA binding motifs, 3 double-stranded RNA binding motifs and a deaminase domain. The short form which starts at methionine 296, lacks the N-terminal Z α domain.

A number of different variants of ADAR1 are produced in cells. The DRBM for example show some variation arising from differential splicing that may affect substrate specificity. A dramatic variation appears due to a splicing event that replaces the exon containing the methionine necessary to initiate translation of full length ADAR1. Instead a methionine at position 296 is used to produce a short form of ADAR1 (compare sequence in Genbank accession number X79448 with that in X79449). The shorter form has only one copy of the Z-DNA binding motif and binds ZDNA with a lower affinity when compared to the longer form. The splicing event that leads to the formation of the shorter form of ADAR1 is sensitive to the presence of interferon, which causes increased production of the long form of the enzyme. The Z-DNA binding properties of ADAR1 can thereby be modulated, although the physiological significance of this event remains to be determined.

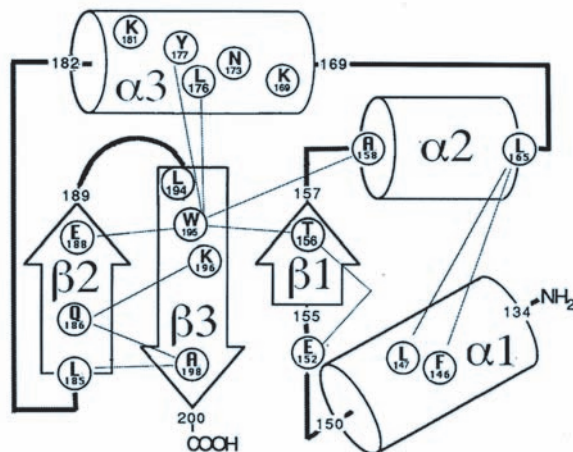


Figure 12. The topology and candidate Z-DNA contacts of Zα. The data show that the topology determined by NMR and the location of candidate contacts of Zα with Z-DNA determined by mutagenesis are in some respects similar to those of histone H5 and HNF-3γ. The position of α-helices and β-strands are indicated by cylinders and arrows connected with thick lines. Numbers correspond to amino acid residues. Long range NOEs are indicated with thin lines and show the interactions between the C-terminal β-sheet and the α-helices of Zα. W195 makes extensive contacts with other amino acids. Residues (K169, N173, Y177, K181) on the face of α3 that putatively contact Z-DNA are indicated.

ADAR1 and DSRNA Editing

ADAR1 belongs to a family of deaminases that modify double-stranded mRNA by catalyzing the hydrolytic deamination of adenine to form inosine, which is subsequently translated as guanosine. When this change occurs in the first two positions of a codon, an amino acid different from the one encoded by the gene may be placed at that site. Twenty six codons specifying 12 amino acids can be substituted in this manner. An illustrative example is editing of the GluR-B receptor subtype RNA which specifies a component of the AMPA sensitive glutamic acid receptor. Glutamic acid is the major excitatory neurotransmitter in mammalian brains. As a result of editing, a glutamine (CAG) in the exon specifying second transmembrane domain of the GluR-B subunit is replaced by arginine (CGG). This modification changes the electrophysiological properties of the assembled receptor, reducing the calcium conductance of the ion

channel. Whether this change is brought about by ADAR1 or another member of the ADAR family, such as ADAR2, is currently controversial. Other putative substrates for ADAR1 and ADAR2 are known. The list currently includes multiple sites in glutamic receptor sub-units GluR-A,B,C,D,5 and 6, the serotonin-2C receptor, a liver α 2,6 sialotransferase, and the squid voltage-gated potassium receptor. Addition substrates are likely to exist as inosine is found in brain mRNA at a frequency of one in every 17000 ribonucleotides. A method has been recently described to allow detection of such edited messages. Members of the ADAR family are also ubiquitous in metazoa. These results suggests that A to I editing of evolutionary significance, producing phenotypic variation by altering the linear flow of information from DNA to RNA.

DSRNA Editing and Z-DNA

In the cases so far examined, the double-stranded RNA editing substrate is formed by folding the 3' intron back onto the exon to basepair with the site that is edited. The involvement of introns requires that editing occur soon after transcription of RNA, and before splicing. This sequence of events provides a rationale for the recognition of Z-DNA by ADAR1. Z-DNA *in vivo* is a transcription-dependent structure and will form when appropriate sequences are present behind (5' to) a moving RNA polymerase. Transcription-induced Z-DNA may serve to localize the editing activity of ADAR1 to a particular region within a gene, preventing indiscriminate modification of other double-stranded RNAs (Figure 13). This mechanism allows editing to initiate as the transcript is produced; allowing ADAR1 to act before the intron is removed by the splicing apparatus. Recognition of Z-DNA by ADAR1 may also block the gene from further transcription until editing of the RNA is complete. Furthermore, it is possible to imagine that the extent of adenosine to inosine modification at a particular site is related to the amount of Z-DNA formed, depending upon the ease with which surrounding sequences adopt the ZDNA conformation as well as the local superhelical density.

Recent results have allowed this model to be refined further. We have shown that recognition of Z-DNA by Z α is conformation specific rather than sequence specific.

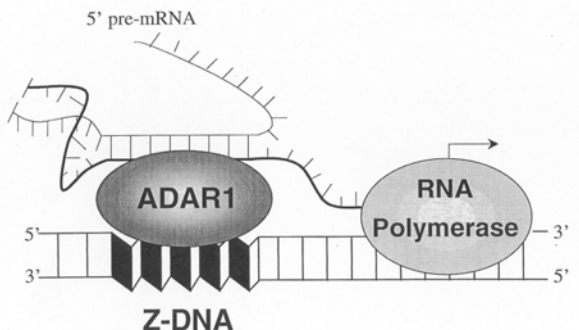


Figure 13. Model for regulation of ADAR1 activity by Z-DNA. In vivo, Z-DNA can be stabilized by the negative supercoiling generated by an RNA polymerase moving through a gene. Transcription also gives rise to regions of double-stranded RNA (dsRNA), formed when a nascent RNA transcript (pre-mRNA) folds back on itself. The RNA editing enzyme, dsRNA adenosine deaminase type 1 (ADAR1) has been shown to bind both Z-DNA and dsRNA with nanomolar affinity. It is proposed that binding to Z-DNA allosterically activates editing by ADAR1, initiating modification of a transcript as it forms and before splicing has occurred. This enzyme causes the hydrolytic deamination of adenine within the dsRNA to form inosine, which is subsequently translated as guanine. Editing thus changes the read-out of a gene. Several editing sites may exist in a particular pre-mRNA. ADAR1 thus utilizes the structural information encoded in Z-DNA and dsRNA to alter the linear flow of information from DNA to RNA.

A rather dramatic example of the ability of $Z\alpha$ to bind different Z-DNA sequences is shown in Figure 14. In this circular dichroism experiment, short **oligonucleotides** with two 6 basepair binding sites for $Z\alpha$ are used. The first site is $d(CG)_3$ and is used to initiate Z-DNA formation by the oligonucleotide. The second site can be varied to test for interaction of $Z\alpha$ with the Z-DNA conformation of that sequence. Two such substrates are shown in Figure 14. In Figure 14A both binding sites are $d(CG)_3$. Under low salt conditions, the oligonucleotide is in the B-DNA conformation (dotted line). However in the presence of $Z\alpha$ (solid line), under these same low salt conditions, the spectrum inverts to give one characteristic of Z-DNA, as shown by comparison with the spectrum obtained when the oligonucleotide is placed in 4M NaCl (dashed line). In Figure 14B the second site is $d(TA)_3$. The low salt (dotted line) and high salt spectrum of the oligonucleotide (dashed line) are shown. In 4M NaCl, the spectrum only undergoes partial inversion, indicating that even under these stringent conditions the $d(TA)_3$ site cannot fully adopt the Z-DNA conformation. However, in the presence of $Z\alpha$ under low salt conditions,

inversion of the spectrum is complete showing that $Z\alpha$ binds to the $d(TA)_3$ as well as the $d(CG)_3$ site and stabilizes it in the Z-DNA conformation. This result is confirmed by bandshifting experiments that demonstrate that both sites on the oligonucleotide are occupied by $Z\alpha$. Similar results are obtained when other sequences are used in the second site, indicating that $Z\alpha$ is specific for the Z-DNA conformation rather than for a particular sequence. Thus $Z\alpha$ can recognize any sequence when conditions are such that it forms Z-DNA.

Oligonucleotides

are short DNA or RNA molecules, oligomers that have a wide range of applications in genetic testing, research, and forensics.

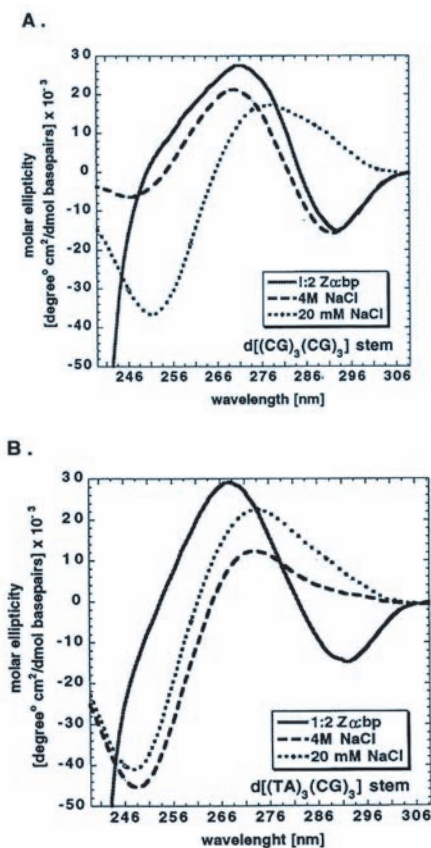


Figure 14. CD titration of $Z\alpha$ complexed to DNA hairpins with two binding sites. The DNA hairpins $d[(CG)_6T_3(CG)_6]$ (panel A.) and $d[(TA)_3(CG)_3T_3(CG)_3(TA)_3]$ (panel B) were titrated with $Z\alpha$

peptide in 50 mM Tris.HCl, 50 mM NaCl, 0.1 mM Na₂EDTA (pH 7.4) at 30°C in a Aviv 60DS spectrometer. Spectra obtained using a ratio of 1 mole of Z α to 2 moles of basepairs are shown (solid lines). Reference spectra obtained in the absence of protein (dotted line) and in 4M NaCl (dashed line) are also shown. In 4M salt hairpins d[(CG)₆T₃d(CG)₆] forms Z-DNA, while d[(TA)₃(CG)₃T₃(CG)₃(TA)₃] undergoes only a partial transition. However, Z α can stabilize d(TA)₃ in the Z-DNA conformation as shown by the solid line in panel B. The CD signal produced by Z α alone is equivalent to the baseline in the region of 250 to 300 nm. The protein alone has a strong negative component below 240 nm.

This result raises the question of how Z α targets a particular Z-DNA forming regions within a gene, rather than all Z-DNA forming regions in the genome. A number of models can be proposed. ADAR1 may for example travel with the polymerase as it elongates transcripts, dissociating when it sees a Z-DNA sequence. However, immunofluorescence studies fail to show a co-localization between Z α and RNA polymerase making this model unlikely (Herbert, unpublished data). We have shown that high affinity binding to Z-DNA requires two Z α molecules. Other protein partners may confer on Z α sequence-specific recognition of Z-DNA. Z β may be one such domain and there is some in vitro evidence to support this proposal. Alternatively ADAR1 may bind to a specific B-DNA sequence and wait until a passing polymerase induces the local formation of Z-DNA before it becomes active. In this case, the conformation specific recognition of ZDNA may induce allosteric changes in ADAR1 that initiate binding of dsRNA and induce catalytic function. This mechanism allows Z-DNA to be captured as it forms and before it is dissipated by topoisomerases and other proteins that relieve superhelical stress. Current work using UV crosslinking techniques using this technique supports the idea that ADAR1 binds DNA in vivo, although the full complement of sequences bound are not yet known.

Other Z-DNA Binding Proteins

The winged-helix-turn-helix family of proteins to which Z α belongs is large and is involved in many aspects of growth and development. It is possible that other members of this family recognize Z-DNA in a manner similar to Z α , allowing their activities to be regulated by transcription.

Many different roles in the spatial and temporal co-ordination of the cell's molecular machinery are possible.

One interesting element may be E3L protein from vaccinia virus, which shows some sequence similarity to Z α in its N-terminus. The carboxy terminus of E3L also contains a dsRNA binding domain that is essential to interferon resistance, making it similar in domain structure to ADAR1. It is also expressed in a long and a short form that lacks the N-terminal region. We have confirmed that the N-terminal domain binds to Z-DNA, but its function remains to be established.

Other proteins may exist that bind to Z-DNA with lower affinity than ADAR1. It has been demonstrated that peptides in which every second residue is lysine will stabilize Z-DNA in vitro at micromolar concentrations. This provides a simple protein motif with which to recognize Z-DNA. This motif exists in a number of proteins, but it remains to be shown that such proteins interact with Z-DNA. In addition, evidence has been presented to show that topoisomerase II from *Drosophila*, humans and calf thymus recognize a number of different DNA conformations, including Z-DNA. However, the domain interacting with Z-DNA has not yet been biochemically defined, nor has direct biophysical evidence been provided proving that this protein binds to Z-DNA rather than some other non-B-DNA conformation present in these polymers.

Assays for Z-DNA

There are a large number of physical and chemical differences between Z-DNA and B-DNA that have been determined using physical-chemical approaches such as X-ray crystallography, circular dichroism, ultraviolet and Raman spectroscopy, nuclear magnetic resonance, and electron spin resonance. These physical-chemical approaches, especially X-ray crystallography, have provided a picture of Z-DNA at the atomic level. In other assays, changes in the supercoiling of plasmids due to Z-DNA formation can be detected by agarose gel electrophoresis. Differential reactivity of Z-DNA with DNA methylases, restriction endonucleases, and a variety of chemicals has provided very powerful assays for Z-DNA that have been applied in vivo. Antibodies against Z-DNA have also been extremely useful as probes for Z-DNA both in vitro and in situ.

Z-DNA Antibodies

A number of different polyclonal and monoclonal antibodies have been raised against Z-DNA. Brominated poly(dG-dC)·poly(dG-dC) is typically used to elicit the antibody response. Different antibodies against Z-DNA recognize different structural features of the Z-DNA helix. *For example*, some recognize the base pairs that are on the outside of the DNA helix whereas others recognize the sugar residues.

The binding of antibodies to Z-DNA can be detected by a variety of methods. One involves the binding of DNA to nitrocellulose. Double stranded DNA does not normally bind to a nitrocellulose filter. However, when a protein is tightly bound to DNA, the protein will stick to nitrocellulose and trap the DNA with it. By using radioactively labeled DNA, the extent of Z-DNA formation can be quantitated by measuring the amount of radioactivity retained on a nitrocellulose filter in the presence of Z-DNA antibody. Antibody binding to DNA is very tight and relatively insensitive to changes in the ionic environment. Thus, antibodies are useful over the wide range of salt concentrations and in the presence of the various mono-, di-, and trivalent cations that stabilize Z-DNA.

Utilizing Antibodies to localize a Z-DNA Site within a Plasmid

Z-DNA antibodies bound to a $(GC)_n$ Z-DNA region within a supercoiled plasmid. In addition, they used this method to identify a naturally occurring Z-DNA forming sequence within the plasmid. Proteins can be “fixed” onto DNA by incubation in the presence of glutaraldehyde which forms covalent bonds, or cross-links, between the DNA and the protein. Following cross-linking with glutaraldehyde, the plasmid was cut with a restriction enzyme (HaeIII) that produces 22 fragments of various sizes. The DNA sample was then passed through a nitrocellulose filter. DNA fragments without covalently bound antibody passed through the filter whereas DNA fragments with bound antibody were retained. The DNA samples that passed through the filter were analyzed by electrophoresis on an acrylamide gel and compared with a HaeIII digestion pattern of “control” plasmid (not exposed to the antibody-glutaraldehyde treatment).

The antibody-treated fragments that passed through the nitrocellulose revealed “missing bands” or bands reduced in intensity, called “reduced bands.” The “missing bands” and “reduced bands” that were retained on the nitrocellulose filter contained Z-DNA. This type of experiment can be used to localize Z-DNA to a small region of a DNA molecule.

Antibodies would not induce Z-DNA and in time, without cross-linking, antibodies would fall off the DNA as the Z-DNA region reformed into B-DNA.

In the presence of 3-4 M NaCl or DNA supercoiling, Z-DNA is stabilized relative to the B conformation. However, Z-DNA-forming regions are in an equilibrium between the B and Z forms. In supercoiled plasmid, the equilibrium is shifted well toward the Z form. Relaxation of DNA supercoils by digestion with *HaeIII* causes regions of Z-DNA to be converted back to the B form. Some antibodies can bind to DNA inducing or stabilizing regions of Z-DNA in relaxed DNA.

Relaxation of Negative Supercoils as an Assay for Z-DNA

In plasmids containing regions of alternating $(GC)_n$ or $(GT)_n$ DNA supercoiling will drive the B-to-Z transition and stabilize Z-DNA within these purine-pyrimidine runs. The formation of ZDNA within a plasmid results in the relaxation of about 1.78 negative supercoils for every 10 bp of B-DNA that form Z-DNA. Two-dimensional agarose gels, can be used to visualize the Z-DNA transition. The critical super-helical density at which Z-DNA forms is determined from the topoisomer number for which migration in the first dimension is slower than for the preceding topoisomer. As with cruciforms, the number of super-helical turns removed provides an indication of the length of the Z-DNA region.

Chemical Probes of Z-DNA

Numerous chemical probes have been applied as assays for Z-DNA. The basis for these assays lies in the differential chemical reactivity of the Z and B forms of the double helix. In Z-DNA, the N7 and C8 positions of guanine are on the outside of the helix. Consequently, these positions are more reactive to some chemicals in the Z form than in the B form.

Chemicals that specifically react with unpaired bases will frequently react at the B-Z junctions. The assays used to identify the sites of chemical reaction, can also be applied to the analysis of Z-DNA. The sites of modification can be mapped by cleavage of the DNA with S1 nuclease. In addition, sites of chemical modification can be mapped by chemical cleavage at the site of the modified base or by a primer extension assay.

Many of these chemical probes are not specific for Z-DNA, but can be used to analyze cruciforms, intramolecular triplexes, or other non-B-DNA structures.

- **Bromoacetaldehyde, Chloroacetaldehyde, Osmium Tetroxide, and Hydroxylamine:** Bromoacetaldehyde (BAA), chloroacetaldehyde (CAA), osmium tetroxide (OsO_4), and hydroxylamine react specifically with unpaired bases. These reagents will chemically modify the B-Z junctions.
- **Diethylpyrocarbonate:** Diethylpyrocarbonate (DEPC) reacts with purines at the N7 position forming a carbethoxy group. DEPC will react with purines in B-form DNA; however, reactivity is enhanced in Z-DNA. In $(\text{GC})_n$ Z-DNA regions, guanines are hyperreactive. In $(\text{GT})_n$ Z-DNA regions, guanines (in the GT strand) are not hyperreactive; however, adenines (in the CA strand) are hyperreactive. The specificity of these reactions has been used as a chemical probe for DNA conformation.
- **Aminofluorenes:** The aminofluorene derivatives constitute a group of chemical carcinogens that covalently modify DNA. N-2- Acetylaminofluorene (AAF) covalently binds to the C8 position on guanosine. Although this position is relatively protected in B-form DNA, AAF reacts with B-form DNA. When a region of $(\text{GC})_n$ or $(\text{GT})_n$ exists in the Z-DNA conformation, AAF does not covalently bind to the C8 position of guanosine despite the fact that this atom is relatively exposed to the solvent. In a typical B-form DNA helix, a bulky modification on the C8 position of guanosine is difficult to incorporate into a B-form helix. In fact, the AAF-modified guanosine is probably bulged out of the helix. Within an alternating purine-pyrimidine region, the AAF-modified G can easily exist in the syn configuration in a Z-DNA helix. Modification of B-form

poly(dG-dC)·poly(dG-dC) with AAF facilitates its transition to Z-DNA.

Aminofluorene derivatives react in a hypersensitive fashion with guanosines at B-Z junctions. The structure at the junction must facilitate the reactivity with aminofluorene derivatives. The characteristic reactivity of guanosine in B-DNA compared with the selective hyperactivity of B-Z junction regions in Z-DNA provides an assay for Z-DNA.

- **Diepoxybutane:** Diepoxybutane is a relatively simple compound that covalently binds to the N7 position of guanosine. It forms only monoadducts to B-form DNA. However, in the Z-DNA conformation, interstrand cross-links are formed between guanosines of adjacent base pairs in opposite strands. This would seem to be an ideal assay for Z-DNA if it could be applied in vivo. Unfortunately, diepoxybutane is relatively reactive with the N7 of guanosine, making the glycosidic bond labile. On formation of an apurinic site, the phosphodiester bond is susceptible to hydrolysis which leads to the introduction of nicks in the DNA. If Z-DNA is stabilized by supercoiling in cells, chemicals that nick DNA will likely lead to the loss of Z-DNA.
- **Phenanthroline Metal Complexes:** Chiral (tris)-phenanthroline metal complexes have been developed as probes of DNA structure. These compounds contain four modified phenanthroline rings coordinated to a transition metal atom. The stereochemistry of the metal complexes and the ability of the phenanthroline rings to intercalate into DNA allow compounds to be designed that bind to specific shapes of DNA. For example, structures with a right-handed propeller twist of the phenanthroline moiety can bind to righthanded B-DNA. One reagent, A-tris(4,7-diphenyl-1,10-phenanthroline) cobalt(III), which has a left-handed propeller-like twist, binds specifically to left-handed Z-DNA. (Other compounds have been designed that bind selectively to A-DNA and to cruciform structures.) An additional advantage of these reagents is that the use of cobalt as the transition metal makes these stereospecific DNA binding chemicals photoactive. Irradiation of DNA in the presence of A-tris(4,7-diphenyl-

1,10-phenanthroline) cobalt(III) results in the introduction of nicks in the DNA at regions of alternating purine-pyrimidine runs that have the potential to form Z-DNA. This cleavage provides a potential assay for Z-DNA.

- **Psoralen Photobinding:** The psoralen derivative 4,5',8-trimethylpsoralen has provided an extremely useful assay for DNA that is applicable in living cells. Psoralens bind to B-form DNA by intercalation and, on absorption of 360-nm light, form monoadducts and interstrand cross-links. The structure of Z-DNA is not conducive to cross-link formation, since the base-stacking interactions are sheared and the twist angles are irregular. Thus, the 5,6 double bonds of pyrimidines in Z-DNA are not positioned correctly for cross-linking to occur. Indeed, psoralen and other intercalating drugs bind poorly to Z-DNA (binding of psoralen and ethidium bromide have been shown to actually drive the B-to-Z equilibrium toward B-DNA).

The sequence GAATT(CG)₆TA(CG)₆AATTC was designed as a ZDNA-forming sequence that was assayable using psoralen. Psoralen preferentially photobinds to the 5' TA at the center of the 26-bp Z-DNA-forming region when in the B-form but not when it exists as Z-DNA. The level of psoralen photobinding can be measured with base-pair resolution by an exonuclease III mapping procedure (Figure 15) or by a primer extension assay. A "standard relationship" has been determined between the superhelical density of the DNA, which defines the fraction of the Z-DNA forming sequence stabilized as Z-DNA, and the relative rate of photobinding to the 5' TA in the Z-DNA-forming sequence. The extent of photo binding to the Z-DNA 5' TA is measured relative to a nearby 5' TA that can only exist as B-DNA. Analysis of this relative ratio in living cells will provide an estimate of the effective level of supercoiling and level of existence of Z-DNA.

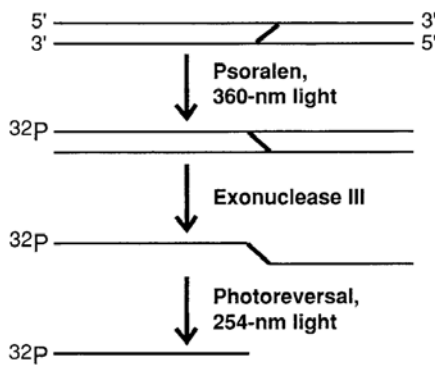


Figure 15. Psoralen exonuclease III assay for Z-DNA.

The reactivity of the EcoRI sites flanking the Z-DNA-forming region also provides an indication of the conformation of the DNA. The central AATT sequences are not reactive with psoralen when they exist as B-DNA. However, when Z-DNA forms, the EcoRI sites exist as B-Z junctions and the TT dinucleotides are hyperactive with psoralen. In the Z-DNA-forming sequence $\text{GAATT}(\text{CG})_6\text{TA}(\text{CG})_2(\text{TG})_8\text{AATTCC}$, psoralen photo binding occurs to only one strand of each B-Z junction. The reactivity of the junctions is a function of supercoiling (and the extent of ZDNA formation), providing additional “standard relationships” that can be used to estimate the extent of Z-DNA formation in living cells.

The psoralen photo binding assay has several advantages as an *in vivo* probe for Z-DNA compared with other Z-DNA assays. First, this assay can be done at levels of photo addition of about one psoralen adduct per 8000 bp of DNA. Although psoralen is a potent mutagen, >95% of an *E. coli* culture survives this level of DNA damage. Thus, the assay is truly done in “living cells.” Second, the photo binding of psoralen to Z-DNA actually drives the $\text{B} \rightleftharpoons \text{Z}$ equilibrium toward B-DNA. Consequently, the assay itself cannot introduce the Z-DNA conformation. This is not true for some assays involving chemical probes or Z-DNA antibodies. Third, the assay can be done under a variety of growth conditions (in different media and at various temperatures). Fourth, the photo binding protocol is very rapid. Fifth, this approach can be applied to DNA in living eukaryotic cells, since psoralens are freely permeable to eukaryotic cells and nuclei.

Assays Involving the Restriction/Modification System

Restriction endonucleases and their corresponding methylases usually recognize 4, 6, or 8 bp of DNA. The endonucleases introduce a nick into each of the two DNA strands and the methylases methylate the DNA at the same sequences. Methylation then prevents subsequent digestion by the corresponding endonuclease. The endonucleases and methylases recognize the B form of DNA and, in general, do not recognize the site in the Z-DNA form or in a B-Z junction.

The validity of this approach was first demonstrated by Singleton using the restriction endonuclease BamHI. The BamHI recognition site (GGATCC) was placed adjacent to a run of alternating GC base pairs in plasmid DNA. This site was cut by the enzyme in relaxed DNA. When the DNA was supercoiled, Z-DNA formed in the GC sequence and the BamHI site existed as a B-Z junction; BamHI cutting was reduced by more than 80%. The HbaI restriction enzyme fails to cleave its recognition site (GCGC) when the site exists as Z-DNA. In addition, the HbaI methylase, designated MHbaI, fails to methylate cytosine when the recognition site exists in the Z-DNA conformation.

Application of Z-DNA Assays for Measuring the Level of Unrestrained Torsional Tension in *E. coli*

The analyses of the level of Z-DNA in *E. coli* provide information about the level of unrestrained DNA supercoiling in bacterial cells because the formation of Z-DNA under physiological conditions (in the absence of Z-DNA binding proteins) requires negative DNA supercoiling. Since the level of Z-DNA formation is dependent on the level of supercoiling, determination of the level of Z-DNA in vivo provides an indication of the level of unrestrained torsional tension maintained in DNA inside cells. Most of our understanding of the energy within the helix in cells comes from quantitative measurement or simple detection of alternative DNA conformations in vivo.

The various Z-DNA assays and probe sequences used have provided different estimates of the level of supercoiling in DNA in vivo. Sinden and Kochel and Zheng et al., using psoralen photo binding, estimated a (J in vivo of -0.035 for wild-type cells and -0.048 for cells defective

in topoisomerase I. The experiments of Jaworski et al. (1987) were consistent with DNA at a level of supercoiling in vivo of ($J = -0.025$).

Utilizing the linking number assay, the shortest Z-DNA-forming sequence that produced a σ in vivo of -0.035 for wild-type cells and -0.048 for cells defective in topoisomerase I. The experiments of Jaworski were consistent with DNA at a level of supercoiling in vivo of $\sigma = -0.025$.

Utilizing the linking number assay, the shortest Z-DNA-forming sequence that produced a bimodal distribution in vivo formed Z-DNA at $\sigma = -0.025$ in vitro. Thus, a super-helical density in vivo of $\sigma = -0.025$ was suggested from these experiments. The Z-DNA-forming sequence $(CG)_6TA(CG)_6$ studied by Zheng did not produce a bimodal distribution although $> 40\%$ of the sequence existed as Z-DNA in *E. coli* as determined by the psoralen assay. Keep in mind that the assays employed or subtle procedural differences used may have an influence on the data obtained and the conclusions reached. Moreover, the linking number assay is indirect since it requires purification of plasmid from cells and analysis of the plasmid on agarose gels. One assumes that changes in linking number do not occur on lysis of the cells or during subsequent purification steps. However, there is really no independent way to confirm that linking number changes do not occur. The O_5O_4 experiments of Rahmouni and Wells suggested an in vivo negative superhelical density in the range of $\sigma = -0.038$. This is substantially lower (more negative) than previous estimates from the Wells laboratory based on the MEcoRI assay or linking number assays, but agrees with the results using the trimethylpsoralen assay.

Using the O_5O_4 assay, the level of Z-DNA in different locations of a plasmid in situ. The local level of supercoiling can be influenced by transcription. Specifically, DNA upstream of divergently transcribed genes can be more negatively supercoiled when transcription occurs. DNA downstream of divergently transcribed genes can be less negatively (perhaps even positively) supercoiled during transcription. Various $(CG)_n$ Z-DNA forming sequences were cloned into plasmid pBR322 upstream or downstream from two divergently transcribed genes (the Tet gene and the Amp gene). Upstream from the genes, a $(CG)_7$ sequence readily formed Z-DNA. Downstream from the genes, a 40-bp Z-DNA forming sequence could not be detected to exist as Z-DNA. These experiments

provided an elegant demonstration of transcription influencing levels of local supercoiling. The existence of Z-DNA in vivo can depend on the length (and sequence) of the potential Z-DNA-forming region as well as its location within a chromosome. Moreover, these experiments demonstrate that Z-DNA formation, in some cases, can be coupled to transcription.

REVIEW QUESTIONS

1. What is Z-DNA?
2. Discuss about the helix geometry of Z-DNA.
3. Explain about functional consequences of Z-DNA formation.
4. What role does recognition of Z-DNA play in the biology of ADAR1?
5. Describe the Z-DNA antibodies.

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CHAPTER 6

TRIPLEX DNA

Triplex-DNA is a DNA structure in which three oligonucleotides wind around each other and form a triple helix. In triple-stranded DNA, the third strand binds to a B-form DNA (via Watson–Crick base-pairing) double helix by forming Hoogsteen base pairs or reversed Hoogsteen hydrogen bonds.

After studying this chapter, you will understand the core concept of:

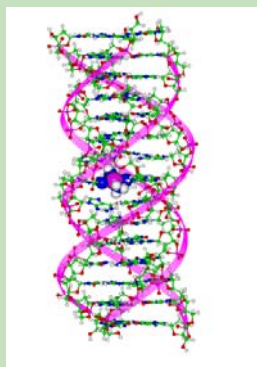
- Triplex DNA structures
- Formation of stable DNA triplexes

INTRODUCTION

Triplex DNA technology has been considered as an attractive antigen strategy for the treatment of genetic-based diseases. There are two classes of triplex DNA: intermolecular and intramolecular formations. An intermolecular triplex refers to triplex formation between a duplex and a different (third) strand of DNA. The third strand can either be from a neighboring chromosome or a triplex forming oligonucleotide (TFO). Intramolecular triplex DNA is formed from a duplex with homopurine and homopyrimidine strands with mirror repeat symmetry.

The degree of supercoiling in DNA influences the amount of intramolecular triplex formation that occurs. There are two different types of intramolecular triplex DNA: H-DNA and H*-DNA. Formation of H-DNA is stabilized under acidic conditions and in the presence of divalent cations such as Mg^{2+} . In this conformation, the homopyrimidine strand in the duplex bends back to bind to the purine strand in a parallel fashion. The base triads used to stabilize this conformation are T-A*T and C-G*C⁺.

The cytosine of this base triad needs to be protonated in order to form this intramolecular triple helix, which is why this conformation is stabilized under acidic conditions. H*-DNA has favorable formation conditions at neutral pH and in the presence of divalent cations. This intramolecular conformation is formed from the binding of the homopurine and purine strand of the duplex in an antiparallel fashion. It is stabilized by T-A*A and C-G*G base triplets.



CORE CONCEPT: TRIPLEX DNA STRUCTURES

A DNA triplex is formed when pyrimidine or purine bases occupy the major groove of the DNA double Helix forming Hoogsteen pairs with purines of the Watson-Crick base pairs. Intermolecular triplexes are formed between triplex forming oligonucleotides (TFO) and target sequences on duplex DNA. Intra molecular triplexes are the major elements of H-DNAs, unusual DNA structures, which are formed in homopurine-homopyrimidine regions of supercoiled DNAs. TFOs are promising gene-drugs, which can be used in an anti-gene strategy, that attempt to modulate gene activity in vivo. Numerous chemical modifications of TFO are known. In peptide nucleic acid (PNA), the sugar phosphate backbone is replaced with a protein-like backbone.

Triplex DNA formation depends upon the presence of a polypurine-polypyrimidine tract, and a third singlestranded DNA element. The third strand can come from the same molecule, for example from the other half of a mirror repeat sequence to form an intramolecular triplex, or from a different molecule to form an intermolecular triplex structure. This latter configuration has been leveraged to alter gene expression using triplex forming oligonucleotides. Lastly, triplex structures can form during the replication of mirror repeat sequences, called replicative triplex structures in this review for clarity.

Identification of Triplex DNA

DNA triplex formation occurs under favorable thermodynamic conditions. For an intramolecular triplex to form, the energy driving formation is the negative superhelical tension. When such energy is applied, the $T_{1/2}$ of formation can be as little as two minutes, though the rate of formation would be dependant on the length and composition of the mirror repeat, pH, supercoiling, temperature, and the presence of stabilizing agents like

divalent cations and polyamines. The formation of intermolecular triplex would depend also on the concentration of the duplex and singlestranded components, and the replicative structures would depend on the ability of other proteins to compete for binding to both the duplex and single-stranded components. There are a variety of methods to characterize triplex DNA.

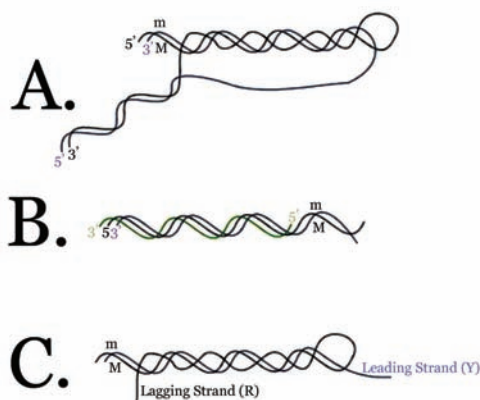


Figure 1. Types of triplex structures. A. Intramolecular triplex. B. Intermolecular triplex. C. Replicative form of triplex. 'M' denotes major groove, 'm' denotes minor groove.

Two-dimensional gel electrophoresis

When an intramolecular triplex forms in a plasmid, approximately one superhelical turn is absorbed for every 11 base pairs of the repeat participating in the triple-stranded structure. Subtle differences in energy absorption occur depending on whether the 3' or 5' pyrimidine strand fold into the major groove and whether or not the third strand is linked or unlinked at the tip.

Chemical Modification of unpaired strand

Base pairing significantly influences the chemical reactivity of nitrogenous bases. When one strand of a mirror repeat half binds the major groove, the complementary single-strand is exposed to the solvent phase and vulnerable to chemical modification (Figure 1A). **Chemical agents** can be used to probe for unpaired bases: chloroacetaldehyde and bromoacetaldehyde for cytosines and adenosines, diethylpyrocarbonate

for unpaired purines, and osmium for thymine. These modified bases can be identified by cleavage methods such as with hydrazine followed by piperidine, or by primer extension methods.



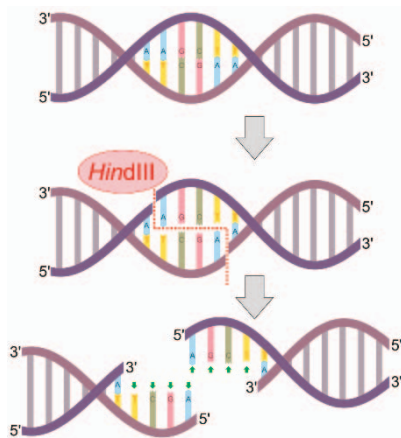
Chemical agent is a substance which is intended for use in military operations to kill, seriously injure, or incapacitate people because of its physiological effects.

Antibody recognition of triplex DNA conformation

The mammalian immune system can form antibodies that recognize DNA structures, for example patients with systemic lupus erythematosus and rheumatoid arthritis can develop antibodies to Z-DNA. Studies using murine monoclonal antibodies to triplex DNA suggest both A·T and C·G rich triplex structures are present in chromatin.

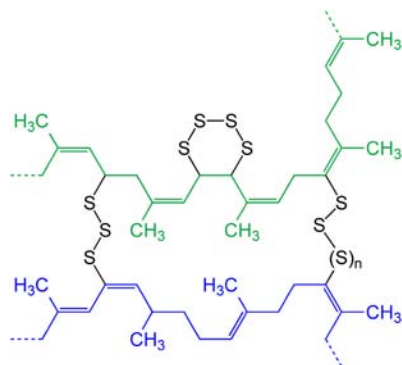
Nucleases

Single-strand specific nucleases, like the S1, P1, and mung bean nucleases, are efficient at cleaving singlestranded DNA at low pH. This single-strand specificity can be used to map the location of triplex structures by cleaving the associated single-strand (Figure 1C) in combination with restriction endonucleases.



Chemical Cross-linking

While single-stranded DNA can be identified by its reactivity to different chemical agents or nucleases, 4, 5', 8-trimethylpsoralen photobinding is dependent on B-form DNA. Psoralen intercalates into the B-form of DNA and can be activated to bind to 5' TA. Because intercalation fails to occur in single-stranded DNA, detection of perturbations in B-form DNA has been used to characterize triplex structures.



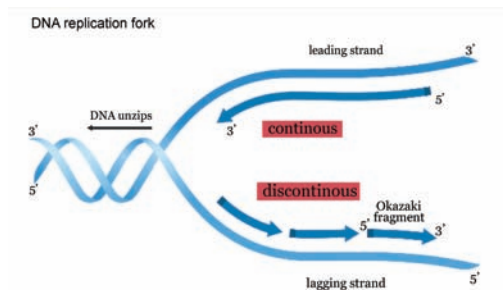
Role of Mirror Repeats in Disease

Polypurine·polypyrimidine tracts in the genome represent potential causes as well as therapeutic targets to ameliorate disease. Mirror repeat polypurine·polypyrimidine tracts may lead to disease because of mutagenesis or alterations in gene expression. Genome instability is associated with double-strand breaks. The formation of an intramolecular

triplex in vivo may be subject to cleavage by a host of processes including a variety of repair mechanisms leading to double strand breaks and instability or by formation of the replicative triplex structure that leads to fork collapse and subsequent double-strand breaks. Alternatively mirror repeat polypurine-polypyrimidine may be mutagenic because they can be decorated by DNA repair proteins that lead to targeted genetic destabilization.

Possible role of replication blockade and mutagenesis

Triplex DNA can adversely affect DNA replication and potentially lead to replication fork collapse. There are two features of polypurine-polypyrimidine tracts that may further potentiate their effects on replication. The replication primers created by the polymerase α holoenzyme usually start with purines, making the purine stand a very unlikely template since pyrimidines would be needed for the primer base pairing. In addition, the binding of replication protein A, the singlestranded binding protein in mammalian cells, to pyrimidines is 50 fold better than to purine. These two features mean that there is a reduced probability of the polypurine strand being bound to a primer or to replication protein A, and this 'nakedness' facilitates the probability of the purine strand binding in the major groove to form triplex DNA at neutral pH. Baran even postulated that such regions may function as replication terminators. If the fork collapse with a double-strand break occurs, then a recombinational repair may facilitate mutagenesis, for example by templating from a similar sequence elsewhere in the genome as suggested by evidence of gene conversion events. Such double-strand breaks resulting from triplex structures may be repaired either by homologous recombination or non-homologous end-joining and may result in translocation events.



Possible role of intramolecular triplex DNA as scaffold for repair proteins

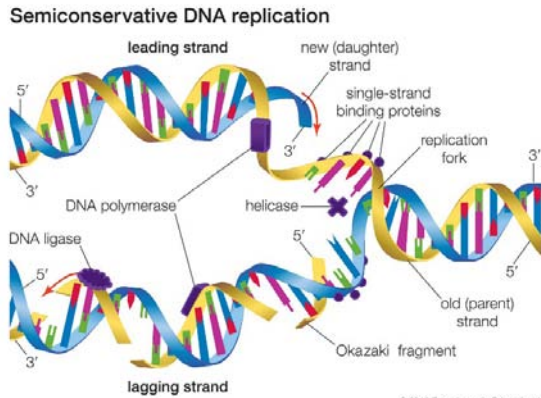
If an intramolecular triplex forms, driven by negative superhelical tension, the single-stranded polypyrimidine strand (Figure 1) may facilitate recombination without replication by acting as an annealing target leading to the development of a D-loop (32), or by acting as the invading strand to another region of homology.

Because polypyrimidine strands are the preferred substrate for replication protein A (RPA) binding and this strand will be single-stranded in intramolecular triplex structures at neutral pH, persistent RPA binding may lead to RPA hyper-phosphorylation that may trigger repair reactions. The intramolecular structure also may be targeted by nucleases resulting in single- or double-strand breaks and subsequent mutagenesis or recombination. The triplex structure itself may trigger repair-mediated mutagenesis. Synthetic oligonucleotides have been extensively investigated as a means of sequence-targeting mutagenesis.

Nucleotide excision repair (NER), damaged bases are cut out within a string of nucleotides, and replaced with DNA as directed by the undamaged template strand.

Nucleotide excision repair (NER) proteins appear to bind to intermolecular triplex structures, and are involved in the mutagenesis and recombination that occurs in cells. In bacterial cells, NER proteins UvrB and UvrC were required for triplex-induced cell growth retardation.

The ability for polypurine-polypyrimidine tracts to engage in the formation of different forms of triplex structures may also effect gene expression. Such tracts are over represented in the 5' end of genes and have been postulated to potentiate or suppress transcription. Lee. postulated that the alternative secondary structure formation may act as a molecular switch to turn genes off, and such appears to be the case at the γ -globin locus.



Diseases Associated with Mirror Repeats

There are a number of inherited as well as acquired human diseases that are associated with triplexforming DNA (Table 1). Both inherited and acquired diseases may require somatic mutation for phenotypic expression.

Table 1. Human diseases associated with triplex-forming DNA.

Disease	OMIN#
Autosomal Dominant Polycystic Kidney Disease (ADPKD)	#173900
Tuberous Sclerosis Complex (TSC)	#191100
Lymphangioliomyomatosis (LAM)	#606690
Friedreich's ataxia	#229300
Follicular Lymphoma	*136440
Hereditary Persistence of Fetal Hemoglobin (HPFH)	*142470

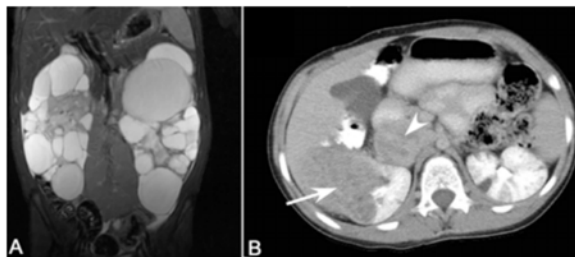
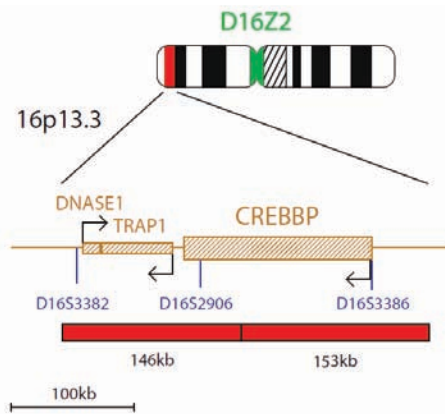


Figure 2. Manifestations of ADPKD and TSC. A. ADPKD in a child with TSC. Note numerous bilateral renal cysts (white). B. Large right angiomyolipoma (arrow) with invasion of the vena cava (arrowhead).

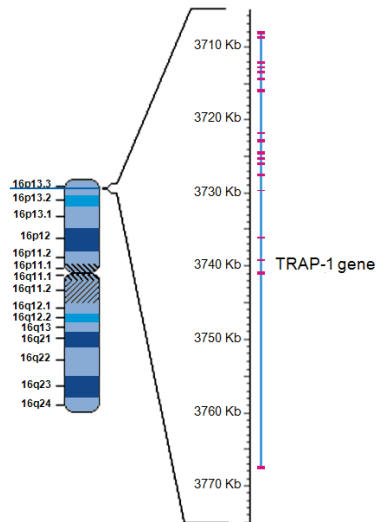
Chromosome 16p13.3 and autosomal dominant renal disease

Both tuberous sclerosis complex (TSC) and autosomal dominant polycystic kidney disease (ADPKD) dramatically affect renal architecture and function as the patient ages (Figure 2). Renal manifestations of TSC include angiomyolipomata, cysts, and, rarely, malignant tumors. ADPKD is associated with the development of renal cysts and approximately 50% of patients will require renal replacement therapy by the age of sixty years, making up 10% of the end stage renal disease population.



Both the ADPKD-related renal cysts and TSC-related angiomyolipomata are associated with a remarkable variability of expressivity. Angiomyolipomata are hamartomas consisting of abnormal vessels, smooth muscle cells and adipose tissue. Typically, angiomyolipomata are present in the young and continue to grow during adulthood. These lesions distort renal architecture and can compromise renal function. Because of the dysmorphic vasculature, aneurysms can develop that rupture and bleed, resulting in significant morbidity and even death. A more dramatic, though rare, renal manifestation of TSC is polycystic kidney disease. The TSC2 and PKD1 genes lie closely adjacent in a tail-to-tail orientation at 16p13.3. Frequently, mutational analyses in such polycystic kidney variant TSC patients reveal a deletion involving both the TSC2 and PKD1 genes. A large number of such patients have new mutations or exhibit mosaicism and these children have rapid development of cysts (Figure 2A). Renal failure is a long-term complication of TSC and end stage renal disease is the leading life

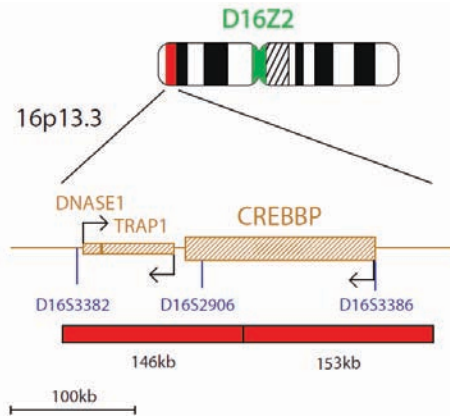
threatening complication of TSC in the adult.



Analysis of somatic and germline mutations in the TSC2 and PKD1 genes, as well as the comparison of physical to recombinational distances of 16p13, identify recombination as a mechanism of mutagenesis in this region. In heteroallelic individuals the normal TSC2 and PKD1 genes are the targets of somatic second mutation resulting in a cell lacking the functional protein (two hit hypothesis). Evidence supporting the two hit hypothesis for both TSC and ADPKD has come from the analysis of DNA and protein from affected tissues. Henske demonstrated an absence of tuberlin, the protein product of the TSC2 gene, in spindle and epithelioid cells from giant cell astrocytomas and in the smooth muscle, blood vessels, and adipose tissue in angiomyolipomata from patients with TSC. The loss of tuberlin immunostaining correlated with the demonstration of loss of heterozygosity (LOH) at the TSC2 locus in these tissues such that only the mutant allele could be identified. In angiomyolipomata, LOH has been found at the TSC2 locus as well as at the TSC1 locus suggesting that deletion or mitotic recombination as a second hit may be widely distributed. In a study of 130 sporadic TSC

The ability of triplex formation to modulate gene expression or alter the genetic code has lead to the therapeutic considerations of administering oligonucleotides that, once in place, may even be cross-linked to induce effects.

cases, 68% had TSC2 mutations, 22% had no identifiable mutation, and 10% had TSC1 mutations. Dabora demonstrated that de novo mutations in TSC2 are six times as frequent as in TSC1.



Similarly, lymphangioleiomyomatosis, a pulmonary disease of woman, can occur sporadically or in association with TSC. In the sporadic form of this disease, approximately half of the women also have renal angiomyolipomata. Microsatellite marker analysis revealed that in the sporadic form of lymphangioleiomyomatosis there is also TSC2 LOH in the angiomyolipomata, while LOH at the TSC1 locus has not been identified. In addition, Lininger examined benign and malignant mammary papillary neoplasms and found that loss of heterozygosity was frequently found at 16p13, while other regions, such as p53, INT-2, and several other loci on chromosome 16 were preserved suggesting that 16p13 may be somehow prone to LOH.

Mutational analysis of both germline and somatic mutations in the PKD1 gene support a role for recombination. Germline mutations in the PKD1 gene are associated with more than 85% of the cases of ADPKD. The proposed mutagenic mechanism involves recombination, such as gene conversion, to repair doublestrand breaks following replication fork blockade. Rossetti screened for mutations in 131 unrelated patients with ADPKD and found that mutations were more than twice as frequent in the 3' half of the gene compared with the 5' half. Furthermore the mutation frequency was higher on either side of the

peculiar polypurine-polypyrimidine tract in intron 21 than at the 5' end of the gene. They suggested that the tract may have a long-range effect that was reflected in the higher rate of mutation overall in the 3' end of the gene. Besides finding more mutations in the vicinity of the PKD1 polypurine-polypyrimidine tracts, Rossetti also found several mutations that could be attributed to germline gene conversion.

Somatic mutations may also implicate the polypurine-polypyrimidine tract. Cystogenesis in ADPKD is thought to result from the inheritance of a dysfunctional PKD1 allele coupled with a somatic mutation in the PKD1 allele from the non-affected parent occurring in the renal tubular cell. Such somatic mutations can be multiple and reflect possible gene conversion as a mechanism of somatic mutagenesis. Gene conversion uses homologous sequences as targets to facilitate repair. Watnick surmised from their mutational analysis of cystic tissue that gene conversions occurred and that the templating occurred at 16p13.1 49. This region has more than three transcriptionally active homologous (HG) genes with significant homology to PKD1, and the mutations in the PKD1 gene in the cysts identical to the sequence from one of the centromeric homologous genes. These mutations suggested that gene conversion had rendered the functional PKD1 gene mutant, allowing for cysts to develop. Gene conversion notwithstanding, many lines of evidence support the two hit hypothesis for ADPKD. Analysis of the cells lining the cystic dilatation revealed only the mutant allele for PKD1, precluding the ability of such cells to correctly synthesize polycystin-1. Such somatic mutagenesis leading to polycystic disease is also seen in PKD1 knockout mice. Two of eight mice heterozygous at the PKD1 locus by gene knock out, developed significant renal cystic disease with cysts devoid of polycystin-1 by immunostaining, indicating that the wild-type allele had undergone a somatic mutation.

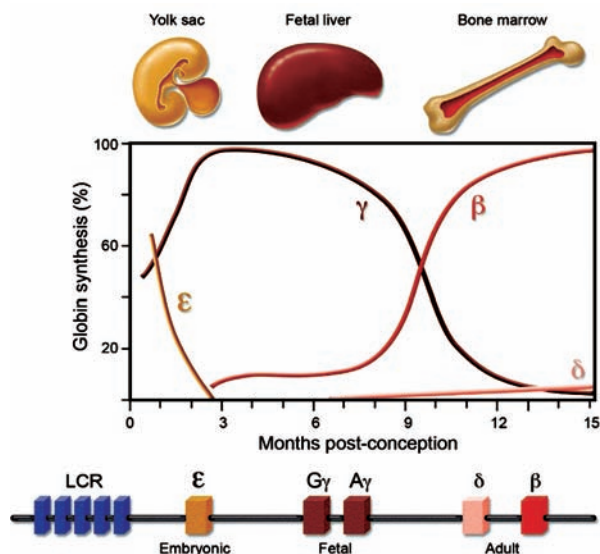
Analysis of physical and recombinational distances of 16p13 also point to a recombination as a cause for mutation in this region. The HG genes have ~97% homology to the first 34 exons of PKD1 and containing the polypurine-polypyrimidine tract. Callen demonstrated that this region is a hot-spot for meiotic recombination by comparing the genetic distance with the physical map. One interpretation of the Callen work is that the polypurine-polypyrimidine tract promotes meiotic recombination

because these sequences block replication and induce recombination. Because this effect is sequence specific, it may, likewise, also promote mitotic recombination.

DNA that can form triplex structures such as the PKD1 polypurine-polypyrimidine tract, is associated with blockade of the mammalian replication fork and may lead to mutations in a mouse model. Such polypurine-polypyrimidine tracts block DNA replication by forming a triplex structure between the mirror repeat template strand and the nascent strand that sequesters the 3' end of the nascent DNA leading strand and prevents fork extension. We demonstrated that the PKD1 gene contains polypurine-polypyrimidine tracts that form triplex structures, and strongly interfere with replication in prokaryotic models and in human replication systems. Bacolla studied the stability of the PKD1 polypurine-polypyrimidine tract in a bacterial model and concluded that the alternative DNA conformations could also trigger genomic rearrangements through recombination-repair activities.

Hereditary persistence of fetal hemoglobin

A myriad of physiological and gene expression changes occur to accommodate extrauterine survival, including down regulation of fetal genes. Patients who are homozygous or compound heterozygous for the β -globin allelic variant (Hb S) exhibit sickle cell disease. Symptoms are most often identified within the first six months of life, but there is considerable variability in the disease. The patient's genotype is the major determinant for disease severity, and patients with Hb SS are most severely affected. Several therapies for sickle cell disease, for example hydroxyurea, function by increasing the levels of fetal hemoglobin (Hb F) in affected individuals. Increased expression of Hb F ameliorates the disease phenotype. Some patients have a genetic predisposition to unusually high levels of Hb F due to the failure to down-regulate the γ -globin chains. This condition, called hereditary persistence of fetal hemoglobin, occurs in 1/188,000 patients with sickle cell disease, and these patients have a less severe disease.



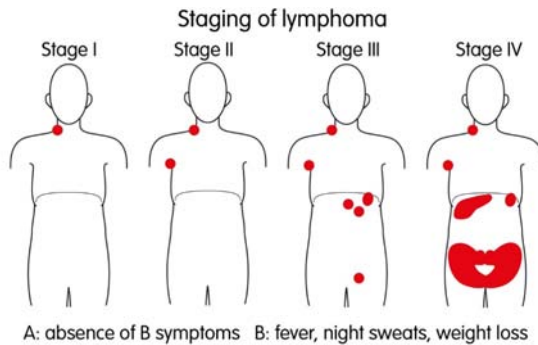
O'Neill used S1 nuclease to map a 250 base pair polypurine-polypyrimidine tract upstream from the human δ -globin gene. Ulrich characterized the secondary DNA structure that lies upstream from the $A\gamma$ and $G\gamma$ -globin genes. They identified that four out of five HPFH mutations destabilize triplex formation using S1 nuclease and oligonucleotides to map the structure. Structure dependence of the sequence was already well established, but this was the first example of such point mutations effecting disrupting triplex structure that also had a biologic outcome effecting human health. Bacolla used **osmium tetroxide**, chloroacetaldehyde, and diethyl pyrocarbonate to more clearly map out the triplex structures that could form and again noted the influence of negative supercoiling, low pH, and magnesium ions on structure formation.

Osmium tetroxide is an osmium coordination entity consisting of four oxygen atoms bound to a central osmium atom via covalent double bonds.

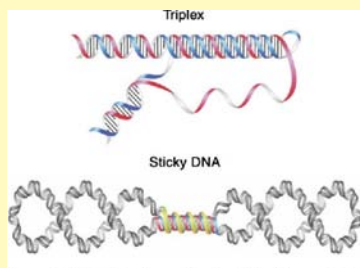
Lymphoma

Certain sequences adversely affect DNA replication, repair, or recombination and may lead to mutations such as translocations. The most common chromosomal translocation in human cancer is the translocation

that occurs in almost all follicular lymphomas. This translocation appears to result from recombination between the *bcl-2* gene and V(D)J subexons during V(D)J recombination and juxtaposes the *bcl-2* locus on chromosome 18 to the intron enhancer of the immunoglobulin heavy chain locus on chromosome 14. This positioning leads to over expression of the antiapoptotic protein, Bcl-2, and hence, to follicular lymphoma. In most follicular lymphoma patients, the break at chromosome 18 occurs within a small 150-bp region designated as the major breakpoint region (Mbr).



Within this 150-bp Mbr region, there are three translocation hotspots. The precision and distribution profile within the *bcl-2* Mbr strongly suggest a DNA structural effect. Recently, the *bcl-2* Mbr was identified as being polypurine·polypyrimidine in nature and possibly capable of forming triplex DNA structures. The breaks at the V(D)J sites on chromosome 14 involve cleavage at a pair of signal sequences (12- or 23- signal) by the RAG complex (RAG1, RAG2, and HMG1).



CORE CONCEPT: FORMATION OF STABLE DNA TRIPLEXES

DNA triple helices offer new perspectives towards oligonucleotide-directed gene regulation. Triple helix forming oligonucleotides, which bind to double-stranded DNA, are of special interest since they are targeted to the gene itself rather than to its mRNA product (as in the antisense strategy). However, the poor stability of some of these structures might limit their use under physiological conditions. Specific ligands can intercalate into DNA triple helices and stabilize them. This review summarizes recent advances in this field while also highlighting major obstacles that remain to be overcome, before the application of triplex technology to therapeutic gene repair can be achieved.

Triple-helical nucleic acids can be generated by binding an oligonucleotide within the major groove of duplex DNA and are stabilized by the formation of hydrogen bonds between the third-strand bases and exposed groups on the purine strand of duplex DNA. The third strand can be oriented in either the parallel or anti-parallel direction, relative to the purine strand, although parallel triplexes are usually more stable, especially at low pH. In the parallel motif, GC is recognized by protonated cytosine, while AT is bound by thymine, generating C+.GC and T.AT triplets. These structures, which were first noted with polynucleotides in 1957, and subsequently with TFOs (triplex-forming oligonucleotides) targeted to sites within duplex DNA in 1987, offer the possibility of designing oligonucleotides which bind to duplex DNA with considerable sequence specificity. However, triplehelix formation with natural nucleotides is limited by (i) the requirement for low pH (necessary for formation of the C+.GC triplet), (ii) the requirement for homopurine target sequences, and (iii) the relatively low affinity due to charge repulsion between the polyanionic strands. We have prepared heavily modified oligonucleotides (while retaining the phosphodiester

backbone) in order to overcome these limitations, which include the addition of positive charges to the sugar and/or base, the inclusion of cytosine analogues, the development of nucleosides for recognition of pyrimidine interruptions and the attachment of one or more cross-linking groups (psoralen). By these means we are able to generate triplexes, which have high affinities at physiological pH at sequences that contain pyrimidine interruptions.

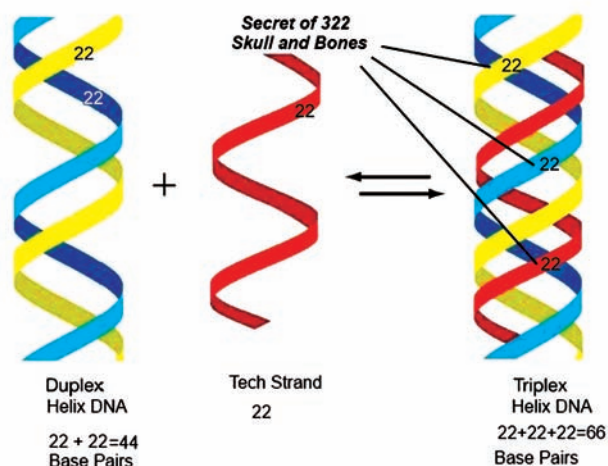
Formation of triple helix DNA

Triple helix formation is a result of oligonucleotides binding with a high specificity of recognition to the major groove of double helical DNA by forming Hoogsteen type bonds with purine bases of the Watson-Crick base pairs, the compound rationally designed for artificial regulation of gene expression. Triplex formation requires Mg^{2+} ions whereas it is inhibited by K^+ ions.

In the triple helix or antigene strategy, the oligonucleotide binds in the major groove of double-stranded DNA via Hoogsteen hydrogen bonding to form a triple helix. TFOs bind homopurine-homopyrimidine sequences in double-stranded DNA. There are four structural motifs for triplex formation that have been described based on the third-strand composition and its orientation relative to the purine-rich strand of the duplex. Purine motif TFOs (those that are comprised of G and A) form $G^*G:C$ and $A^*A:T$ triplets and bind in antiparallel orientation with regard to the purine strand of the duplex. On the other hand, pyrimidine motif TFOs (C/T) form triplexes in parallel orientation and generally only at low pH due to the necessity of the cytosine bases being protonated in order to form Hoogsteen bonds; they form $C^+*G:C$ and $T^*A:T$ triplets. Finally, mixed purine and pyrimidine TFOs bind in either parallel or antiparallel orientation and form $G^*G:C$ and $T^*A:T$ triplets. The orientation in which the mixed motif TFOs bind, is dependent upon the number of GpA and ApG steps in the homopurine tract. The antiparallel orientation is favored by a greater number of steps, while a low number of steps favor the parallel orientation.

Once the best motif for binding a particular target sequence is established, problems with natural phosphodiester oligonucleotides limit the success of the antigene approach and the therapeutic applications

of oligonucleotides in general. Oligonucleotides with the natural phosphodiester backbone are susceptible to endo and exonucleases. The predominant activity that degrades oligonucleotides is 3'-exonuclease activity, but endonuclease activity has also been observed in some settings. Thus, for application as therapeutics *in vivo* TFOs must be able to resist both exonuclease and endonuclease activity in order to reach their target. A backbone modification that confers nuclease resistance but allows binding to double-stranded DNA with high affinity is required for the *in vivo* applications of TFOs.



Phosphorodiamidate morpholino oligomers are modified backbone oligonucleotides that have been investigated as antisense agents. Morpholino oligonucleotides have an uncharged backbone in which the deoxyribose sugar of DNA is replaced by a six-membered ring and the phosphodiester linkage is replaced by a phosphorodiamidate linkage (**fig. 3**). Morpholino oligonucleotides are resistant to enzymatic degradation and appear to function as antisense agents by arresting translation or interfering with pre-mRNA splicing rather than by activating RNase H. They have been successfully delivered to tissue culture cells by methods that physically disrupt the cell membrane, and one study comparing several of these methods found that scrape-loading was the most efficient method of delivery; however, because the morpholino backbone is uncharged, cationic lipids are not effective mediators of morpholino oligonucleotide uptake in cells. A recent report demonstrated triplex formation by a morpholino oligonucleotide and, because of the non-ionic

backbone, these studies showed that the morpholino oligonucleotide was capable of triplex formation in the absence of magnesium.

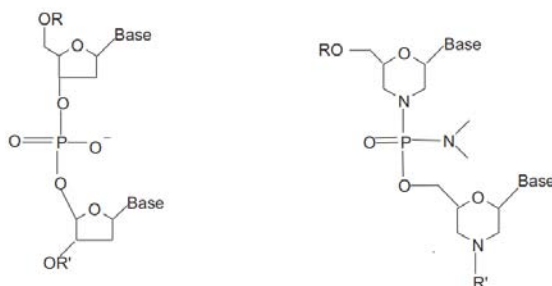


Figure 3: Structural presentation of phosphodiester DNA and morpholino

Cations have been shown to play an important role in triple helix formation. When phosphodiester oligonucleotides are used as TFOs, magnesium is generally required for triplex formation with purine and mixed motif TFOs; it also speeds the reaction and stabilizes the triplex formed with pyrimidine motif TFOs. Other divalent cations have been shown to function in the same capacity as magnesium with regard to triplex formation. Magnesium occurs at a concentration of ~ 0.8 mM in the cell and ~ 1.5 mM in the blood, but most *in vitro* triplex reactions are performed in 5–10 mM MgCl_2 . Potassium occurs in the cell at a concentration of ~ 140 mM, and at 4 mM in the blood. High concentrations of potassium can inhibit triplex formation with guanine-rich oligonucleotides designed as TFOs by favoring other secondary DNA structures, such as dimers and quadruplexes. It will be necessary to overcome the limited ability of phosphodiester TFOs to form a triplex in low magnesium and high potassium for them to be effective under physiological conditions. In a recent study of morpholino TFOs, triplex formation was demonstrated in the absence of magnesium and in the presence of potassium. These properties make morpholino TFOs good candidates for further study as antigene therapeutics.

Molecular modeling

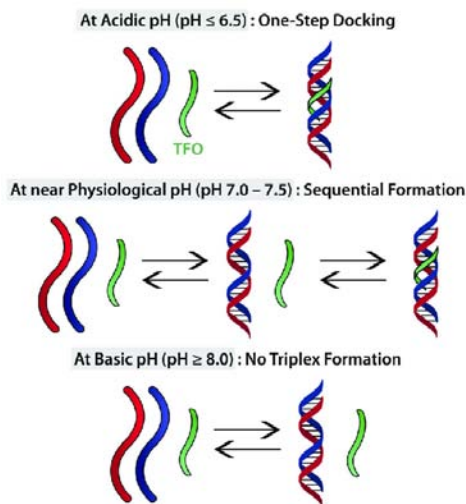
DNA triplex structure can be constructed by molecular modeling techniques by using coordinates that correctly take into account the sugar

conformation of (T,C)-motif triple helices. This structure is closer to a B-form DNA as reported by NMR studies as compared to the structure proposed, based on fiber X-ray diffraction. The JUMNA program allows constructing DNA structures according to their helical parameters. An intercalation site can be easily created in the triplex by doubling the rise parameter for two adjacent T·A×T base triplets (rise = 6.8 Å), and subsequently decreasing the twist parameter between these two triplets from 34° to 16° to reduce bond distance constraints.

Using molecular modeling, one can demonstrate the possibility of forming a parallel triple helix in which the single strand interacts with the intact duplex in the minor groove, via novel base interactions.

JUMNA uses a mixture of helical and internal coordinates (valence and dihedral angles) to describe nucleic acid flexibility. The helical parameters position each 3' monophosphate nucleotide with respect to a fixed-axis system. Junctions between successive nucleotides are maintained with quadratic restraints on the O5'-C5' distances. In addition to a reduced number of variables with respect to Cartesian coordinate programs, the choice of physically meaningful variables allows large, concerted conformational moves during minimization, together with an efficient control of the structure and easy introduction of constraints or restraints. Available tools include both adiabatic mapping and combinatorial searches with respect to chosen structural parameters. Particularities of the Flex force field include the presence of a specific term to account for the angular dependence of hydrogen bonding and the possibility of electrostatic energy screening with a sigmoidal dielectric function, $\epsilon(R) = D - (D - D_0)/2[(RS)^2 + 2RS + 2] \exp(-RS)$, where R is the distance between two charges. The slope S , the plateau value at long distance D , and the initial value D_0 of the function are adjustable, with default values of 0.16, 80, and 1, respectively, using two assumptions already employed for constructing the minor-groove triple helix. First, the base triplets are restrained to be coplanar to avoid any possible interbase triplet interactions. Such interactions easily form during the construction of stretched helices but cannot play a role in recognition or strand exchange, since these processes are independent of the overall sequence. It has been checked out that the optimized structure of the minor-groove triplex is independent of these restraints. The second assumption, in line with the stoichiometry of RecA/ DNA complexes, which shows three nucleotides

per RecA monomer, is the use of trinucleotide helical symmetry. For this reason preliminary studies have been limited to sequences with trinucleotide repeat.



Specific restraints or constraints are needed for triplex construction and manipulation. These include the “plateau” restraints and the trinucleotide symmetry constraints. The “plateau” restraint maintains the co-planarity of the bases forming a triplet, while allowing the rotations and displacements required for base pair switching. The trinucleotide symmetry constraint implies the equivalence of the variables describing each successive group of three nucleotides. Stretching dsDNA, so that the twist decreases and the minor groove opens have been achieved by restraining the distance between the terminal O3' atoms of the trinucleotide symmetry unit. This restraint has been slightly modified because the O3'-O3' distance can be altered by a lateral displacement of the backbones during strand exchange. In a recent work, only the component of the O3'-O3' vector parallel to the helix axis was restrained. The restraints on the groove width, calibrated with the help of numerical Poisson-Boltzmann electrostatic calculations, were used to avoid groove narrowing due to the lack of explicit solvent molecules.

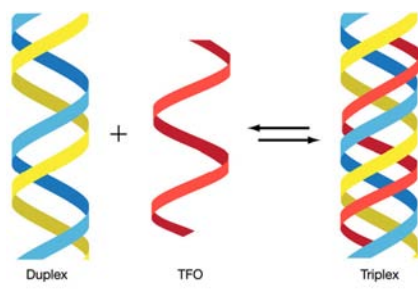
Base pair switching is studied by base rotation, using the approach defined by Bernet. This involves a restraint applied to the angle θ between the glycosidic bond (purine: C1'-N9 or pyrimidine: C1'-N1) and the vector joining the two C1' atoms of a base pair, projected on

the plane perpendicular to a local helical axis. θ has a value of 55° in canonical B-DNA. Modeling base pair switching for a chosen base involves an adiabatic variation of θ from 65° to 10° by steps of 2° while maintaining both “plateau” and stretching restraints.

Obstacles and limitations encountered in triple helix formation

Biological applications of TFOs are compromised by fundamental biophysical considerations, as well as limitations imposed by physiological conditions. Triplex formation involves the approach and binding of a negatively charged third strand to a double-negatively charged duplex. **Neutralization** of charge repulsion is typically provided experimentally by levels of Mg^{++} (5-10 mM) that are much higher than what is thought to be available in cells. Furthermore, triplex formation involves conformational changes on the part of the third strand, and some distortion of the underlying duplex. Pyrimidine motif triplexes are unstable at physiological pH because of the requirement for cytosine protonation that occurs at relatively acidic pH ($pK_a = 4.5$). This is necessary for the second Hoogsteen hydrogen bond, although the resultant positive charge apparently makes the more important contribution to triplex stability.

Neutralization is a chemical reaction in which an acid and a base react quantitatively with each other.



Pyrimidine motif triplexes containing adjacent cytosines are often less stable than those with isolated

cytosines. Traditionally this has been ascribed to charge-charge repulsion effects, although a recent study suggests incomplete protonation of adjacent cytosines may be the critical factor. In addition, purine motif third strands (which are G rich) may form G tetrads in physiological levels of K^+ , which inhibit triplex formation. All these factors impose kinetic barriers on triplex formation and reduce the stability of triplexes once formed (most triplexes, even under optimal conditions *in vitro*, are less stable than the underlying duplex).

Increasing affinity

Nucleic acids triple-helices are usually much less stable than their duplex counterparts on account of their slow rate of formation (typically three orders of magnitude slower than duplexes) and electrostatic repulsion between the three polyanionic strands. The T·AT triplet is less stable than C⁺·GC and requires millimolar concentrations of magnesium for effective formation. Several strategies have been employed to enhance the stability of this triplet, including increasing the aromatic surface area and altering the backbone conformation to N3–P5, PNA (peptide nucleic acid) or LNA (locked nucleic acid) (third strands with N-type sugar conformations are more stable than S-type and RNA-like third strands bind well to duplex DNA targets). We have sought to increase stability by the addition of positively charged groups to the sugar, the base or both. 5-Propargylamino-dU (Figure 4B) produces a large increase in affinity ($T_m \approx 4^\circ\text{C}$ per substitution), similar to that of 2'-aminoethoxy-T, while combining both modifications in the analogue BAU (bisamino-U) produces a dramatic increase in stability with an increase in T_m of up to 8°C per substitution. The additive effect of these two substitutions occurs because they contact different phosphates; the 2'-aminoethoxy group interacts with the adjacent phosphate in the duplex purine strand, whereas the 5-substitution binds to a phosphate in the TFO itself. Triplexes formed with these charged analogues no longer require the addition of divalent cations.

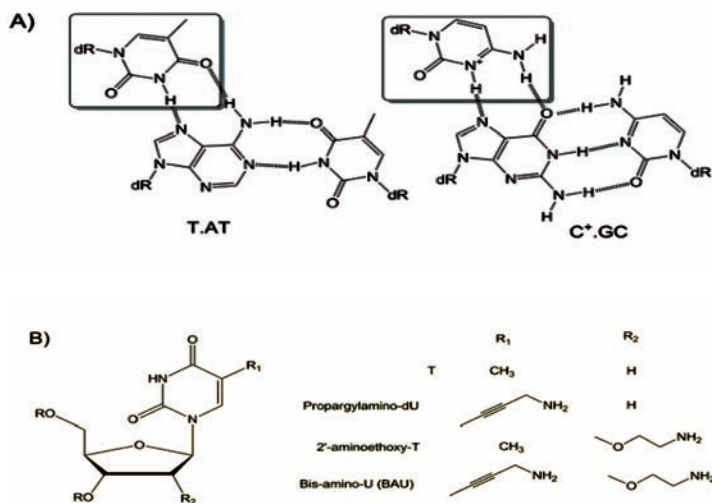


Figure 4. Chemical structures of base triplets. (A) T.AT and C+.GC triplets. The third-strand base is shown highlighted by the rectangle. (B) T, propargylamino-U, 2'-aminoethoxy-T and BAU.

Inclusion of BAU has a large effect on triplex kinetics, and melting curves with TFOs that contain only a single substitution show considerable hysteresis, even at slow rates of heating and cooling. Analysis of the hysteresis profiles reveals that even a single substitution with this analogue within an 18-mer TFO produces a 10-fold decrease in the rate of dissociation, with little change in the association rate. Inclusion of a second BAU residue further enhances the stability and reduces the rate of dissociation. In this instance, the effect is greatest when the two modified nucleotides are closer together. The selectivity of BAU for AT base pairs is also reflected in a much slower rate of dissociation from this compared with other base pairs.

We have used BAU in combination with other derivatives, described below, to achieve triplex recognition of mixed DNA sequences at physiological pH and it is the most stabilizing derivative that we have developed. However, synthesis of this phosphoramidite is not trivial and, as far as we are aware, this analogue has not subsequently been used by other groups. In order to simplify the synthesis we have examined the effects of dimethylaminopropargyl dU on triplex stability (since this does

Oligonucleotides

are short DNA or RNA molecules, oligomers, that have a wide range of applications in genetic testing, research, and forensics.

not need protecting groups during **oligonucleotide** synthesis and it can be used with other derivatives that require mild deprotection conditions). This derivative still enhances triplex stability, although to a lesser extent than other analogues that bear positive charges. This represents a useful nucleotide analogue that combines the ease of synthesis with triplex stabilization.

For oligonucleotides that contain multiple substitutions with these analogues, their positions and arrangement within the TFO are important. For multiple additions of BAU, clustering these towards one end of the TFO produced a much smaller increase in affinity than when they were evenly distributed throughout the oligonucleotide. A similar effect has been noted with the dimethylaminopropargyl derivative, although in this case the lower stabilization of adjacent residues may be the result of steric clash between methyl groups on adjacent residues.

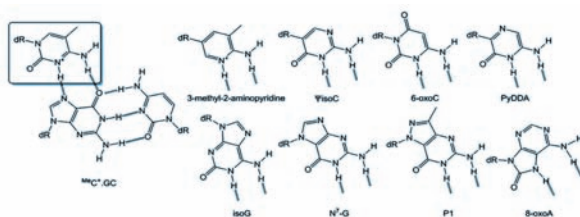


Figure 5. Third-strand bases for recognizing GC base pairs. The $\text{MeC}^+ \cdot \text{GC}$ triplet is shown on the left with the third-strand base highlighted in the rectangle. The other structures show a selection of base analogues that have been used for recognition of G. The upper row shows those based on a pyrimidine nucleus: methylaminopyridine, ψ -isoC, 6-oxoC, pyDDA; whereas the lower row is based on a purine nucleus: isoG, N⁷-G, P1 and 8-oxoA.

In some instances, these modifications produce triplexes that are now more stable than the underlying duplex and the apparent T_m in melting experiments is limited by the duplex stability. In these instances,

when working with short duplex targets, the apparent triplex stability depends on the sequence and length of the flanking duplex sequences. For a 14-mer parallel TFO, the T_m at pH 5.0 was 10°C lower with an intermolecular 14-mer duplex target than with an intramolecular duplex, or one that was flanked by six GC base pairs at either end.

The addition of intercalating or cross-linking groups to TFOs has also been used to increase TFO affinity. These groups are usually attached to the 5'-end of the TFO, although they can also be added to the 3'- or both ends, generating a triplex staple. Intercalating compounds such as acridine or anthraquinone can be attached at either or both ends of the oligonucleotide. Addition of intercalators at both 3'- and 5'-ends of an oligonucleotide can produce triplexes with melting temperatures above 40°C at pH 7 even though they do not contain any triplex-stabilizing base analogues. It has been suggested that 5'-attachment of an intercalators enhances stability more than 3'-attachment, but this may simply reflect the observation that intercalation at YpR is favoured over RpY. Psoralen can also be attached at either end or at an internal site of a TFO and UV irradiation of these complexes produces interstrand cross-links at adjacent TpA steps. Photoreaction of TFOs containing two psoralens, located at the 5'- and 3'-ends, has been used to create crosslinks at both termini of the TFO (triplex staples), producing complexes that have no free single-stranded ends [28].

The use of simple intermolecular duplex targets may therefore underestimate the triplex stabilization that is produced by some nucleotide analogues. Target sites that are contained within longer, more stable, duplexes will also more closely reflect the situation *in vivo* when the targets will be flanked by the remainder of the genomic DNA.

Applications of Triple Helix DNA

The formation of intermolecular DNA triple helices offers the possibility of designing compounds with extensive sequence recognition properties, which may be useful as antigene agents or tools in molecular biology. During the past decade, a new approach using DNA analogues, as therapeutic agents, is emerging in medicinal chemistry. This is based on regulating expression of genes of disease-related proteins/enzymes

by blocking their transcription (antigene) or translation (antisense) (**fig. 6**). It is affected through sequence-specific binding of complementary oligonucleotides to either DNA duplex via triplex formation to inhibit production of mRNA or interfere in the translation of the latter to proteins. Since oligonucleotides do not enter cells easily and are amenable to destruction by cellular nucleases, a variety of chemically modified analogues of oligonucleotides are being designed, synthesized and evaluated for development as therapeutic agents.

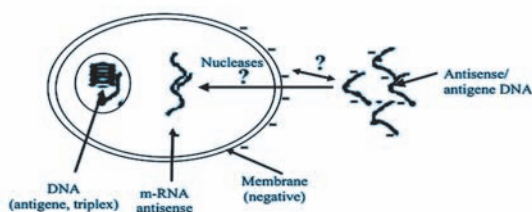
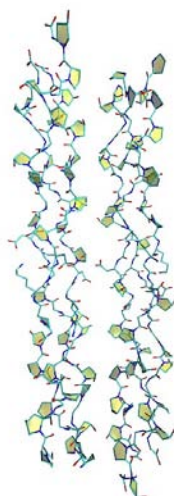


Figure 6. Principles of antigene and antisense therapeutics

The specific recognition of homopurine-homo pyrimidine regions in duplex DNA by triplex-forming oligonucleotides (TFOs) provides an attractive strategy for genetic manipulation, with the ultimate goal of repairing genetic defects in human cells. The ability to target mutations may prove useful, as a tool for studying DNA repair, and as a technique for gene therapy and genetic engineering.



Efficient tools based on triple helices were developed for various biochemical applications such as the development of highly specific artificial nucleases. The antigene strategy remains one of the most fascinating fields of triplex application to selectively control gene expression (**Table 2**). Targeting of genomic sequences is now proved to be a valuable concept on a still limited number of studies; local mutagenesis is in this respect an interesting application of triplex-forming oligonucleotides, on cell cultures.

Table 2: Antigene Nucleic Acid Studies within Eukaryotic Cells⁸⁰

Target gene	Cell line	Oligomer size, and modifications
Transfected genes	HSB2 cells (T-cell)	15-mer acridine
CAT gene/ IL -2R promoter	HeLa cells	orporsalen linked
CAT gene/(6-16) IRE	cv-1 cells	21-mer
CAT gene/tk promoter		38-mer, cholesterol
PRE upstream		
Endogenous genes		
IL-2R	Human lymphocytes	28-mer
c-myc	HeLa cells	27-mer
SV 40 T Ag	Tsa 8 cells	15,20-mer PNAs
Antivirals		
SV 40	CV-1 MT4	8-mer, acridine
HIV-1		31,38-mers

Antigene strategies focus primarily on gene targeting by homologous recombination or by triple helix-forming oligodeoxynucleotides. For many technical reasons, including very limited gene accessibility within the highly malariacompacted, protein-wrapped chromosomal structure, the clinical application of these methods has not progressed rapidly. Kielkopf et al have recently described an alternative approach, using polyamides that can diffuse into the nucleus and recognize specific DNA sequences. Although very exciting, this methodology is still in its infancy and its ultimate clinical utility remains unknown.

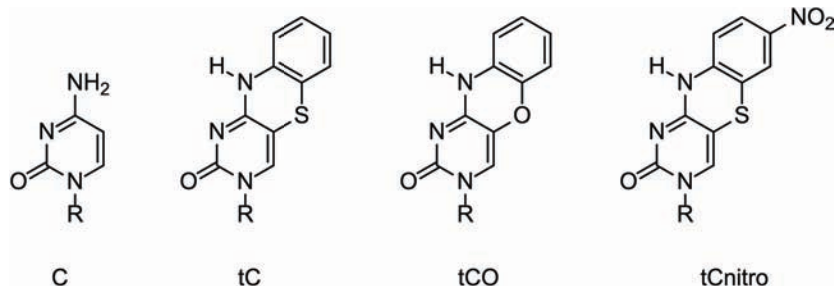
Secondary Binding Sites

We have examined whether the addition of positive charges affects the stringency of triple-helix formation. TFOs with single substitutions of BAU bind best when this base is placed opposite AT as expected. The

next best combination is BAU.GC, and although this is more stable than T.GC, the discrimination between AT and GC is not affected. Triplets with BAU placed opposite TA or CG are not stabilized relative to T.CG and T.TA, demonstrating that BAU has enhanced discrimination for AT (and GC) over CG and TA relative to T. Oligonucleotides that contain multiple substitutions with these charged analogues do show some weaker interaction with secondary sites. In particular, TFOs that contain $(\text{BAU})_n$ tracts can cause the oligonucleotide to bind to A_nT_n tracts even though the surrounding duplex sequence does not match the remainder of the TFO sequence.

Cytosine analogues

The pH-dependence of the C+.GC triplet presents a major limitation to triplex formation at physiological conditions, which is especially pronounced when recognizing adjacent GC base pairs. A large number of cytosine analogues with elevated pKa values have been developed to overcome this problem. Early studies used 5-methylcytosine, which has a slightly higher pKa than C, although the improved binding probably arises from the stacking of the additional methyl groups within the major groove (in the same way as T.AT is more stable than U.AT). A selection of analogues that have been examined is presented in Figure 5. The ones that are based on the pyrimidine nucleus are generally isostructural with C and form triplets that are isomorphic with T.AT. Of these analogues, only 2-aminopyridine [31–33] retains the positive charge (the nucleoside has a pKa of 5.93 compared with 4.3 for dC) and, unlike the others, it produces triplexes that are as stable as those formed with C at low pH.



Those lacking the stabilizing positive charge may need to be used in combination with other stabilizing analogues, such as 2-aminoethoxy derivatives. For the analogues that are based on a purine nucleus the

glycosidic bond is in a different position, causing distortions in the backbone between X.GC and T.AT triplets. These are therefore best used for recognizing longer G-tracts in a pH-independent fashion. The affinity of C for GC has also been enhanced by appending positively charged groups such as spermine to the base or by attaching it to 2'-OMe (2'-O-methyl) groups or 2'-aminoethoxyribose sugars. These modifications compensate for the lower affinity of unprotonated cytosine. Further increases in stability, extending triplex formation to neutral pH, can be produced by combining these analogues with other base analogues such as 5-propynyl-dU or BAU. We have successfully used 2-aminopyridine (and its 3-methyl derivative) together with modification at the 2'-position of the sugar to produce triplexes that are stable at higher pHs (C. Lou, K.R. Fox and T. Brown, unpublished work). When combined with BAU, this can be used to produce triplexes that form at physiological pH with submicromolar dissociation constants.

Pyrimidine recognition

Since the third strand only makes contacts with the purine base of each base pair, triple-helix formation is generally restricted to homopurine sequences; even single pyrimidine interruptions can abolish triplex formation. Targeting pyrimidine bases is not simple as they present fewer hydrogen-bonding points than purines, and there is some steric interference from the 5-methyl group of thymine. A number of synthetic nucleotides have been developed to overcome this restriction and many of these make hydrogen-bond contacts with both bases of the YR base pair. The synthetic nucleobase S (Figure 7) and its 2'-aminoethoxy derivative is one of the most effective for recognition of TpA, although this has low-sequence discrimination and also binds well to CG, GC and AT. Although it has lower selectivity than formation of the G.TA triplet (the best combination of natural nucleotides for recognition of TA) it binds across this pyrimidine interruption better than any other combination. Indeed, this nucleobase can also be used for producing stable complexes across CG interruptions. We have developed a series of synthetic nucleobases, based on substituted 3H-pyrrolo[2,3-d]pyrimidin-2(7H)-one nucleosides, for recognizing CG inversions (see Figure 7). These generally give good selectivity and affinity, but the relative stability can be affected by the surrounding sequence.

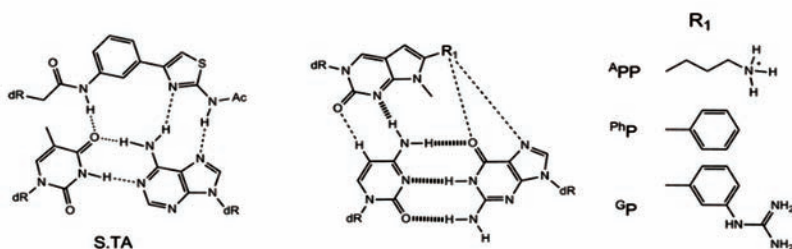


Figure 7. Triplets for recognition of pyrimidine interruptions. The S.TA triplet is shown on the left [38,39]. Right: structure of substituted 3H-pyrrolo[2,3-d]pyrimidin-2(7H)-one nucleosides for recognition of CG [40–42].

Mixed-sequence recognition

One goal of triplex technology is to be able to target any sequence with high affinity at physiological pH. The development of derivatives of C and T for improved recognition of AT and GC means that we are now able to generate high-affinity triplexes at most oligopurine sites at physiological pH. However, the presence of pyrimidine interruptions in the target still presents a challenge. Targets that contain single YR interruptions can be efficiently targeted by using the S or ^ΔPP derivatives in combination with the high-affinity T and C derivatives. This strategy can produce triplexes with low nanomolar dissociation constants at physiological pH. TFOs for targets that contain several pyrimidine interruptions have lower affinity. However, by the combined use of BAU (for strong binding to AT), aminopyridine (for the recognition of GC at high pH), S (for recognizing TA, albeit with low selectivity) and ^ΔPP (for recognition of CG) we have successfully targeted a 19-mer oligopurine sequence containing two CG and two TA interruptions, with micromolar affinity at physiological pH.

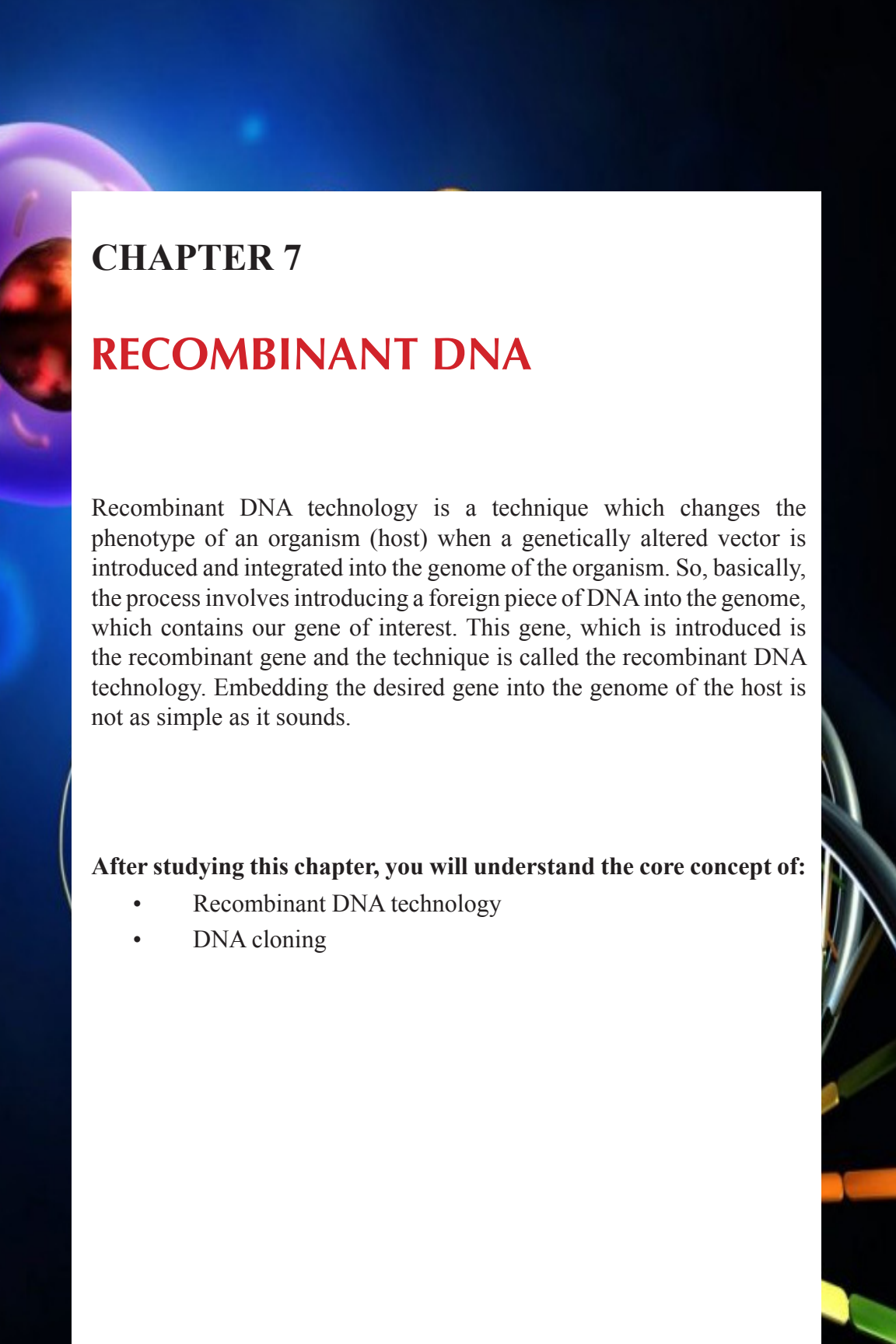
REVIEW QUESTIONS

1. Define the identification process of triplex DNA.
2. What is the role of mirror repeats in disease?
3. Explain the formation strategy of triple helix DNA.
4. Write the applications of triple helix DNA.
5. Discuss on triplex DNA pyrimidine recognition.

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CHAPTER 7

RECOMBINANT DNA

Recombinant DNA technology is a technique which changes the phenotype of an organism (host) when a genetically altered vector is introduced and integrated into the genome of the organism. So, basically, the process involves introducing a foreign piece of DNA into the genome, which contains our gene of interest. This gene, which is introduced is the recombinant gene and the technique is called the recombinant DNA technology. Embedding the desired gene into the genome of the host is not as simple as it sounds.

After studying this chapter, you will understand the core concept of:

- Recombinant DNA technology
- DNA cloning

INTRODUCTION

Recombinant DNA, molecules of DNA from two different species that are inserted into a host organism to produce new genetic combinations that are of value to science, medicine, agriculture, and industry. Since the focus of all genetics is the gene, the fundamental goal of laboratory geneticists is to isolate, characterize, and manipulate genes. Although it is relatively easy to isolate a sample of DNA from a collection of cells, finding a specific gene within this DNA sample can be compared to finding a needle in a haystack. Consider the fact that each human cell contains approximately 2 metres (6 feet) of DNA. Therefore, a small tissue sample will contain many kilometres of DNA. However, recombinant DNA technology has made it possible to isolate one gene or any other segment of DNA, enabling researchers to determine its nucleotide sequence, study its transcripts, mutate it in highly specific ways, and reinsert the modified sequence into a living organism.



CORE CONCEPT: RECOMBINANT DNA TECHNOLOGY

Recombinant DNA (rDNA) technology refers to the process of joining DNA molecules from two different sources and inserting them into a host organism, to generate products for human use. Can you put the DNA molecules in the host organism first and then cut and join them? No! This process involves multiple steps that have to proceed in a specific sequence to generate the desired product.

Recombinant DNA technology involves the selection of the desired gene for administration into the host followed by a selection of the perfect vector with which the gene has to be integrated and recombinant DNA formed. This recombinant DNA, then has to be introduced into the host. And at last, it has to be maintained in the host and carried forward to the offsprings.

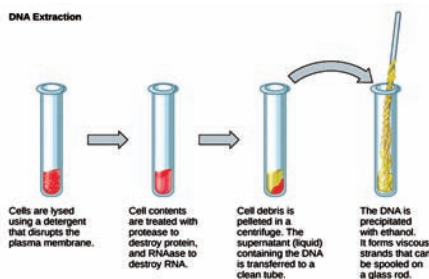
Isolation of Genetic Material

We already know that the genetic material of all living organisms is 'nucleic acid'. In most organisms, it is DNA, whereas in some it is RNA. The first step in rDNA technology is to isolate the desired DNA in its pure form i.e. free from other macromolecules.

However, in a normal cell, the DNA not only exists within the cell membrane, but is also present along with other macromolecules such as RNA, polysaccharides, proteins, and lipids. So, how do we break open the cell and obtain DNA that is free from other macromolecules? We can use the following enzymes for specific purposes:

- **Lysozyme** – to break bacterial cell wall.
- **Cellulase** – to break plant cell wall.
- **Chitinase** – to break fungal cell wall.
- **Ribonuclease** – removes RNA.
- **Protease** – removes proteins (such as histones that are associated with DNA).

Other macromolecules are removable with other enzymes or treatments. Ultimately, the addition of ethanol causes the DNA to precipitate out as fine threads. This is then spooled out to give purified DNA.

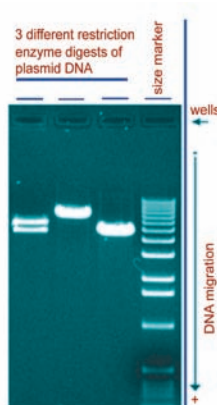


Restriction Enzyme Digestion

Restriction enzymes act as molecular scissors that cut DNA at specific locations. These reactions are called ‘restriction enzyme digestions’. They involve the incubation of the purified DNA with the selected restriction enzyme, at conditions optimal for that specific enzyme.

Agarose gel electrophoresis is a method of gel electrophoresis used in biochemistry, molecular biology, genetics, and clinical chemistry to separate a mixed population of macromolecules such as DNA or proteins in a matrix of agarose, one of the two main components of agar.

The technique – ‘**Agarose Gel Electrophoresis**’ reveals the progress of the restriction enzyme digestion. This technique involves running out the DNA on an agarose gel. On the application of current, the negatively charged DNA travels to the positive electrode and is separated out based on size. This allows us to separate and cut out the digested DNA fragments. The vector DNA is also processed using the same procedure.



Amplification Using PCR

Polymerase Chain Reaction or PCR is a method of making multiple copies of a DNA sequence using the enzyme – DNA polymerase. It helps to amplify a

single copy or a few copies of DNA into thousands to millions of copies. PCR reactions are run on 'thermal cyclers' using the following components:

- **Template** – DNA to be amplified
- **Primers** – small, chemically synthesized oligonucleotides that are complementary to a region of the DNA.
- **Enzyme** – DNA polymerase
- **Nucleotides** – needed to extend the primers by the enzyme.

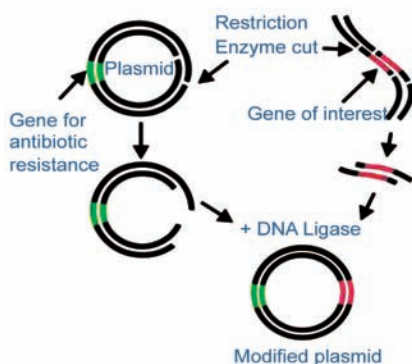


The cut fragments of DNA can be amplified using PCR and then ligated with the cut vector as explained below.

Ligation of DNA Molecules

The purified DNA and the vector of interest are cut with the same restriction enzyme. This gives us the cut fragment of DNA and the cut vector, that is now open. The process of joining these two pieces together using the enzyme '**DNA ligase**' is 'ligation'. The resulting DNA is 'recombinant DNA'.

DNA ligase is a specific type of enzyme, a ligase, (EC 6.5.1.1) that facilitates the joining of DNA strands together by catalyzing the formation of a phosphodiester bond.



Insertion of Recombinant DNA into Host

In this step, the recombinant DNA is introduced into a recipient host cell. This process is 'Transformation'. Bacterial cells do not accept foreign DNA easily. Therefore, they are treated to make them 'competent' to accept new DNA. (The topic – Tools of Biotechnology explains a few ways to make cells competent).

During transformation, if a recombinant DNA bearing a gene for ampicillin resistance is transferred into recipient *E. coli* cells, then the *E. coli* cells also become ampicillin-resistant. This aspect is useful in differentiating transformed cells from non-transformed cells.

For example, if we spread the transformed cells on agar plates containing ampicillin, only the transformed, ampicillin-resistant cells will grow while the untransformed cells will die. Therefore, in this case, the ampicillin resistance gene acts as the 'selectable marker'.

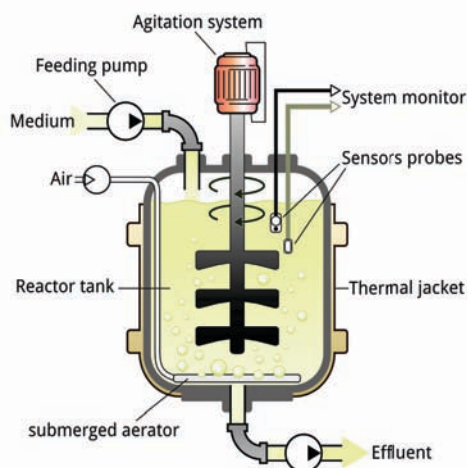
Obtaining Foreign Gene Product

The recombinant DNA multiplies in the host and is expressed as a protein, under optimal conditions. This is now a recombinant protein. Small volumes of cell cultures will not yield a large amount of recombinant protein. Therefore, large-scale production is necessary to generate products that benefit humans. For this purpose, vessels called bioreactors are used.

Bioreactors are large containers with a continuous culture system, where the fresh medium is added from one side and used medium is taken out from another side. Bioreactors can process about 100-1000 litres of cell cultures. A bioreactor provides optimum conditions (temperature, oxygen, pH, vitamins etc.) to biologically convert raw materials into specific proteins, enzymes etc.

‘Stirred-tank bioreactor’ is the most common type of bioreactor. It is usually cylindrical and has the following parts:

- Agitator system – to stir the contents evenly
- Oxygen delivery system – to introduce air into the system
- Foam control system
- Temperature control system
- pH control system
- Sampling ports – to take out small amounts of culture



Downstream Processing

Before the protein is marketed as a final product, it is subjected to downstream processing which includes:

- Separation and purification.
- Formulation with suitable preservatives.
- Clinical trials to test the efficacy and safety of the product.

- Quality control tests.

Process of Recombinant DNA Technology

Recombinant DNA technology popularly known as genetic engineering aims at synthesizing recombinant DNA also known as recombinant deoxyribonucleic acid which contains DNA from two different sources. In order to produce recombinant DNA, following are required:

If the rDNA sequences encode a gene that is expressed, then the presence of RNA and/or protein products of the recombinant gene can be detected, typically using RT-PCR or western hybridization methods

Gene of interest, which is to be cloned

- **Molecular scissors**, to cut out the gene of interest
- **Molecular carrier (vector)**, on which gene of interest could be placed
- **An expression system**, on which the gene of interest along with the vector is then introduced as a result of which a specific product is made

How to get a gene

In the process of recombinant DNA technology, first of all we shall have to get a gene which is to be cloned. There are two ways to get the gene of interest.

- *To Isolate It from The Chromosome:* Genes can be isolated from the chromosomes by cutting the chromosomes on the flanking sites of the gene using special enzymes known as restriction endonucleases.
- *Chemical Gene Synthesis by Messenger RNA (mRNA):* If the genes are small or for some reason we cannot isolate from chromosomes, they can also be synthesized chemically in the laboratory. To make a gene chemically in laboratory, messenger RNA (mRNA) is used. We shall use the **reverse transcriptase** enzyme method. Reverse transcriptase is the enzyme which forms DNA from an RNA template in reverse

transcription. This DNA molecule is called complementary DNA (cDNA) and is mainly associated with retro viruses.

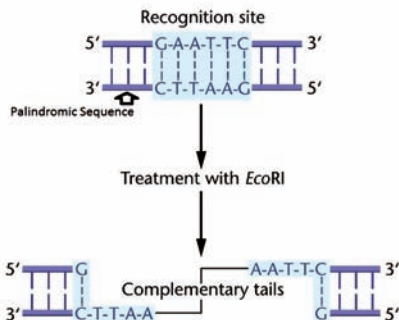
Molecular Scissors (Restriction Endonucleases)

Using of Molecular Scissors is the second step in making recombinant DNA. Molecular Scissors also known as Restriction Endonuclease is the enzyme which cleaves DNA molecules at specific base sequences producing small gene fragments by breaking internal covalent bonds linking nucleotides, used in recombinant DNA technology and chromosome mapping. They are natural enzymes of bacteria, which they use for their own protection against viruses. The restriction enzyme (Restriction Endonuclease) cuts down the viral DNA, but does no harm to the bacterial chromosome. They are called restriction enzymes or molecular scissors because they restrict the growth of viruses. In 1970, Hamilton D. Smith, at Johns Hopkins University, isolated the first restriction enzyme.

Palindromic Sequences

A palindromic sequence is the nucleic acid sequence (DNA or RNA) which reads the same in both directions. These sequences are recognition sites for enzymes. Bacteria produce a variety of such restriction enzymes, which cut the DNA at very specific sites characterized by specific sequence of four to six nucleotides arranged symmetrically in the reverse order. Such sequences are known as palindromic sequences. So far more than 400 such enzymes have been isolated out of which about 20 are frequently used in recombinant DNA technology.

Reverse transcriptase is an enzyme used to generate complementary DNA (cDNA) from an RNA template, a process termed reverse transcription.



EcoRI (*A Restriction Enzyme*): *EcoRI*, a commonly used restriction enzyme (endonuclease), cuts double stranded DNA when it has this sequence of bases at the cleavage site. Notice there is now a gap into which a piece of foreign DNA can be placed, if it ends in bases complementary to those exposed by restriction enzyme.

Sticky Ends

The single stranded but complementary ends of the two DNA molecules are called sticky ends because they can now bind by the complementary base pairing. They therefore facilitate the insertion of foreign DNA into vector DNA.

Molecular Carrier (Vector)

A vector is the mean by which recombinant DNA is introduced into a host cell. One common type of vector is plasmid. Plasmids were discovered by scientists studying the sex life of the intestinal bacterium *Escherichia coli*.

Plasmids are natural extra chromosomal circular DNA molecules which carry genes for antibiotic resistance and fertility etc. One of the plasmids discovered earlier is PSC 101 has antibiotic resistance gene for tetracycline, whereas PSR 322, has antibiotic resistance gene for tetracycline as well as ampicillin. Inserting gene of interest in tetracycline gene of plasmid (PSR 322) would enable separating out colonies of bacteria in a medium containing ampicillin and vice versa.

Recombinant DNA Preparation

For preparation of recombinant DNA, the plasmids are cut with the same molecular scissors (restriction enzymes), which was used for isolation of the gene. The gene of interest for instance insulin is then joined with the sticky ends produced after cutting the plasmids with the help of another special enzyme called DNA ligase. DNA ligase is a very important enzyme in recombinant DNA technique that facilitates the joining of DNA strands together by catalyzing the formation of a **phosphodiester bond**. It seals the foreign piece of DNA into vector. Now the two different pieces of DNA have joined together which is now known as Recombinant DNA or Chimaeric DNA.

Phosphodiester bond is the linkage between the 3' carbon atom of one sugar molecule and the 5' carbon atom of another, deoxyribose in DNA and ribose in RNA.

Expression of Recombinant DNA

A clone can be a large number of molecules (i.e. cloned genes) or cells (i.e. cloned bacteria) or organisms that are identical to an original specimen. Bacterial cells take up **recombinant plasmid**, especially if they are treated with calcium chloride to make them more permeable. Thereafter, as the cell reproduces, a bacterial clone forms and each new cell contains at least one plasmid. Therefore, each of the bacteria contains the gene of interest, which will express itself and make a product. From this bacterial clone, the cloned gene can be isolated for further analysis or protein product can be separated. Besides plasmids, the DNA of bacterial viruses (for example lambda phage) can also be used as a vector. After lambda phage it attaches to a host bacterium, recombinant DNA is released from the virus and enters the bacterium. Here it will direct the reproduction of many more viruses. Each virus in bacteriophage clone contains a copy of the gene being cloned.

Applications of Recombinant DNA Technology

The three important applications are: (1) Applications in Crop Improvement (2) Applications in Medicines and (3) Industrial Applications.

1. Applications in Crop Improvement

Genetic engineering has several potential applications in crop improvement, such as given below:

1. Distant Hybridization:

With the advancement of genetic engineering, it is now possible to transfer genes between distantly related species. The barriers of gene transfer between species or even genera have been overcome. The desirable genes can be transferred even from lower organisms to higher organisms through recombinant DNA technology.

2. Development of Transgenic Plants:

Genetically transformed plants which contain foreign genes are called transgenic plants. Resistance to diseases, insects and pests, herbicides, drought; metal toxicity tolerance; induction of male sterility for plant breeding purpose; and improvement of quality can be achieved through this recombinant DNA technology. BT-cotton, resistant to bollworms is a glaring example.

3. Development of Root Nodules in Cereal Crops:

Leguminous plants have root-nodules which contain nitrogen fixing bacteria *Rhizobium*. This bacteria converts the free atmospheric nitrogen into nitrates in the root nodules. The bacterial genes responsible for this nitrogen fixation can be transferred now to cereal crops like wheat, rice, maize, barley etc. through the techniques of genetic engineering thus making these crops too capable of fixing atmospheric nitrogen.

4. Development of C_4 Plants:

Improvement in yield can be achieved by improving the photosynthetic efficiency of crop plants. The photosynthetic rate can be increased by conversion of C_3 plants into C_4 plants, which can be achieved either

through protoplasm fusion or recombinant DNA technology C_4 plants have higher potential rate of biomass production than C_3 plants. Most C_4 plants (sorghum, sugarcane, maize, some grasses) are grown in tropical and subtropical zones.

II. Applications in Medicines

Biotechnology, especially genetic engineering plays an important role in the production of antibiotics, hormones, vaccines and interferon in the field of medicines.

1. Production of Antibiotics:

Penicillium and Streptomyces fungi are used for mass production of famous antibiotics penicillin and streptomycin. Genetically efficient strains of these fungi have been developed to greatly increase the yield of these antibiotics.

2. Production of Hormone Insulin:

Insulin, a hormone, used by diabetics, is usually extracted from pancreas of cows and pigs. This insulin is slightly different in structure from human insulin. As a result, it leads to allergic reactions in about 5% patients. Human gene for insulin production has been incorporated into bacterial DNA and such genetically engineered bacteria are used for large scale production of insulin. This insulin does not cause allergy.

3. Production of Vaccines:

Vaccines are now produced by transfer of antigen coding genes to disease causing bacteria. Such antibodies provide protection against the infection by the same bacteria or virus.

4. Production of Interferon:

Interferon's are virus-induced proteins produced by virus-infected cells. Interferon are antiviral in action and act as first line of defense against viruses causing serious infections, including breast cancer and lymph nodes malignancy. Natural interferon is produced in very small quantity from human blood cells. It is thus very costly also. It is now possible to

produce interferon by recombinant DNA technology at much cheaper rate.

5. Production of Enzymes:

Some useful enzymes can also be produced by recombinant DNA technique. For instance, enzyme urikinas, which is used to dissolve blood clots, has been produced by genetically engineered microorganisms.

6. Gene Therapy:

Genetic engineering may one day enable the medical scientists to replace the defective genes responsible for hereditary diseases (e.g., haemophilia, phenylketonuria, alkaptonuria) with normal genes. This new system of therapy is called gene therapy.

7. Solution of Disputed Parentage:

Disputed cases of parentage can now be solved most accurately by recombinant technology than by blood tests.

8. Diagnosis of Disease:

Recombinant DNA technology has provided a broad range of tools to help physicians in the diagnosis of diseases. Most of these involve the construction of probes: short Segments of single stranded DNA attached to a radioactive or fluorescent marker. Such probes are now used for identification of infectious agents, for instance, food poisoning Salmonella, Pus forming Staphylococcus, hepatitis virus, HIV, etc. By testing the DNA of prospective genetic disorder carrier parents, their genotype can be determined and their chances of producing an afflicted child can be predicted.

9. Production of Transgenic Animals:

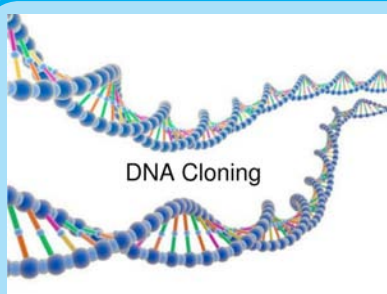
Animals which carry foreign genes are called transgenic animals.

Examples:

Cow, sheep, goat – therapeutic; human proteins in their milk. Fish like common carp, cat fish, salmon and gold fish contain human growth hormone (hGH).

III. Industrial Applications

In industries, recombinant DNA technique will help in the production of chemical compounds of commercial importance, improvement of existing fermentation processes and production of proteins from wastes. This can be achieved by developing more efficient strains of microorganisms. Specially developed microorganisms may be used even to clean up the pollutants. Thus, biotechnology, especially recombinant DNA technology has many useful applications in crop improvement, medicines and industry.

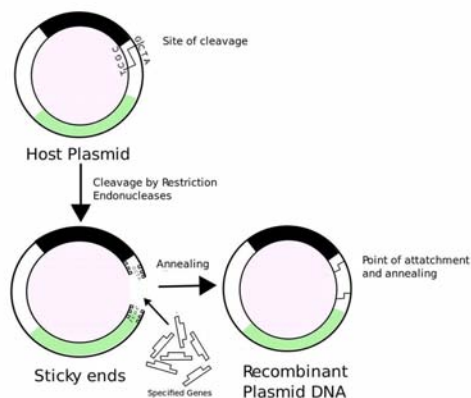


CORE CONCEPT: DNA CLONING

DNA cloning is a molecular biology technique which is used for the creation of exact replicas or clones of a particular gene or DNA using genetic engineering techniques. The DNA comprising of the

target gene(s) is fragmented using restriction enzymes. The cloning is used for making numerous identical copies of a piece of DNA, such as a gene.

A process of producing multiple and identical copies of a particular section of DNA is called as DNA cloning. During this technique, the selected DNA fragment is inserted into a plasmid (the circular piece of DNA) using enzymes. The enzymes used in this process are called as restriction enzymes and DNA ligase enzymes. These restriction enzymes, naturally found in bacteria are used to cut the DNA fragments at specific sequences and DNA ligase enzymes are used to paste or re-join the DNA fragments with complementary ends. The produced molecule of recombinant DNA is introduced into a selected and grown up bacteria. These bacteria reproduce by producing an exact copy of the plasmid and transferring it to their progeny, making copies of the DNA it contains.

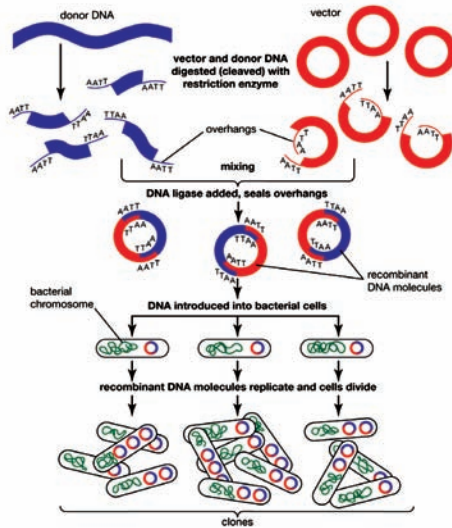


Drug resistance is the reduction in effectiveness of a medication such as an antimicrobial or an antineoplastic in treating a disease or condition.

A clone is a group of individual cells or organisms descended from one progenitor. This means that the members of a clone are genetically identical, because cell replication produces identical daughter cells each time. The use of the word *clone* has been extended to recombinant DNA technology, which has provided scientists with the ability to produce many copies of a single fragment of DNA, such as a gene, creating identical copies that constitute a DNA clone. In practice the procedure is carried out by inserting a DNA fragment into a small DNA molecule and then allowing this molecule to replicate inside a simple living cell such as a bacterium. The small replicating molecule is called a DNA vector (carrier). The most commonly used vectors are plasmids (circular DNA molecules that originated from bacteria), viruses, and yeast cells. Plasmids are not a part of the main cellular genome, but they can carry genes that provide the host cell with useful properties, such as

The DNA test looks for the presence of HIV genetic material using reverse transcription polymerase chain reaction (RT-PCR). Development of the RT-PCR test was made possible by the molecular cloning and sequence analysis of HIV genomes.

drug resistance, mating ability, and toxin production. They are small enough to be conveniently manipulated experimentally, and, furthermore, they will carry extra DNA that is spliced into them.



Creating the Clone

The steps in cloning are as follows. DNA is extracted from the organism under study and is cut into small fragments of a size suitable for cloning. Most often this is achieved by cleaving the DNA with a restriction enzyme. Restriction enzymes are extracted from several different species and strains of bacteria, in which they act as defense mechanisms against viruses. They can be thought of as “molecular scissors,” cutting the DNA at specific target sequences. The most useful restriction enzymes make staggered cuts; that is, they leave a single-stranded overhang at the site of cleavage. These overhangs are very useful in cloning because the unpaired nucleotides will pair with other overhangs made using the same restriction enzyme. So, if the donor DNA and the vector DNA are both cut with the same enzyme, there is a strong possibility that the donor fragments and the cut vector will splice together because of the complementary overhangs. The resulting molecule is called recombinant DNA. It is recombinant in the sense that it is composed of DNA from two different sources. Thus, it is a type of DNA that would be impossible naturally and is an artifact created by DNA technology.

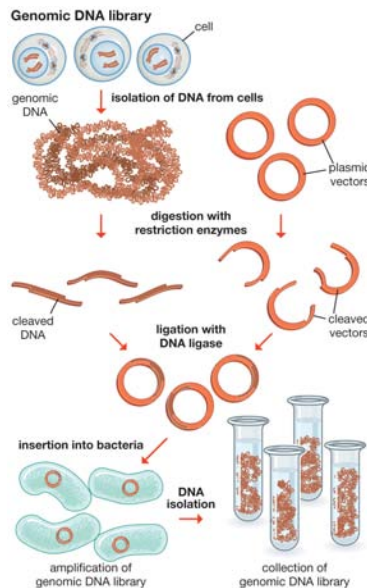
The next step in the cloning process is to cut the vector with the same restriction enzyme used to cut the donor DNA. Vectors have target sites for many different restriction enzymes, but the most convenient ones are those that occur only once in the vector molecule. This is because the restriction enzyme then merely opens up the vector ring, creating a space for the insertion of the donor DNA segment. Cut vector DNA and donor DNA are mixed in a test tube, and the complementary ends of both types of DNA unite randomly. Of course, several types of unions are possible: donor fragment to donor fragment, vector fragment to vector fragment, and, most important, vector fragment to donor fragment, which can be selected for. Recombinant DNA associations form spontaneously in the above manner, but these associations are not stable because, although the ends are paired, the sugar-phosphate backbone of the DNA has not been sealed. This is accomplished by the application of an enzyme called DNA ligase, which seals the two segments, forming a continuous and stable double helix.

The mixture should now contain a population of vectors each containing a different donor insert. This solution is mixed with live bacterial cells that have been specially treated to make their cells more permeable to DNA. Recombinant molecules enter living cells in a process called transformation. Usually, only a single recombinant molecule will enter any individual bacterial cell. Once inside, the recombinant DNA molecule replicates like any other plasmid DNA molecule, and many copies are subsequently produced. Furthermore, when the bacterial cell divides, all of the daughter cells receive the recombinant plasmid, which again replicates in each daughter cell.

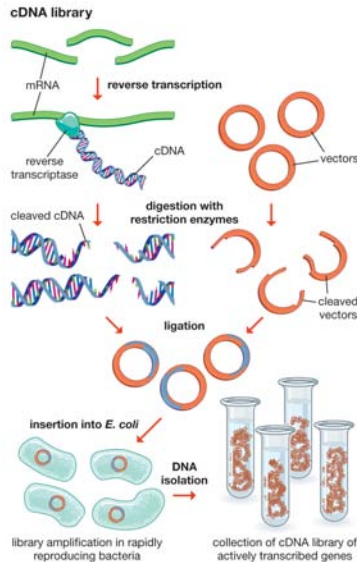
Clone is a group of identical cells that share a common ancestry, meaning they are derived from the same cell.

The original mixture of transformed bacterial cells is spread out on the surface of a growth medium in a flat dish (Petri dish) so that the cells are separated from

one another. These individual cells are invisible to the naked eye, but as each cell undergoes successive rounds of cell division, visible colonies form. Each colony is a cell **clone**, but it is also a DNA clone because the recombinant vector has now been amplified by replication during every round of cell division. Thus, the Petri dish, which may contain many hundreds of distinct colonies, represents a large number of clones of different DNA fragments. This collection of clones is called a DNA library. By considering the size of the donor genome and the average size of the inserts in the recombinant DNA molecule, a researcher can calculate the number of clones needed to encompass the entire donor genome, or, in other words, the number of clones needed to constitute a genomic library.



Another type of library is a cDNA library. Creation of a cDNA library begins with messenger ribonucleic acid (mRNA) instead of DNA. Messenger RNA carries encoded information from DNA to ribosomes for translation into protein. To create a cDNA library, these mRNA molecules are treated with the enzyme reverse transcriptase, which is used to make a DNA copy of an mRNA. The resulting DNA molecules are called complementary DNA (cDNA). A cDNA library represents a sampling of the transcribed genes, whereas a genomic library includes untranscribed regions.



Both genomic and cDNA libraries are made without regard to obtaining functional cloned donor fragments. Genomic clones do not necessarily contain full-length copies of genes. Furthermore, genomic DNA from eukaryotes (cells or organisms that have a nucleus) contains introns, which are regions of DNA that are not translated into protein and cannot be processed by bacterial cells. This means that even full-sized genes are not translated in their entirety. In addition, eukaryotic regulatory signals are different from those used by prokaryotes (cells or organisms lacking internal membranes—i.e., bacteria). However, it is possible to produce expression libraries by slicing cDNA inserts immediately adjacent to a bacterial promoter region on the vector; in these expression libraries, eukaryotic proteins are made in bacterial cells, which allows several important technological applications that are discussed below in DNA sequencing.

Several bacterial viruses have also been used as vectors. The most commonly used is the lambda phage. The central part of the lambda genome is not essential for the virus to replicate in *Escherichia coli*, so this can be excised using an appropriate restriction enzyme, and inserts from donor DNA can be spliced into the gap. In fact, when the phage repackages DNA into its protein capsule, it includes only DNA fragments the same length of the normal phage genome.

Vectors are chosen depending on the total amount of DNA that must be included in a library. Cosmids are engineered vectors that are hybrids of plasmid and phage lambda; however, they can carry larger inserts than either pUC plasmids (plasmids engineered to produce a very high number of DNA copies but that can accommodate only small inserts) or lambda phage alone. Bacterial artificial chromosomes (BACs) are vectors based on F-factor (fertility factor) plasmids of *E. coli* and can carry much larger amounts of DNA. Yeast artificial chromosomes (YACs) are vectors based on autonomously replicating plasmids of *Saccharomyces cerevisiae* (baker's yeast). In yeast (a eukaryotic organism) a YAC behaves like a yeast chromosome and segregates properly into daughter cells. These vectors can carry the largest inserts of all and are used extensively in cloning large genomes such as the human genome.

Isolating the Clone

In general, cloning is undertaken in order to obtain the clone of one particular gene or DNA sequence of interest. The next step after cloning, therefore, is to find and isolate that clone among other members of the library. If the library encompasses the whole genome of an organism, then somewhere within that library will be the desired clone. There are several ways of finding it, depending on the specific gene concerned. Most commonly, a cloned DNA segment that shows homology to the sought gene is used as a probe. For example, if a mouse gene has already been cloned, then that clone can be used to find the equivalent human clone from a human genomic library. Bacterial colonies constituting a library are grown in a collection of Petri dishes. Then a porous membrane is laid over the surface of each plate, and cells adhere to the membrane. The cells are ruptured, and DNA is separated into single strands—all on the membrane. The probe is also separated into single strands and labeled, often with radioactive phosphorus. A solution of the radioactive probe is then used to bathe the membrane. The single-stranded probe DNA will adhere only to the DNA of the clone that contains the equivalent gene. The membrane is dried and placed against a sheet of radiation-sensitive film, and somewhere on the films a black spot will appear, announcing the presence and location of the desired clone. The clone can then be retrieved from the original Petri dishes.

DNA Sequencing

Gene is the basic physical and functional unit of heredity.

Once a segment of DNA has been cloned, its nucleotide sequence can be determined. The nucleotide sequence is the most fundamental level of knowledge of a **gene** or genome. It is the blueprint that contains the instructions for building an organism, and no understanding of genetic function or evolution could be complete without obtaining this information.

Uses

Knowledge of the sequence of a DNA segment has many uses, and some examples follow. First, it can be used to find genes, segments of DNA that code for a specific protein or phenotype. If a region of DNA has been sequenced, it can be screened for characteristic features of genes. For example, open reading frames (ORFs)—long sequences that begin with a start codon (three adjacent nucleotides; the sequence of a codon dictates amino acid production) and are uninterrupted by stop codons (except for one at their termination)—suggest a protein-coding region. Also, human genes are generally adjacent to so-called CpG islands—clusters of cytosine and guanine, two of the nucleotides that make up DNA. If a gene with a known phenotype (such as a disease gene in humans) is known to be in the chromosomal region sequenced, then unassigned genes in the region will become candidates for that function. Second, homologous DNA sequences of different organisms can be compared in order to plot evolutionary relationships both within and between species. Third, a gene sequence can be screened for functional regions. In order to determine the function of a gene, various domains can be identified that are common to proteins of similar function. For example, certain amino acid sequences within a gene are always found in proteins that span a cell membrane;

such amino acid stretches are called transmembrane domains. If a transmembrane domain is found in a gene of unknown function, it suggests that the encoded protein is located in the cellular membrane. Other domains characterize DNA-binding proteins. Several public databases of DNA sequences are available for analysis by any interested individual.

Methods

The two basic sequencing approaches are the Maxam-Gilbert method, discovered by and named for American molecular biologists Allan M. Maxam and Walter Gilbert, and the Sanger method, discovered by English biochemist Frederick Sanger. In the most commonly used method, the Sanger method, DNA chains are synthesized on a template strand, but chain growth is stopped when one of four possible dideoxy nucleotides, which lack a 3' hydroxyl group, is incorporated, thereby preventing the addition of another nucleotide. A population of nested, truncated DNA molecules results that represents each of the sites of that particular nucleotide in the template DNA. These molecules are separated in a procedure called electrophoresis, and the inferred nucleotide sequence is deduced using a computer.

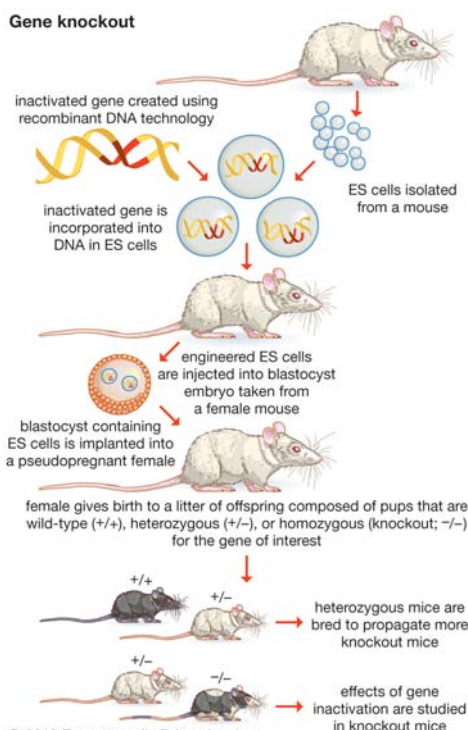
In Vitro Mutagenesis

Another use of cloned DNA is in vitro mutagenesis in which a mutation is produced in a segment of cloned DNA. The DNA is then inserted into a cell or organism, and the effects of the mutation are studied. Mutations are useful to geneticists in enabling them to investigate the components of any biological process. However, traditional mutational analysis relied on the occurrence of random spontaneous mutations—a hit-or-miss method in which it was impossible to predict the precise type or position of the mutations obtained. In vitro mutagenesis, however, allows specific mutations to be tailored for type and for position within the gene. A cloned gene is treated in the test tube (in vitro) to obtain the specific mutation desired, and then this fragment is reintroduced into the living cell, where it replaces the resident gene.

One method of in vitro mutagenesis is oligonucleotide-directed mutagenesis. A specific point in a sequenced gene is pinpointed for

mutation. An oligonucleotide, a short stretch of synthetic DNA of the desired sequence, is made chemically. For example, the oligonucleotide might have adenine in one specific location instead of guanine. This oligonucleotide is hybridized to the complementary strand of the cloned gene; it will hybridize despite the one base pair mismatch. Various enzymes are added to allow the oligonucleotide to prime the synthesis of a complete strand within the vector. When the vector is introduced into a bacterial cell and replicates, the mutated strand will act as a template for a complementary strand that will also be mutant, and thus a fully mutant molecule is obtained. This fully mutant cloned molecule is then reintroduced into the donor organism, and the mutant DNA replaces the resident gene.

Another version of *in vitro* mutagenesis is gene disruption, or gene knockout. Here, the resident functional gene is replaced by a completely nonfunctional copy. The advantage of this technique over random mutagenesis is that specific genes can be knocked out at will, leaving all other genes untouched by the mutagenic procedure.



Importance of DNA Cloning

The DNA molecules produced through the cloning techniques are used for many purposes which includes:

- There is need of more DNA copies for studying about the different gene, its functions, the cause of different diseases and for other research and experiments.
- Cloning livestock, and for reviving endangered or extinct species.
- Gene therapy used to treat some genetic disorders and replacement of a particular gene.
- In pharmaceuticals industries for the production of recombinant protein to produce human proteins with biomedical applications. For example:
- The gene from the human insulin is expressed into the *E. coli* bacteria in order to produce the insulin hormone which is used by diabetics.
- Human growth hormone is used to treat those patients who are unable to synthesize the growth hormone naturally.
- Tissue plasminogen activator is used both in preventing blood clots and treating strokes.
- Gene cloning in agriculture:
- Crop breeding by facilitating the direct changes to the genotype of a plant.
- Altering the characteristics of a plant by providing it with one or more new genes.
- By inactivating one or more of the plant's existing genes rather than removing them.

REVIEW QUESTIONS

1. Describe the recombinant DNA technology.
2. What is the process of recombinant DNA technology?
3. Define the applications of recombinant DNA technology.
4. Elaborate the DNA cloning.
5. What are the importance of DNA cloning?

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Level: Beginner to Advanced
Subject: Life Science

Core Concepts in Biology: Structure Of DNA

It has been argued that the discovery of DNA, as well as our understanding of its structure and functioning, could well be the most important discovery of the last century. It was hoped that understanding of the structure would show that DNA holds the genetic signals that are repeated as chromosomes split and produce two identical copies of themselves. In the late 1940s and early 1950s, numerous research groups in the United States and Europe have made serious efforts— both collaborative and rival— to explain how the atoms of DNA are bound together by covalent bonds and how the resulting molecules are formed in three-dimensional space.

Not surprisingly, there were initially concerns that DNA might have very complex and strange structures that were radically different from one gene to another. Great relief was therefore demonstrated when the basic DNA structure was found to be a double helix. This taught us that all genes have approximately the same three-dimensional structure and that the variations between the two genes exist in the order and number of their four nucleotide blocks along the complementary strands. The effect of the discovery of DNA on scientific and medical advancement has been immense, whether it is the detection of genes that cause major diseases or the creation and manufacture of drugs to treat these devastating diseases. In addition, the discovery of these genes and their subsequent study in terms of therapeutic treatment has eventually influenced and will continue to influence research in the future. DNA was particularly important in the field of forensic science.

The discovery of DNA ensures that it is possible to determine the guilt or innocence of a person who is being investigated for a crime. It also means that scant evidence can still provide valuable information as to the perpetrator of the crime. It is also important that victim identification can occur, especially in cases where the condition of the victim is unrecognizable to family or friends. In this sense, DNA has been important in revolutionizing the entire field of forensic science. This impact is felt within the criminal justice system and contributes to the accurate safeguarding of society.

Core Concepts in Biology Structure of DNA examines the structure of the DNA molecule and explain in general terms how it is able to store hereditary information. Obviously, the DNA and RNA structures are richer and more complex than was initially seen. Yes, there is no single standard DNA and RNA structure. As we will see in this book, there are in fact variations on standard structural themes that emerge from the unique physical, chemical and topological properties of the polynucleotide chain. An important book for living in today's world, this book provides background information that is critical to understanding how genetics is now influencing our everyday lives.

